

Chapter 1. An Introduction to Combustion Chemistry

Yuyang Li

**School of Mechanical Engineering
Shanghai Jiao Tong University**

Statement: This document is only used for the teaching of the Combustion Chemistry course, and is not permitted for the use of any commercial purpose or other non-teaching purposes. Some figures are taken from the internet and the sources are not listed. Any duplication, direct use by other people for teaching, or other behaviors to spread this document should get the permission.

Targets of The Course

- n Understand the fundamental knowledge of combustion chemistry**
- n Have the basic idea how to built, simulate, validate and analyze a kinetic model**
- n Be possible to remember some information of an experimental method when you face its apparatus**
- n Know the state-of-the-art research in this field**
- n Be aware of the current or potential use of this tool in your research**
- n Find some opportunities to perform some novel research in this field if we are lucky**

Outlines

- 1. Introduction**
- 2. A Brief History**
- 3. Research Methods**
- 4. Applications**
- 5. Challenges & Opportunities**
- 6. Recommended Literature**
- 7. Following Chapters**

1. Introduction

I Combustion Around Our Life

Companion in Learning and Romance



Provider of Light, Heat and Joy



Power for Mobility



Furnace for Industry, Pleasure, and Food



Agent of Destruction



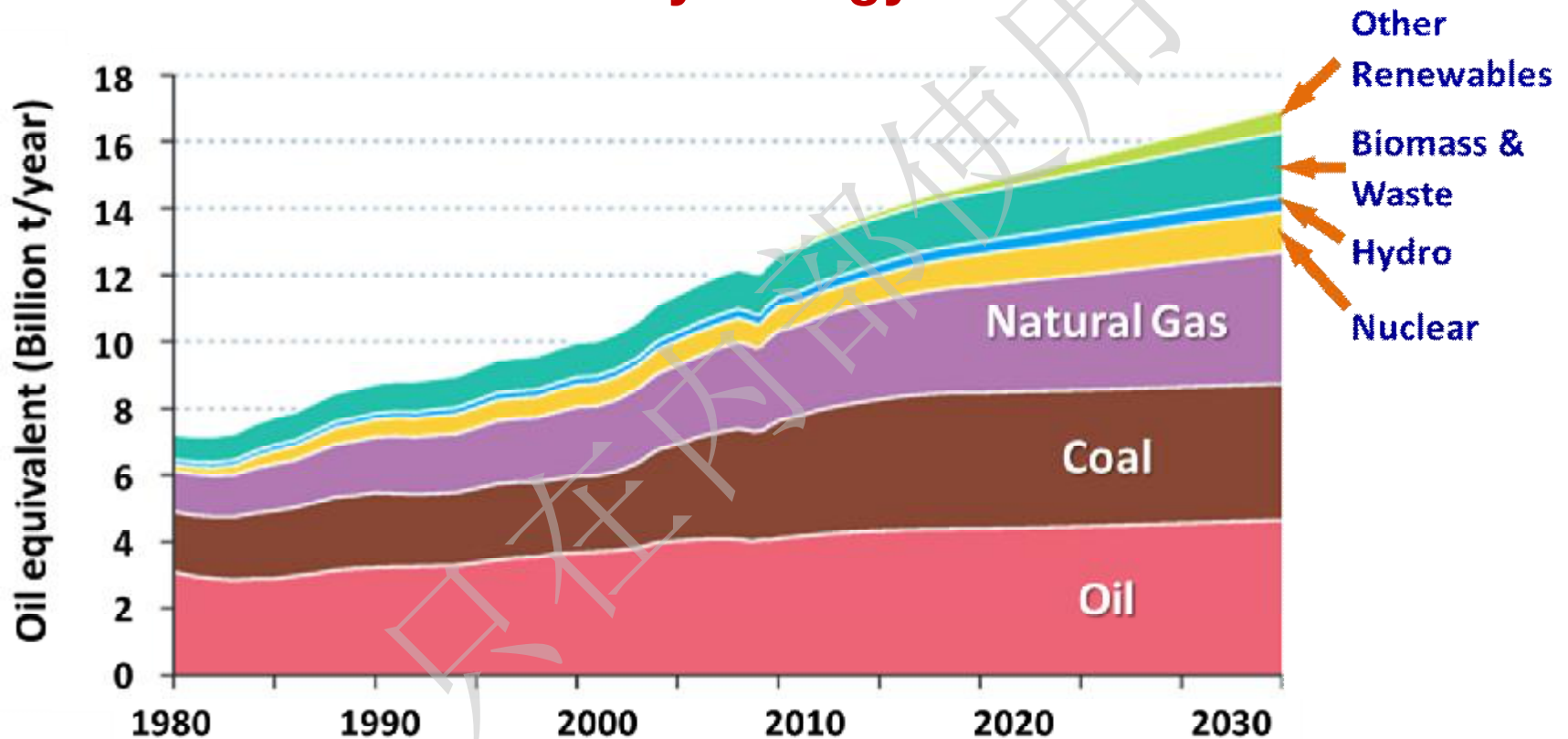
Origin of Environment Problems



1. Introduction

I Combustion, Energy and Fuel

Global Primary Energy Demand

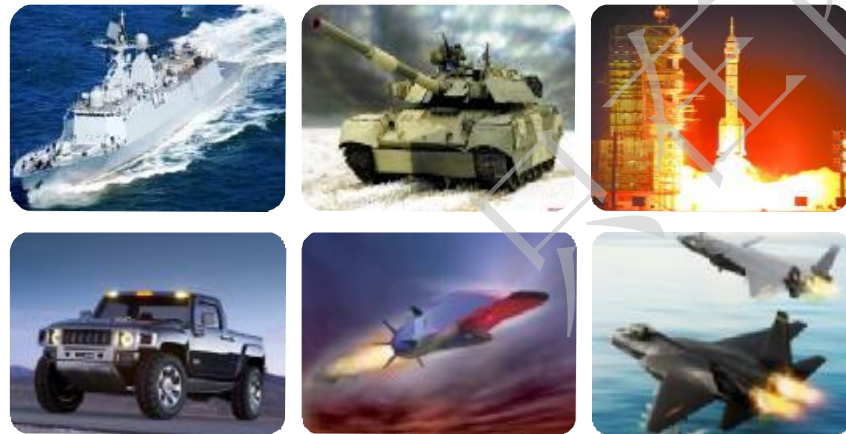
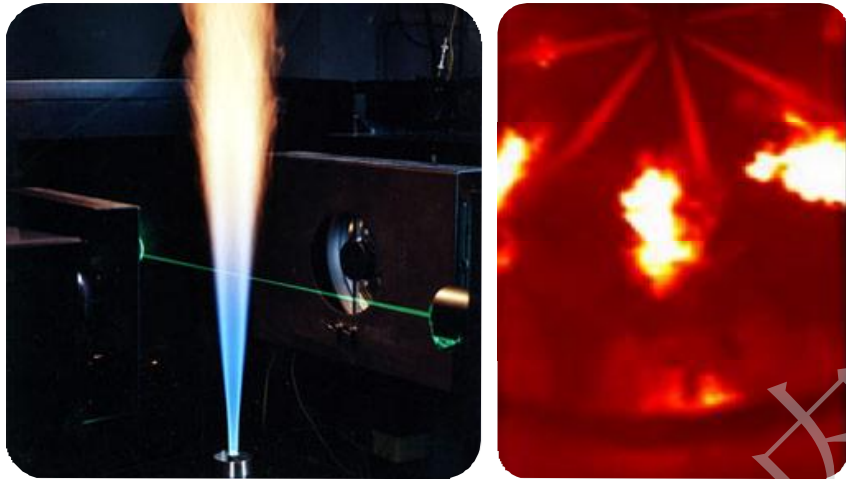


- q Combustion provides >85% of global primary energy consumption
- q This key role will be maintained in the foreseeable future

1. Introduction

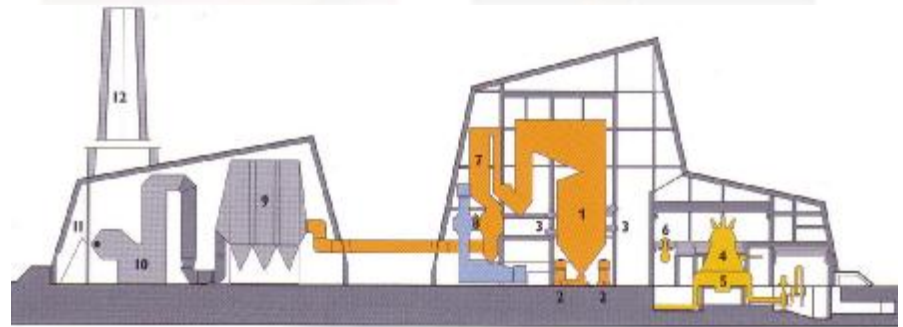
I Combustion, Energy and Fuel

Gas and liquid fuels



Transportation

Coal and other solid fuels

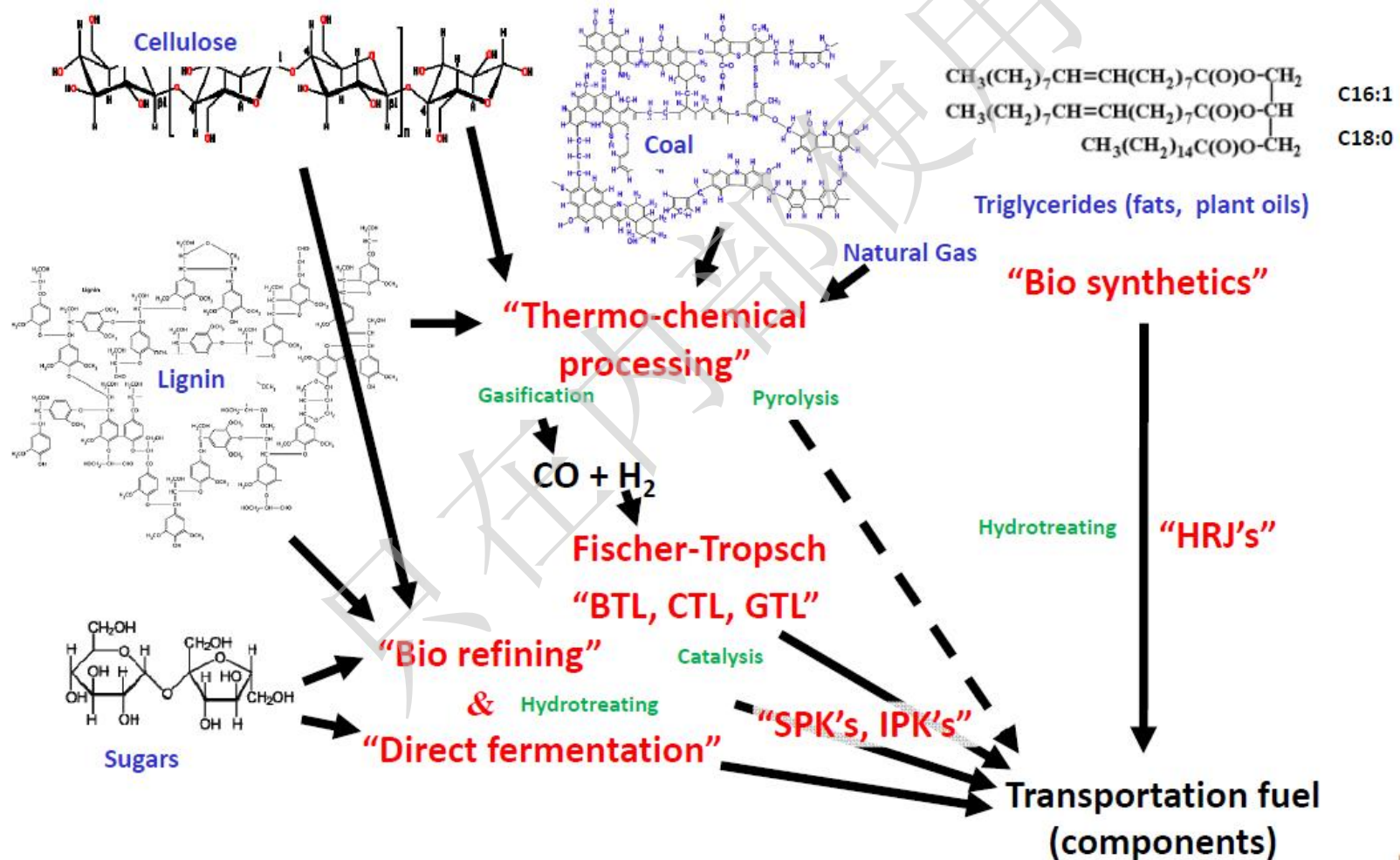


Power plant

1. Introduction

I Combustion, Energy and Fuel

Renewable/Alternative Transportation Fuels



1. Introduction

I Combustion and Environment



Air pollutant emissions from vehicles in China (Year 2017)

Pollutant	Emissions (Mton)
NOx	5.743
UHC	4.071
CO	33.273
PM	0.509
Total	43.597

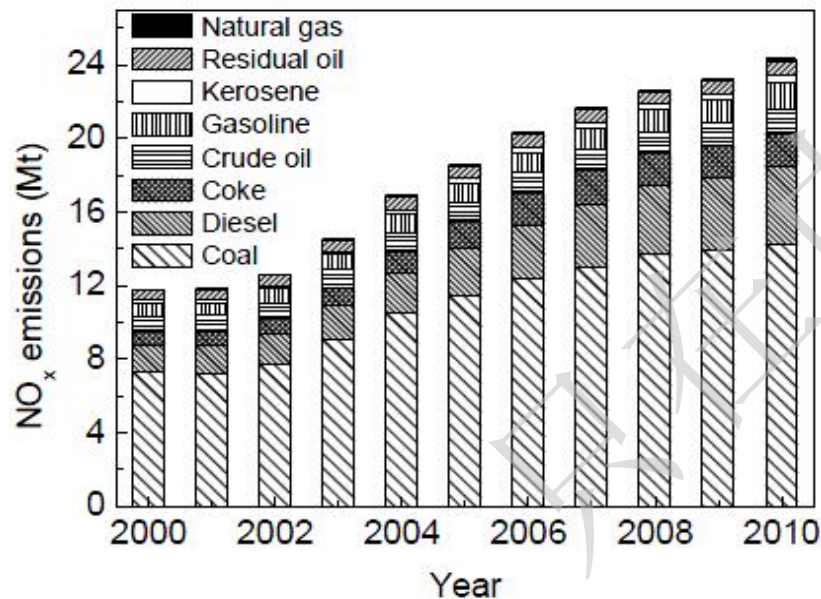
Combustion produces a great amount of air pollutants, such as NOx, SOx, VOCs, PAHs, Soot

1. Introduction

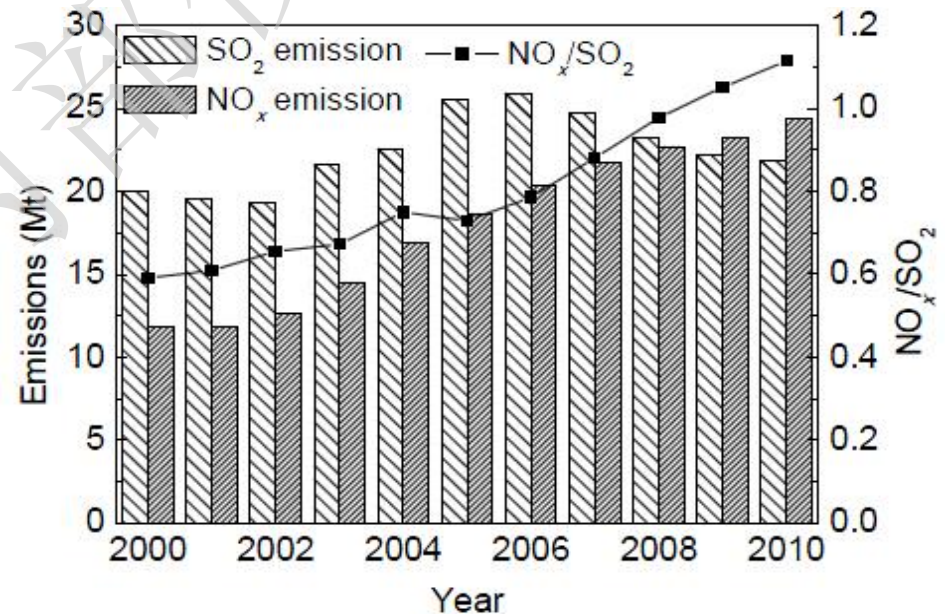
I Combustion and Environment

n Combustion is the main emission source for NO_x and SO_x

NO_x emission from fossil fuel combustion in China



Comparison of the trends of NO_x and SO_x emissions



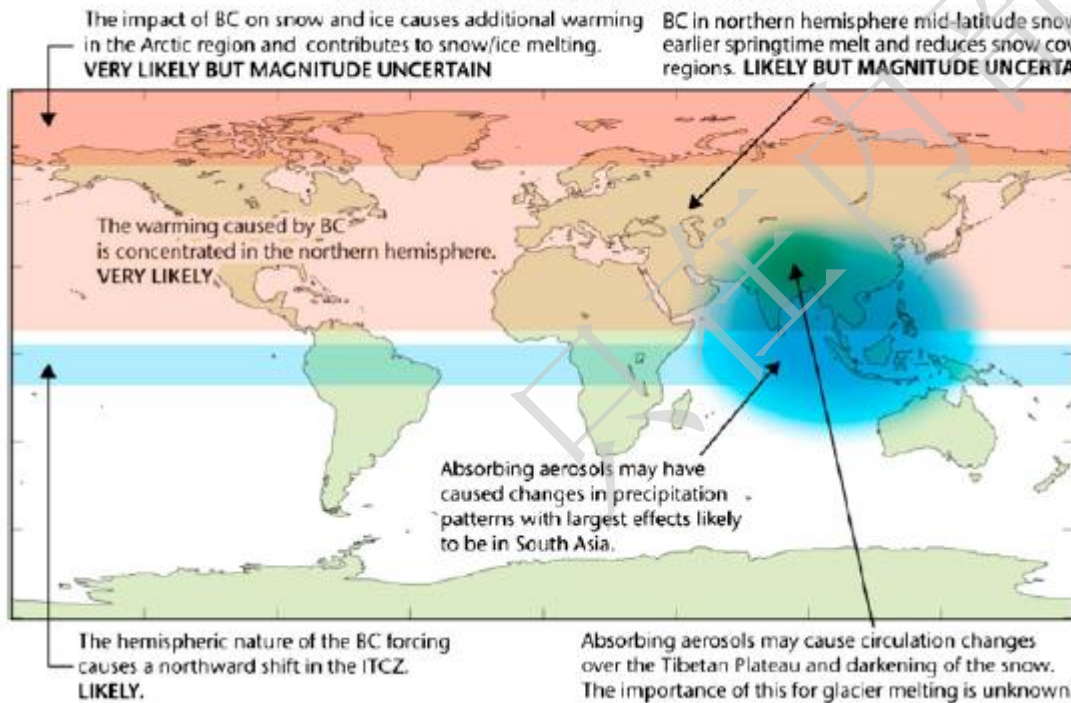
1. Introduction

I Combustion and Environment

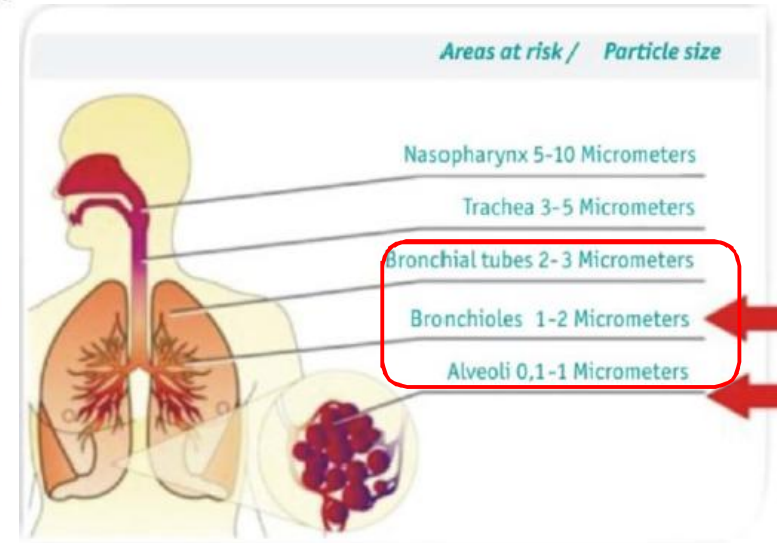
- Combustion is the main emission source for soot (black carbon)
- Soot is known to be carcinogenic and mutagenic and is a major contributor to particulate matter 2.5 (PM_{2.5})

Impact environment

Climate effects of black carbon emissions

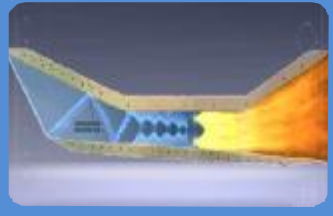


Impact human health



1. Introduction

I Combustion and Power

IC engine	Gas turbine	Aircraft engine	Scramjet engine	Rocket engine
				
				
Emissions	Compression Ratio	Thrust-Weight Ratio	Mach Number	Actual Load
Clean combustion	High density combustion	High temperature rise combustion	Supersonic combustion	Stable combustion

n High performance, high efficiency and low emissions combustion

1. Introduction

I Principle of Combustion

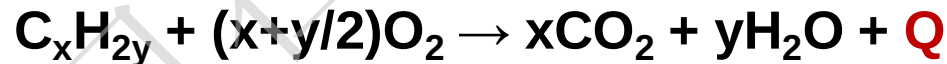


H. W. Emmons
(1912-1998)

Combustion phenomena are basically **chemical reactions** with rapid heat production.

H. W. Emmons, Proc. Combust. Inst. 13 (1971) 1.

- p** Combustion process: convert chemical energy to thermal energy

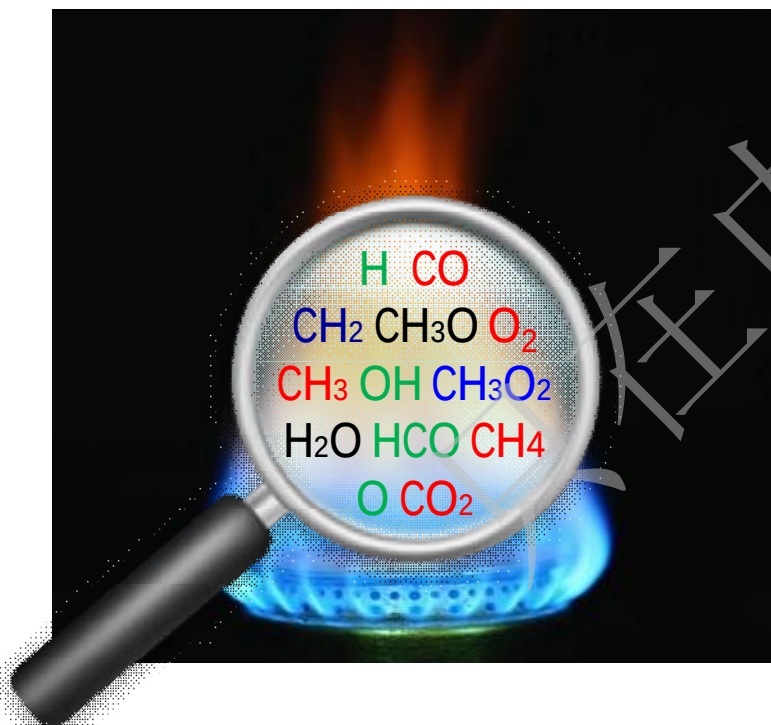


- p** A complicated sequence of **elementary reactions**

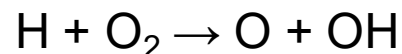
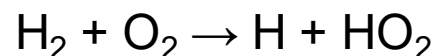
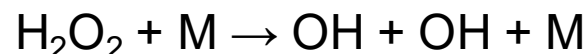
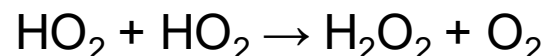
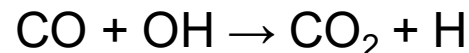
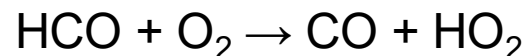
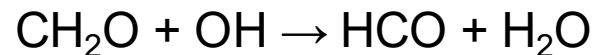
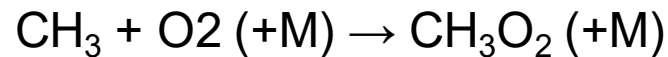
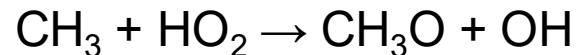
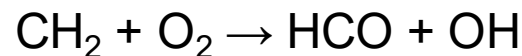
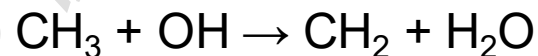
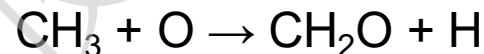
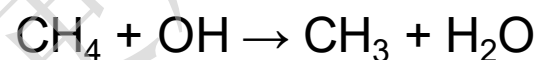
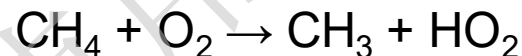
1. Introduction

I Principle of Combustion

Global Reaction



Chain Reactions



1. Introduction

I Principle of Combustion

∅ Thousands of species and tens of thousands of reactions involved in combustion process.

∅ Optical spectroscopy, gas chromatography and mass spectrometry have been used for diagnostics.

- | ignition
- | extinction
- | pollutants
- | propagation
- |



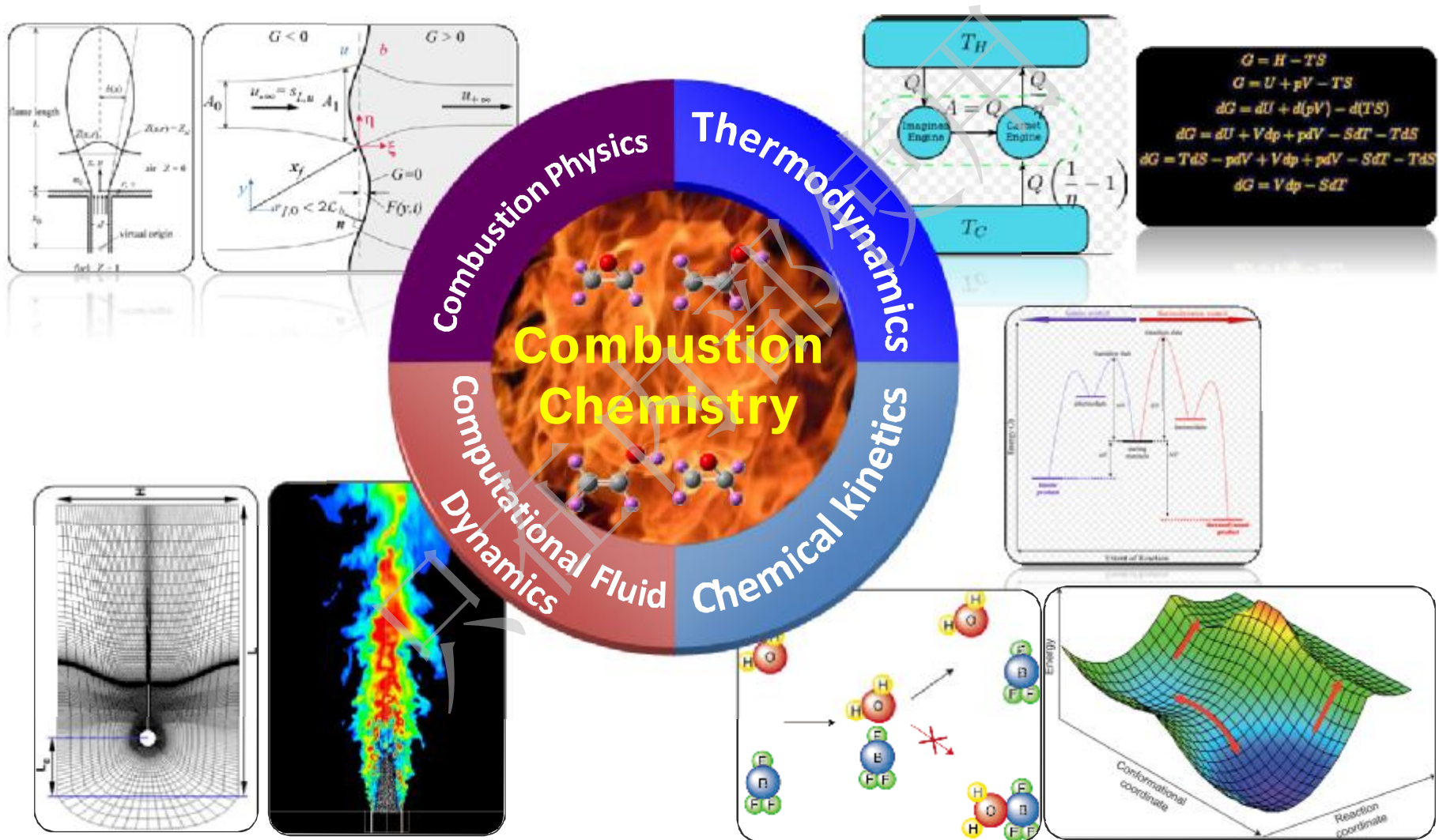
- u reactions
- u turbulent flow
- u mass transport
- u heat radiation

1 Pyrolysis

2 Low-T oxidation

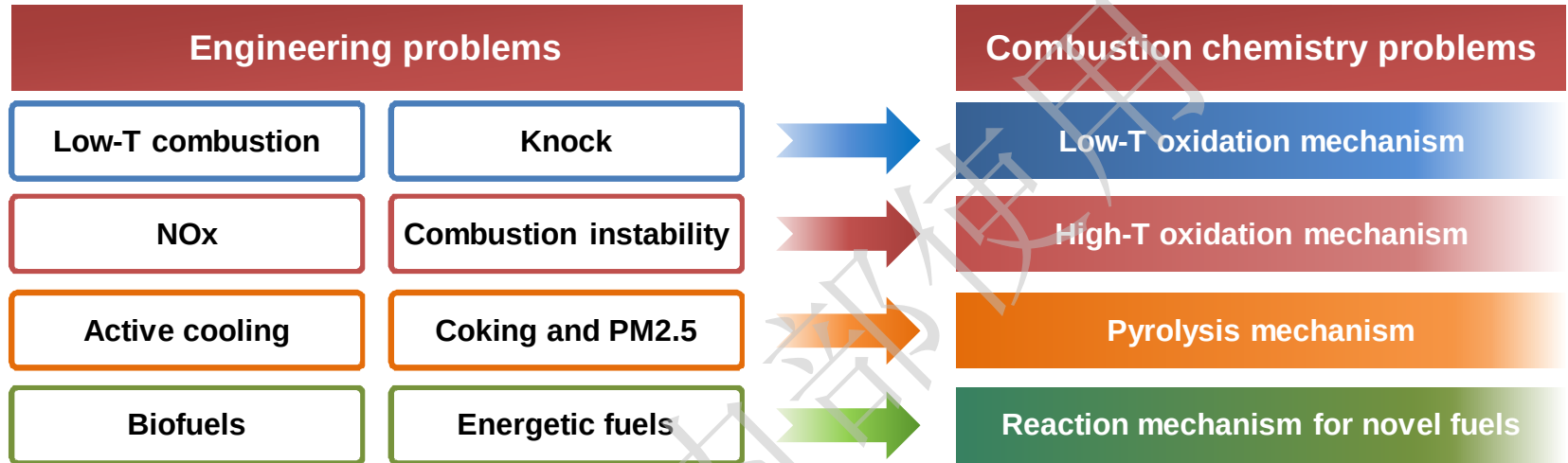
3 High-T oxidation

I Crossing of Chemistry/Fluid Dynamics/Thermophysics

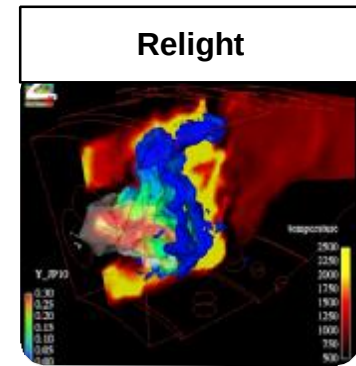
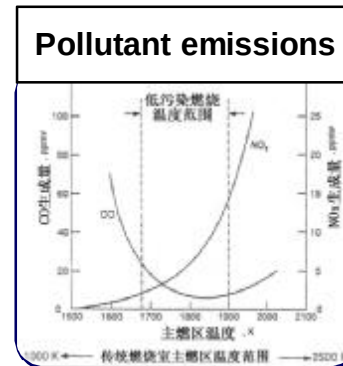
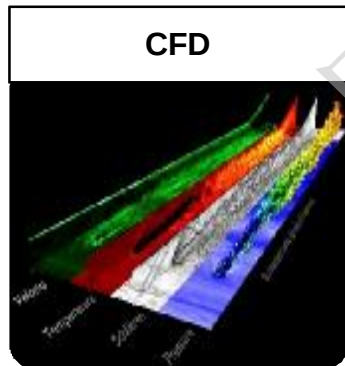


1. Introduction

I Combustion Chemistry in Engineering Problems



Aircraft engine research problems related to combustion chemistry



2. A Brief History

I Ancient Use and Thinking of Combustion

n An irreplaceable technology for the development of human civilization

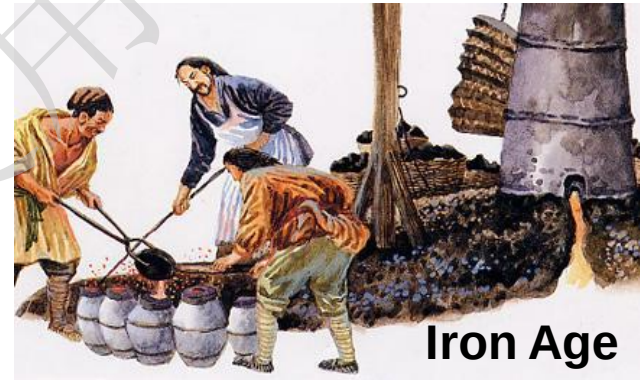
Stone Age



Bronze Age



Iron Age



n A lot of myths about the origin of fire



燧人氏



Prometheus



2. A Brief History

I Ancient Use and Thinking of Combustion

Ancient element concepts

Ancient China



Ancient Greece

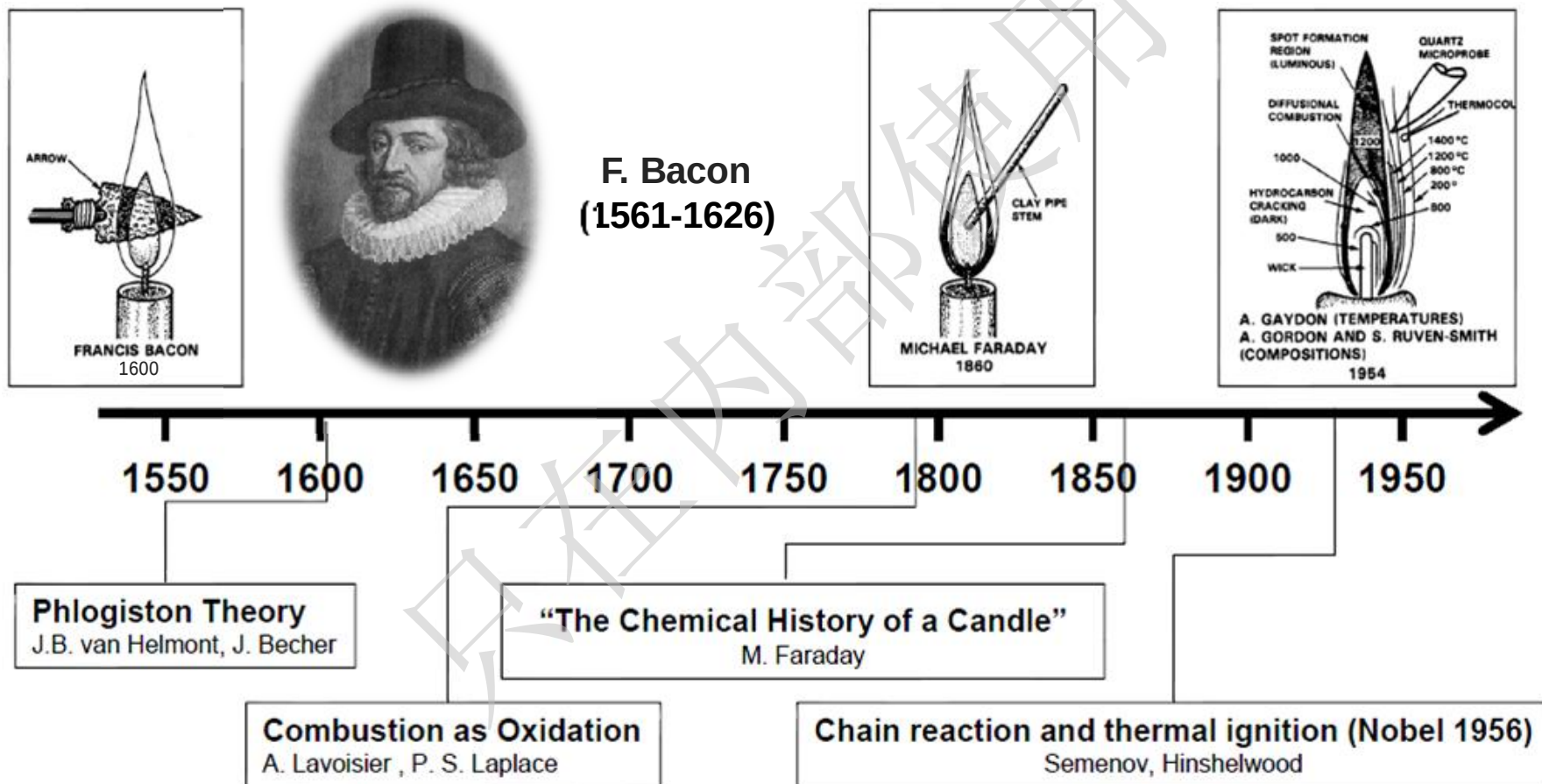


- n Fire is one of the classical elements with hot and dry features
- n The classical elements can be transformed by changing their features

2. A Brief History

I Time Line of Combustion Chemistry Research

n A long way since people would like to know what happened in combustion



→ Theory of gas kinetics, free radicals, elementary reactions and reaction mechanisms for combustion applications

2. A Brief History

I Phlogiston Theory

p 1667, J. J. Becher, Germany

p 1703, G. E. Stahl, Germany

Substances contain phlogiston & dephlogisticated air when burned

charcoal (+ II) + air = fixed air(CO_2) + phlogisticated air
metal (+ II) + air = calx (pure base) + phlogisticated air
charcoal (+II) + calx = metal (+II) + fixed air
metal (+ II) + water = calx + inflammable air (II)
metal (+ II) + acid = salt + inflammable air (II)
mercury (+II) + dephlogisticated air = calx of mercury
inflammable air (+II) + dephlogisticated air = water

J. J. Becher
(1635-1682)

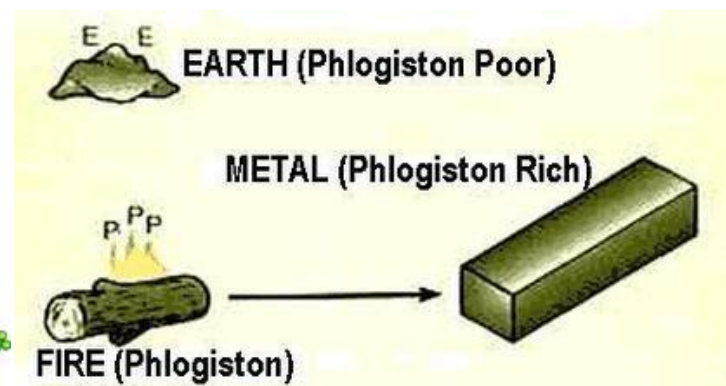


G. E. Stahl
(1659-1734)



Challenges

- n What is the principle of phlogiston?
- n Why some substances gain mass when they burned?
- n Why the volume of air is reduced during combustion?



2. A Brief History

I Oxidation Theory

p 1777, A. Lavoisier, France

- n The gain in weight by combustion is the result of combination with a large quantity of oxygen
- n The mass of the reactants equals the mass of the products

A. Lavoisier
(1743-1794)



*Father of
modern chemistry*

J. Priestley
(1733-1804)



*Discoverer of
oxygen (1774)*



Beginning of combustion science and modern chemistry

2. A Brief History

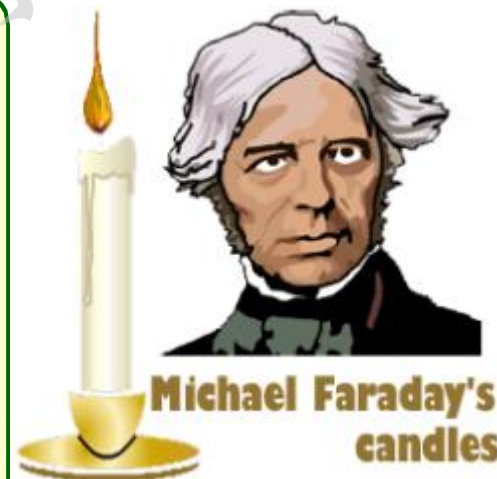
I The Chemical History of a Candle

p 1861, M. Faraday, England

The Chemical History of a Candle

- Lecture I A Candle: The Flame—Its Sources—Structure—Mobility—Brightness
- Lecture II A Candle: Brightness of the Flames—Air Necessary for Combustion—Production of Water
- Lecture III Products: Water from the Combustion—Nature of Water—A Compound—Hydrogen
- Lecture IV Hydrogen in the Candle—Burns into Water—The Other Part of Water—Oxygen
- Lecture V Oxygen Present in the Air—Nature of the Atmosphere—Its properties—Other Products from the Candle—Carbonic Acid—Its Properties
- Lecture VI Carbon or Charcoal—Coal-Gas—Respiration and its Analogy to the Burning of a Candle—Conclusion

M. Faraday
(1791-1867)



2. A Brief History

I Modern Combustion Chemistry

Modern combustion chemistry based on theories of chemical kinetics

1920s



Semenov

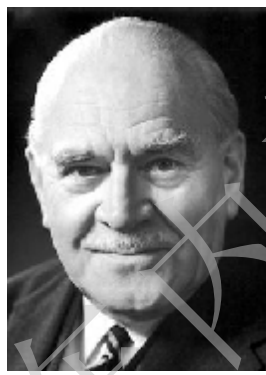


Hinshelwood

Chain reaction and thermal ignition

The Nobel Prize in Chemistry 1956

1940s



Norrish



Porter

Extremely fast chemical reactions and hydrocarbon combustion theory

The Nobel Prize in Chemistry 1967

1940-50s



Zeldovich



Frank-Kamenetskii

Thermal ignition and pollutant formation

2. A Brief History

I Modern Combustion Chemistry

Founders

Chain reaction and thermal ignition

First experimental evidence



N. N. Semenov (Russia, 1896-1986)

- p Phosphorus vapour-oxygen reaction (1926)
- p Explosion occurs, if the partial pressure of O_2 is between two limits
- p Interpretation via a branching chain reaction



C. N. Hinshelwood (England, 1897-1967)

- p H_2 - O_2 reaction (1927)
- p Discovery of the 1st and 2nd explosion limits

The Nobel Prize in Chemistry 1956

"for their researches into the mechanism of chemical reactions"

2. A Brief History

I Modern Combustion Chemistry

Major Contributors

Introduce chemical kinetics to study the ignition, detonation, explosion, flame propagation and pollutant formation

Y. B. Zel'dovich
(1914-1987)

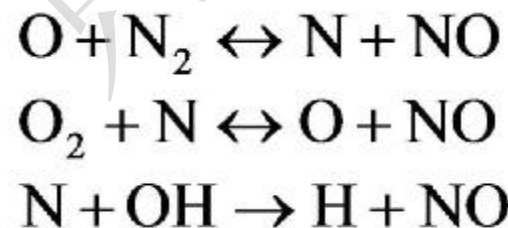
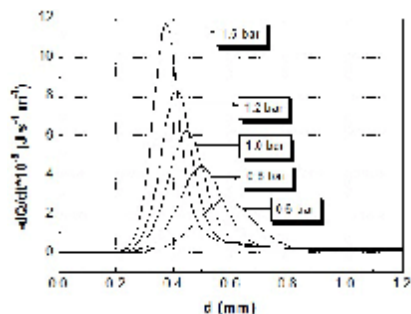
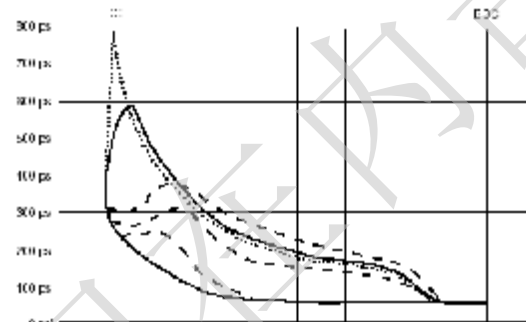
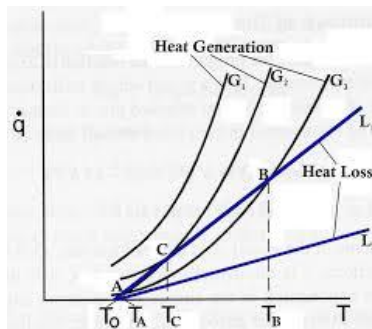


D. A. Frank-Kamenetskii
(1910-1970)



Bernard Lewis
(1899-1993)

Father of *the Combustion Institute*



2. A Brief History

I Modern Combustion Chemistry

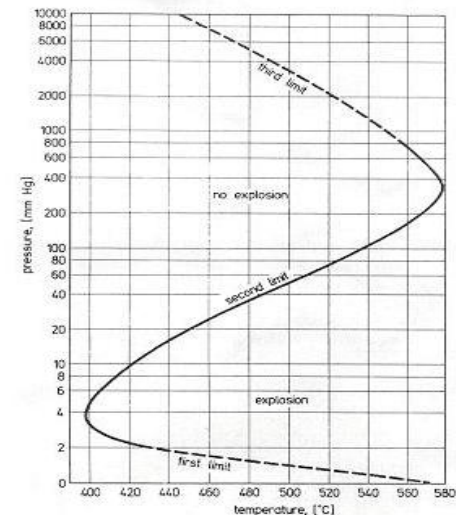
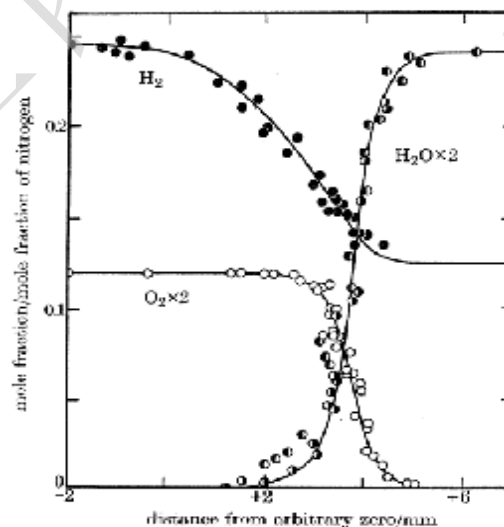
Development of kinetic models for combustion (1960s)

- n Kinetic models widely built to mimic the reaction network in combustion since the pioneering work of Dixon-Lewis in 1967
- n Rapid development in computer-assisted combustion simulation
- n Stiff equation solvers emerged in 1970s



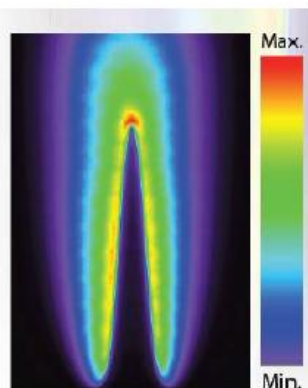
G. Dixon-Lewis
(1922-2010)

		ΔH (kcal/mole)
(i)	$\text{OH} + \text{H}_2 \rightleftharpoons \text{H}_2\text{O} + \text{H}$	- 14.6
(ii)	$\text{H} + \text{O}_2 = \text{OH} + \text{O}$	- 18.0
(iii)	$\text{O} + \text{H}_2 = \text{OH} + \text{O}$	- 1.8
(iv)	$\text{H} + \text{O}_2 + M = \text{HO}_2 + M$	- 47.2
(v)	$\text{HO}_2 + \text{HO}_2 = \text{H}_2\text{O}_2 + \text{O}_2$	- 42.3
(vi)	$\text{HO}_2 + \text{H}_2 = \text{H}_2\text{O}_2 + \text{H}$	+ 14.7
(vii)	$\text{H} + \text{HO}_2 = \text{OH} + \text{OH}$	- 39.2
(viii)	$\text{H}_2\text{O}_2 + M = \text{OH} + \text{OH} + M$	+ 50.3
(ix)	$\text{H} + \text{H}_2\text{O}_2 = \text{H}_2\text{O} + \text{OH}$	- 68.5
(x)	$\text{H} + \text{H}_2\text{O}_2 = \text{HO}_2 + \text{H}_2$	- 14.7
(xi)	$\text{OH} + \text{H}_2\text{O}_2 = \text{HO}_2 + \text{H}_2\text{O}$	- 29.3
(xii)	$\text{H} + \text{HO}_2 = \text{H}_2 + \text{O}_2$	- 57.0
(xiii)	$\text{H} + \text{H} + M = \text{H}_2 + M$	- 104.2
(xiv)	$\text{H} + \text{OH} + M = \text{H}_2\text{O} + M$	- 118.8



2. A Brief History

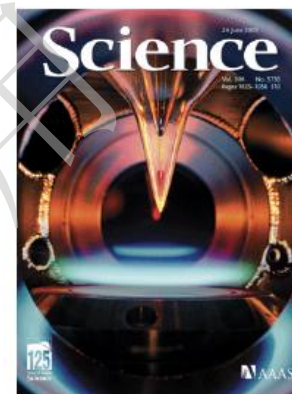
I Combustion Diagnostics & Computation



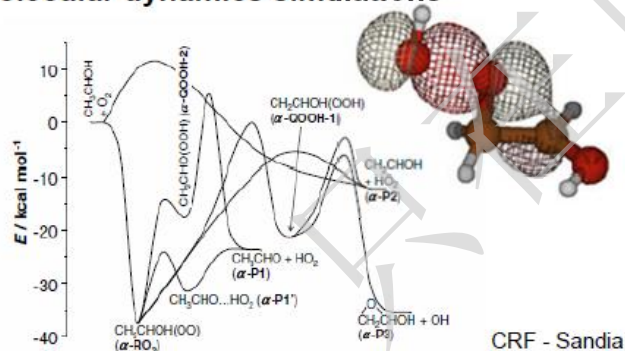
Interference-free composite H-atom LIF image produced with ps excitation in a $\Phi=1.2$ premixed $\text{CH}_4/\text{O}_2/\text{N}_2$ flame.

CRF - Sandia

Molecular-beam flame sampling and synchrotron-photoionization mass spectrometry analysis
[Science - 24 June 2005]

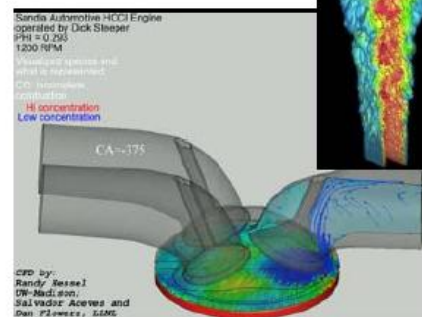


Ab initio electronic-structure calculations and molecular dynamics simulations



CRF - Sandia

CFD calculations: RANS, LES, DNS. Coupling of numerical chemistry and fluid dynamics



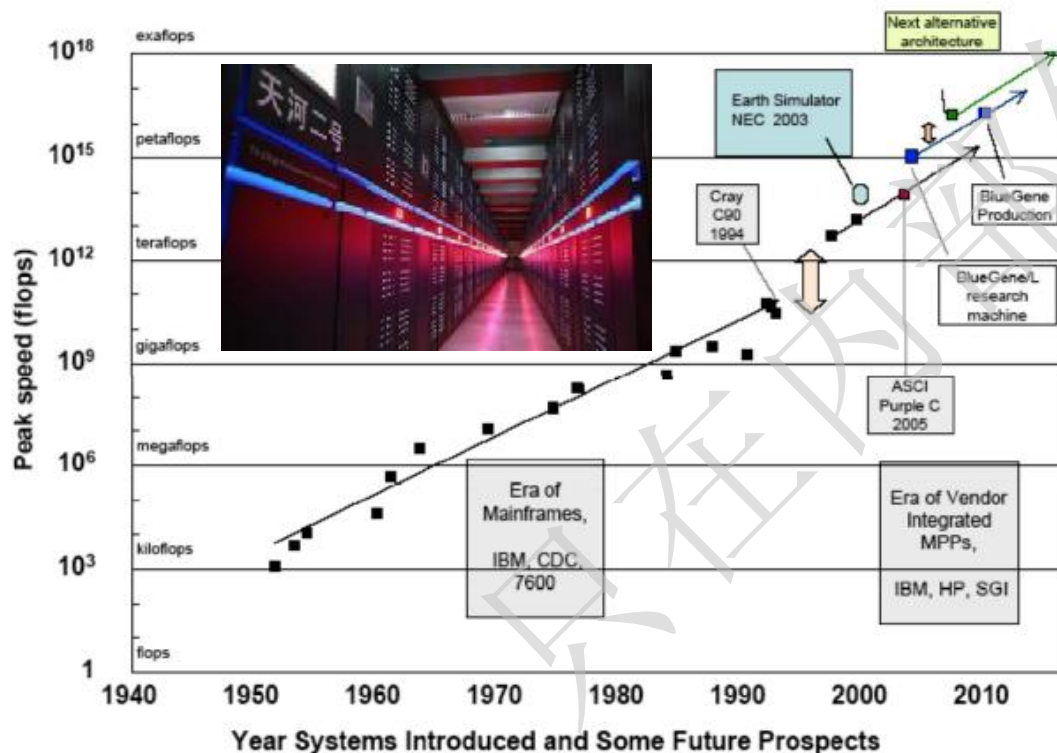
CRF - Sandia

Advanced diagnostic methods and computational chemistry

2. A Brief History

I Benefitting from Growth of Computing Power

Exponential growth in computing power over past 50 years



Gas Phase:

- Multicomponent molecular diffusion
- Complex chemical kinetics (e.g. 50 species, 250 reactions)
- Non-luminous thermal radiation
- UV flame emission

Mass Conservation:

$$\frac{\partial}{\partial t}(\rho_g) + \frac{1}{r^2} \frac{\partial}{\partial r}(r^2 \rho_g v_r) = 0$$

Species Equations:

$$\rho_g \frac{\partial Y_{g,i}}{\partial t} + \rho_g v_r \frac{\partial Y_{g,i}}{\partial r} = -\frac{1}{r^2} \frac{\partial}{\partial r}(r^2 \rho_g Y_{g,i} V_{r,i}) + \omega_{g,i}$$

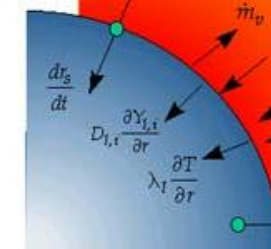
Energy Conservation:

$$\rho_g C_{p,g} \frac{\partial T_g}{\partial t} + \rho_g C_{p,g} v_r \frac{\partial T_g}{\partial r} = \frac{1}{r^2} \frac{\partial}{\partial r}(r^2 \lambda_g \frac{\partial T_g}{\partial r} - q_R) - \rho_g \sum_{i=1}^n (Y_{g,i} V_{r,i} C_{p,g,i}) \frac{\partial T_g}{\partial r} - \sum_{i=1}^n \omega_{g,i} H_{g,i}$$

Droplet Surface:

- Surface regression
- Evaporation of fuel
- Condensation of products
- Radiative heat addition

Net radiative heat flux



Droplet Interior:

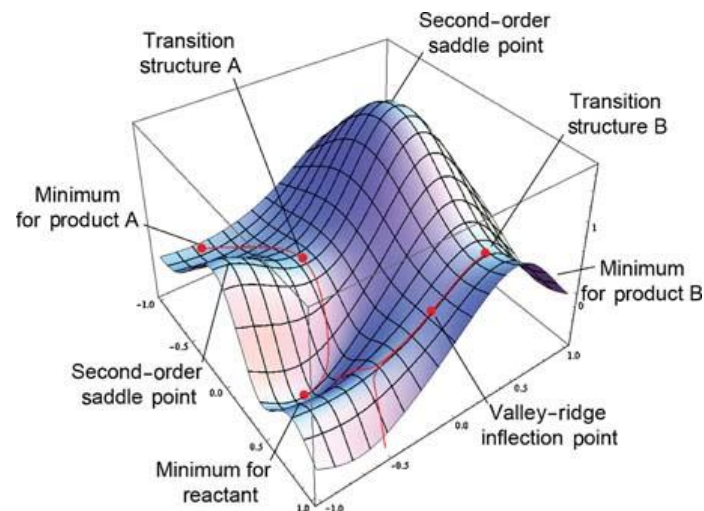
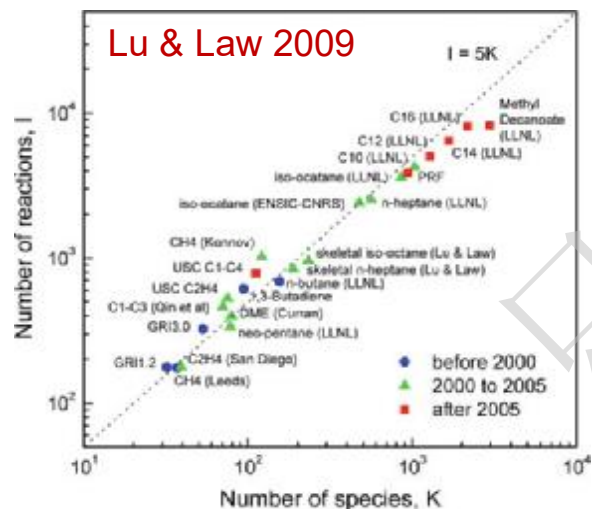
$$\rho_l C_{p,l} \frac{\partial T_l}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r}(r^2 \lambda_l \frac{\partial T_l}{\partial r})$$

$$\frac{\partial Y_{l,i}}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r}(r^2 D_{l,i} \frac{\partial Y_{l,i}}{\partial r})$$

2. A Brief History

I Status and Trends

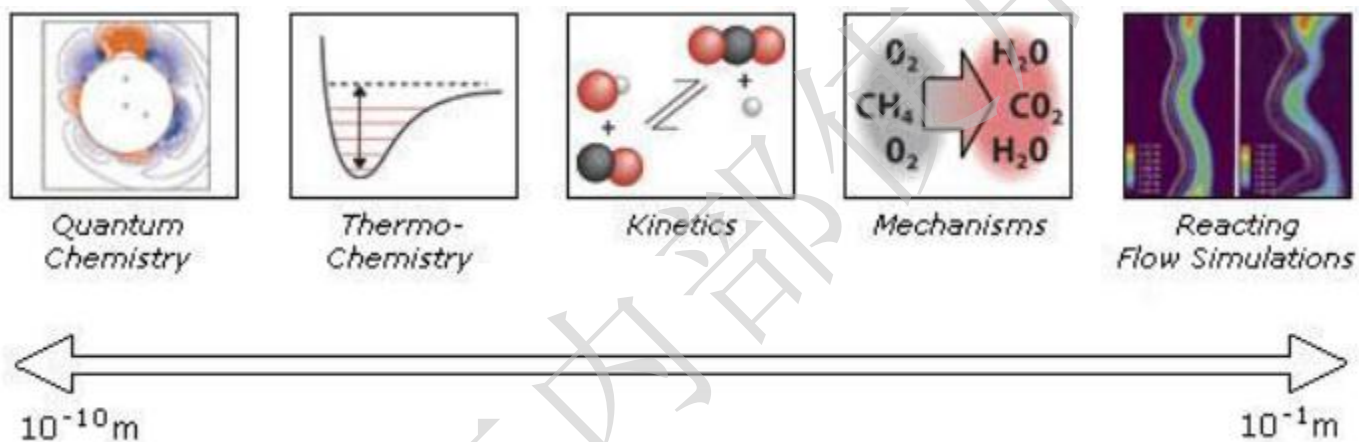
- n Number and size of model grow rapidly, kinetic models become more and more detail with comprehensiveness
- n Beyond hydrocarbon fuels (fossil fuels): oxygenated fuels (biofuels), energetic fuels ...
- n Emphasis both chemical-rigorous and practical use



2. A Brief History

I Status and Trends

Multi-scale research



- n Get accurate rate constants for elementary reactions
- n Get accurate and abundant information about the kinetics and intermediates
- n Address the comprehensiveness of a kinetic model
- n Achieve good prediction ability for reacting flow simulations

2. A Brief History

I Status and Trends

- n Surrogate fuel model**
- n Pollutant formation mechanism**
- n Uncertainty analysis of rate constant and model simulation**
- n Model reduction and dynamic adaptive simulation**
- n Chemistry in non-conventional combustion, such low-T and high-P combustion, catalytic combustion, plasma-assisted combustion, oxygen-enrichment combustion ...**
- n Turbulence-chemistry interaction...**

3. Research Methods

Flow reactor

Jet-stirred reactor

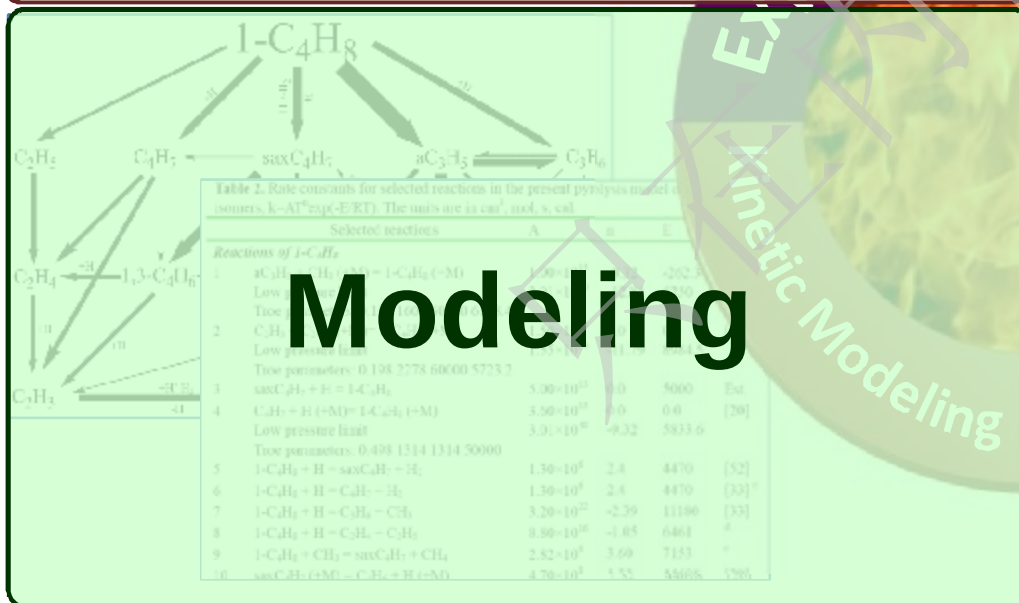
Premixed/nonpremixed flame

Shock tube/RCM

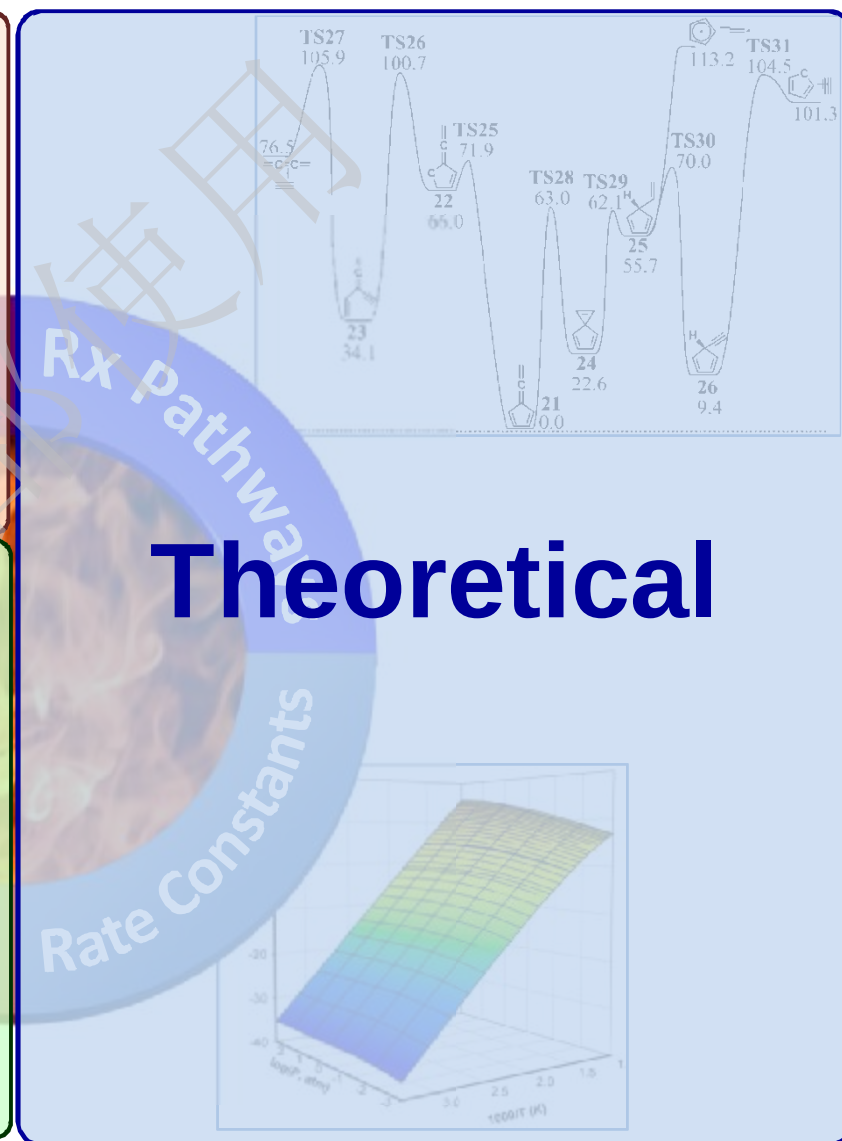
Combustion vessel

...

Experimental



Modeling



Theoretical

3.1. Theoretical methods

I Targets of Theoretical Methods

Targets of theoretical study

- n Potential energy surface: Energy of reaction pathway
- n Rate constant calculation: Kinetic parameters
- n Thermodynamic data calculation: Thermodynamic parameters

Development of quantum chemical computation method:

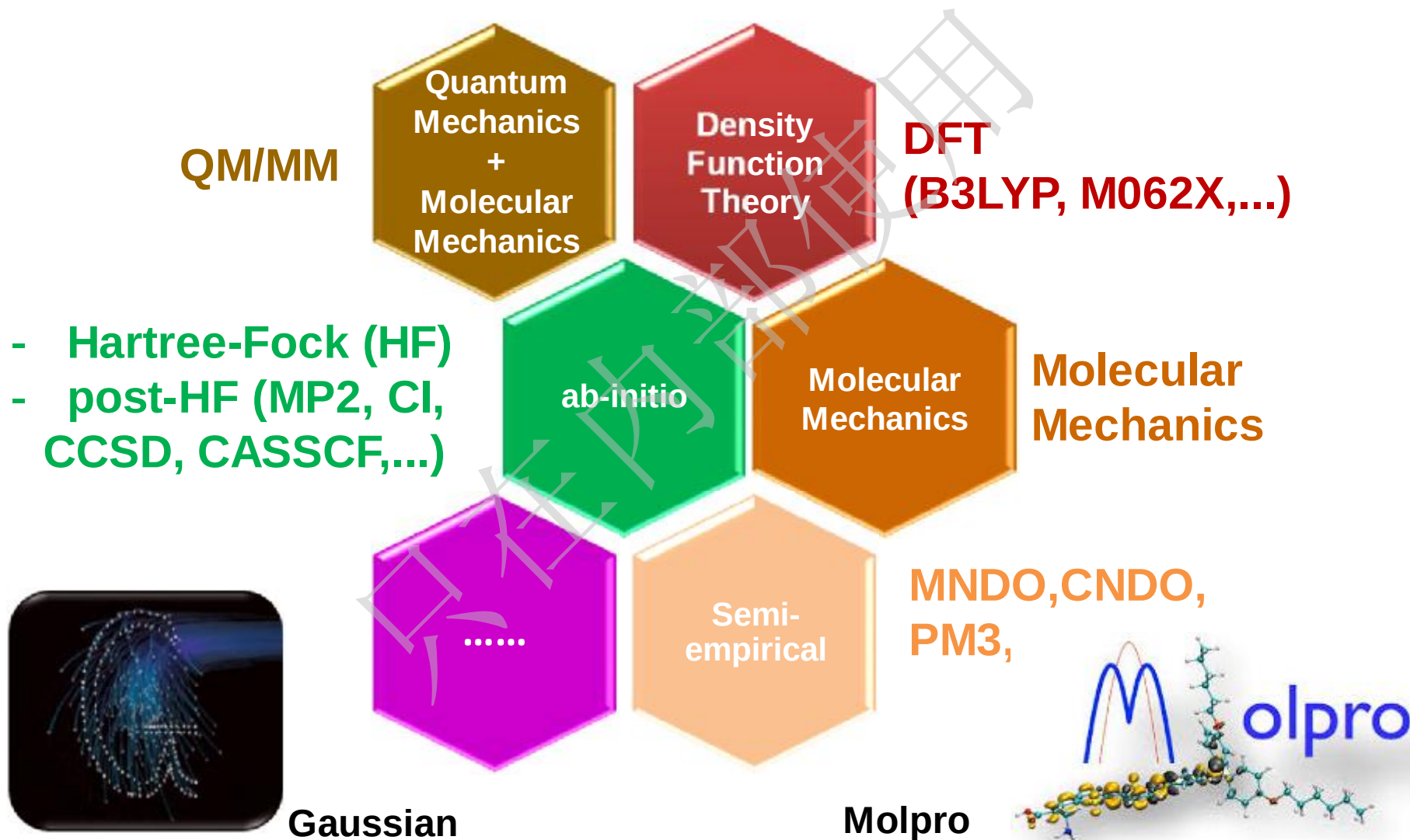


Development of kinetic theory:



3.1. Theoretical methods

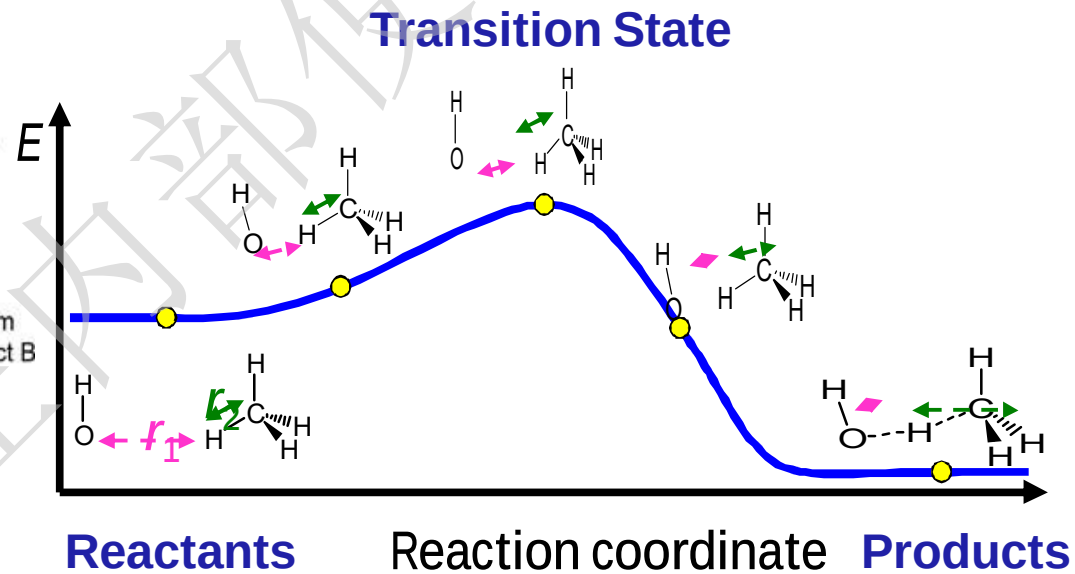
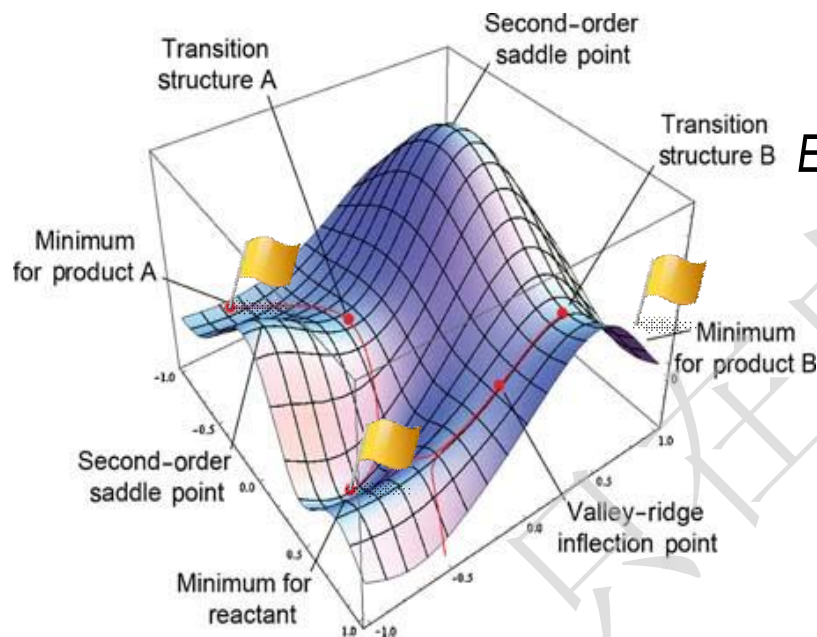
I Quantum Chemistry Methods



3.1. Theoretical methods

I PES and Reaction Pathways

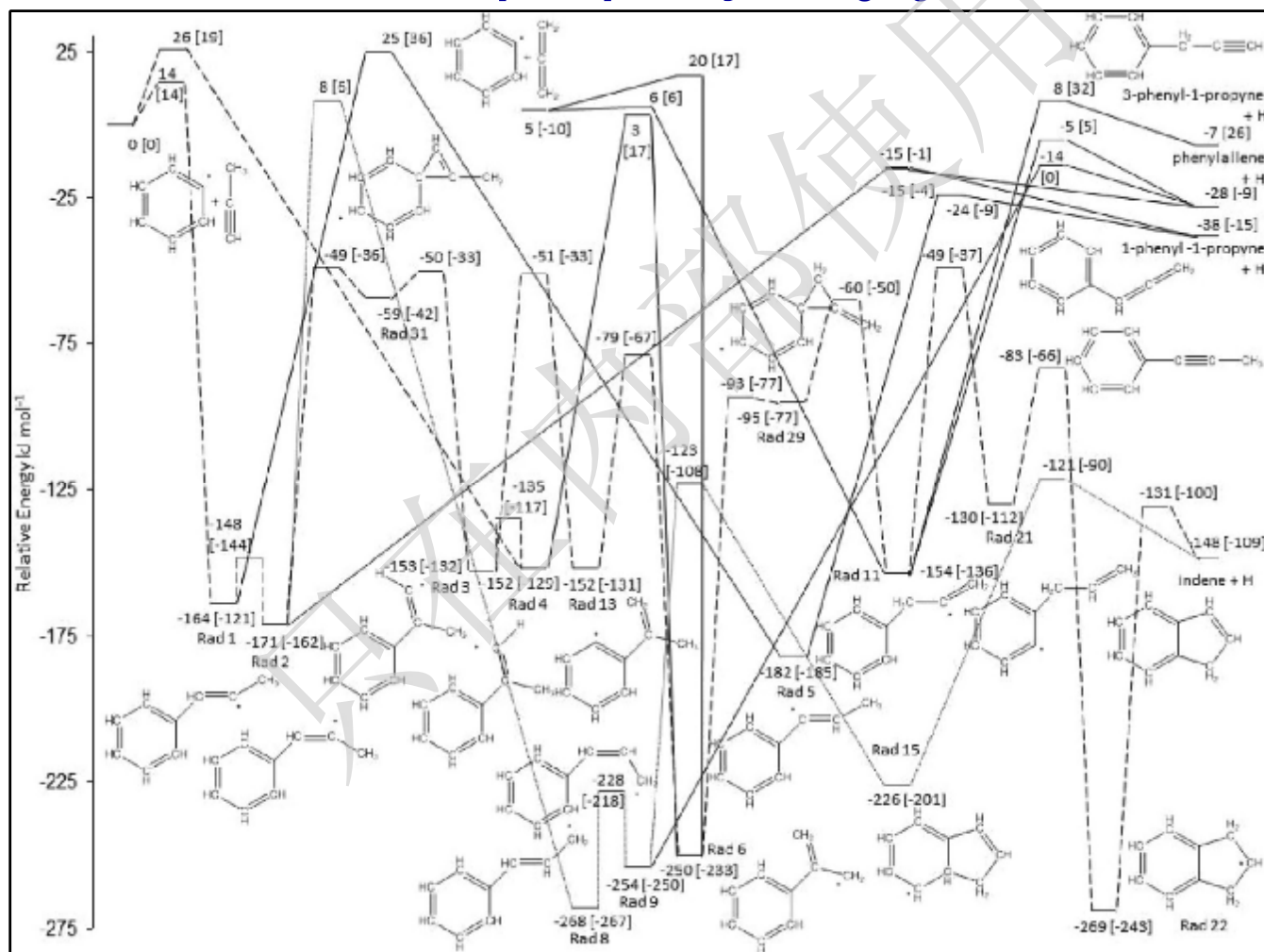
Search all possible pathways in combustion reactions



3.1. Theoretical methods

I PES and Reaction Pathways

An example: phenyl + $C_3H_3 \rightarrow ?$

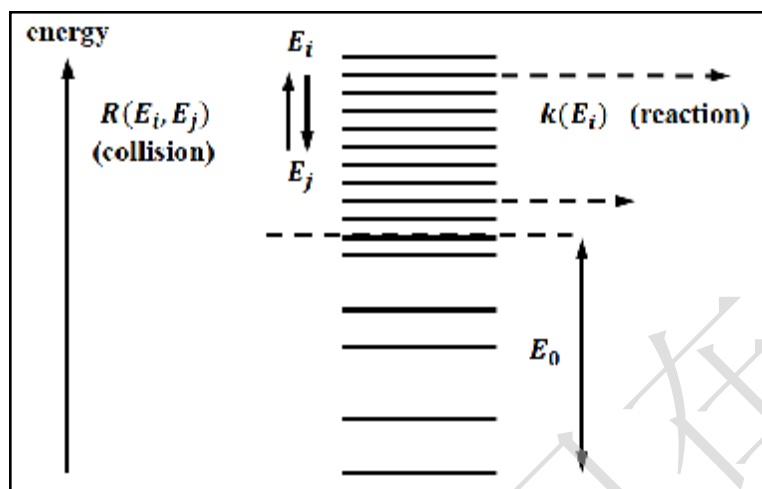


3.1. Theoretical methods

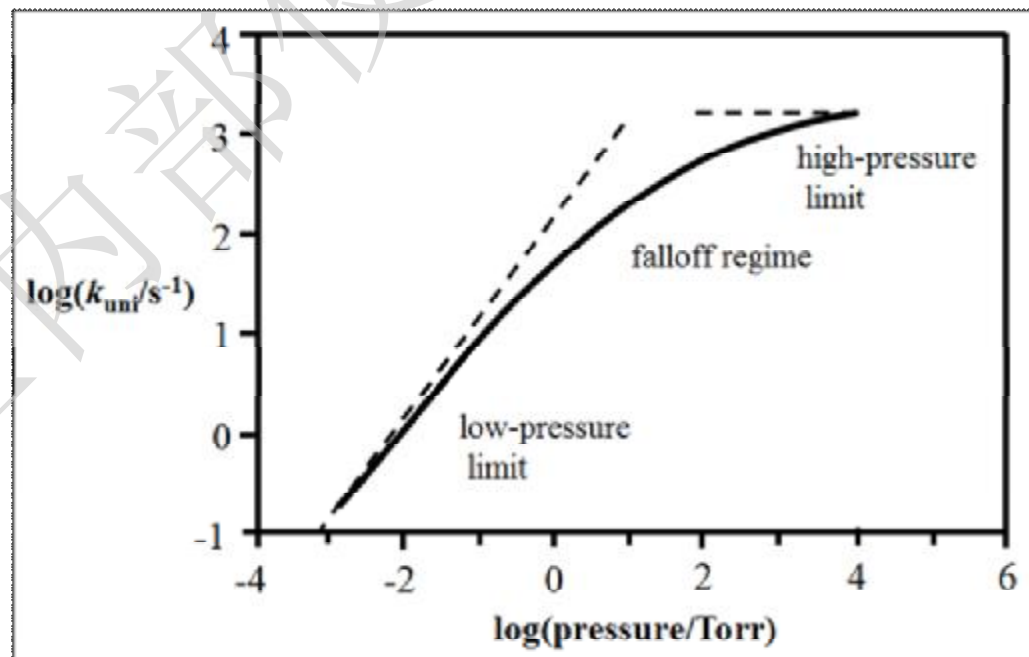
I Kinetics Theory

Unimolecular reactions

Reaction and collision



Fall-off behavior



Lindemann mechanism

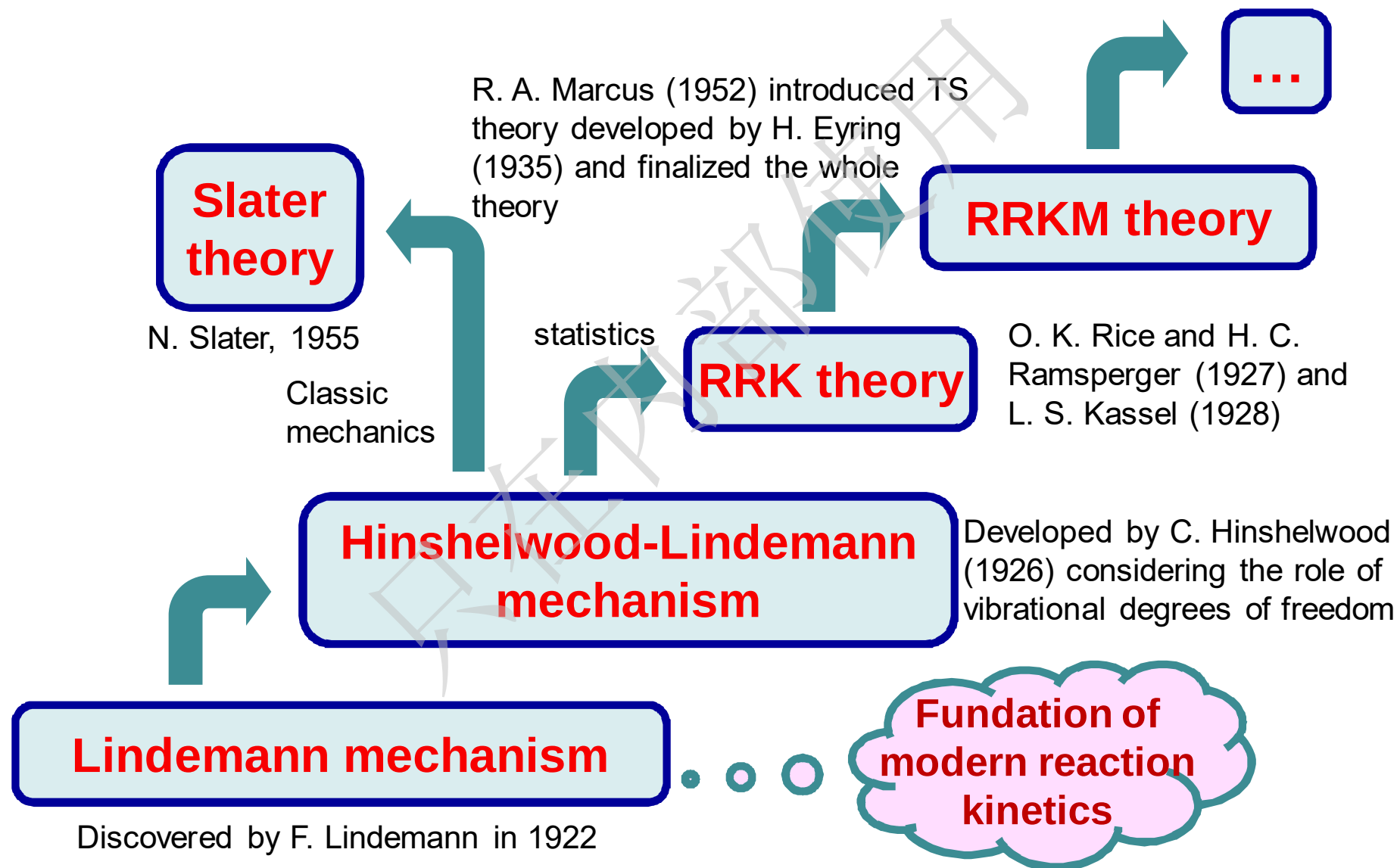
Collision: $A + M \rightleftharpoons A^* + M$ (k_1/k_{-1})

Reaction: $A^* \rightarrow P$ (k_2)

$$\text{Rate constant: } k = k_2 \left(\frac{k_1}{k_{-1}} \right) / \left(1 + \frac{k_2}{k_{-1}[M]} \right)$$

3.1. Theoretical methods

I Kinetics Theory



3.1. Theoretical methods

I Kinetics Theory

n Master Equation method (Solves pressure-dependent cases)

- I Reaction process is separated as a collision step and a reaction step
- I Provides a quantitative description of the interaction between collisional energy transfer and chemical reaction for dissociation, isomerization and association processes in combustion
- I By solving master equation (ME), pressure-dependent rate constants $k(T,P)$ are obtained

n RRKM-ME method

- I Chemical reaction step is treated with the RRKM theory
- I Combining RRKM theory and ME method can predict the rate constants accurately
- I The uncertainty is within a factor of 2

3.1. Theoretical methods

I Codes for Rate Constants Calculation

v PAPER

Predictive Automated Phenomenological Elementary Rates

Stephen J. Klippenstein et al, Argonne National Laboratory

v MESMER

Master Equation Solver for Multi Energy well Reactions

Michael Pilling et al., University of Leeds

v MultiWell

John R. Barker et al., University of Michigan

v ChemRate

Wing Tsang, National Institute of Standard Technology

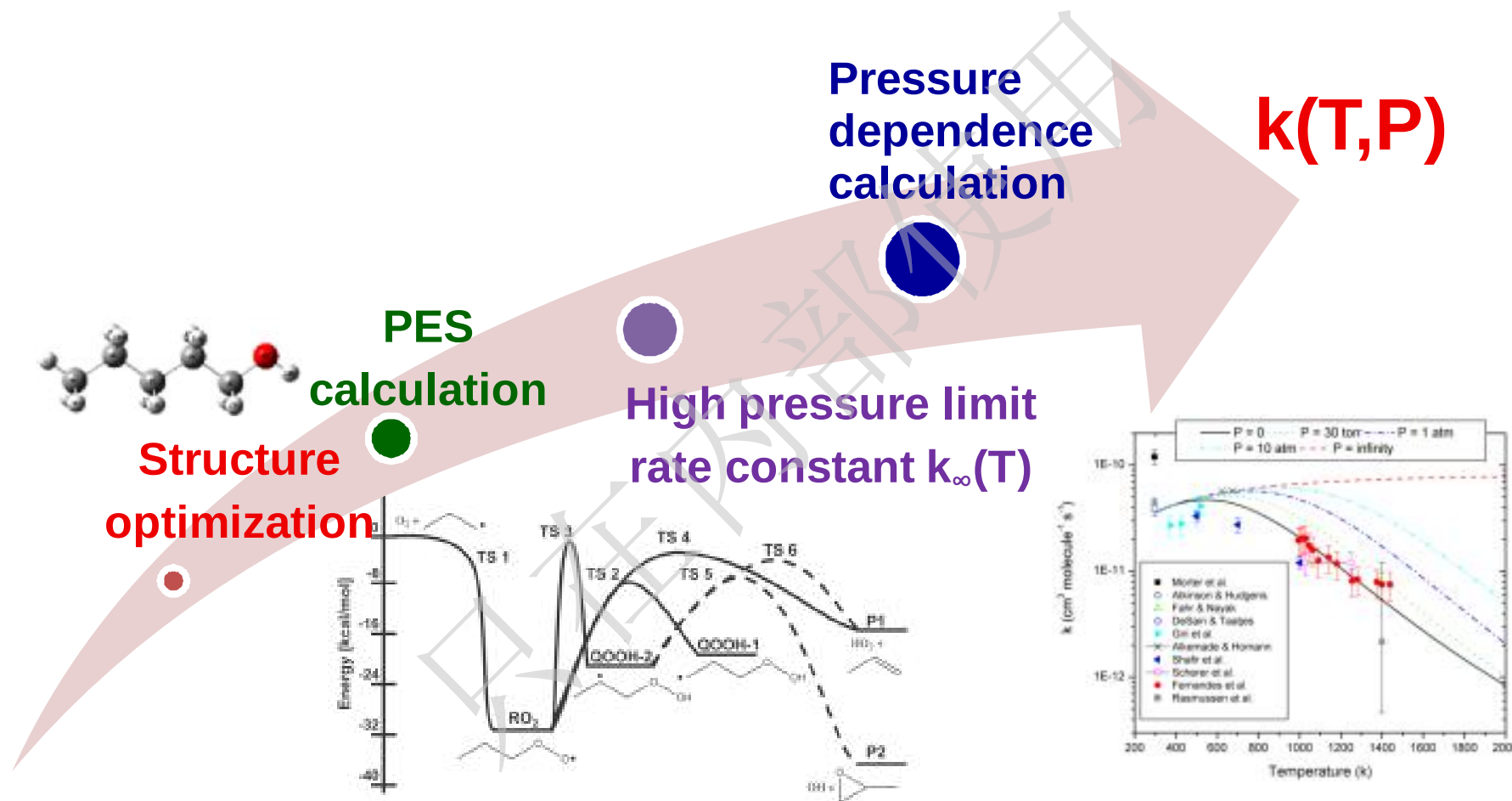
v POLYRATE

Donald L. Truhlar (Minnesota University. USA)

v

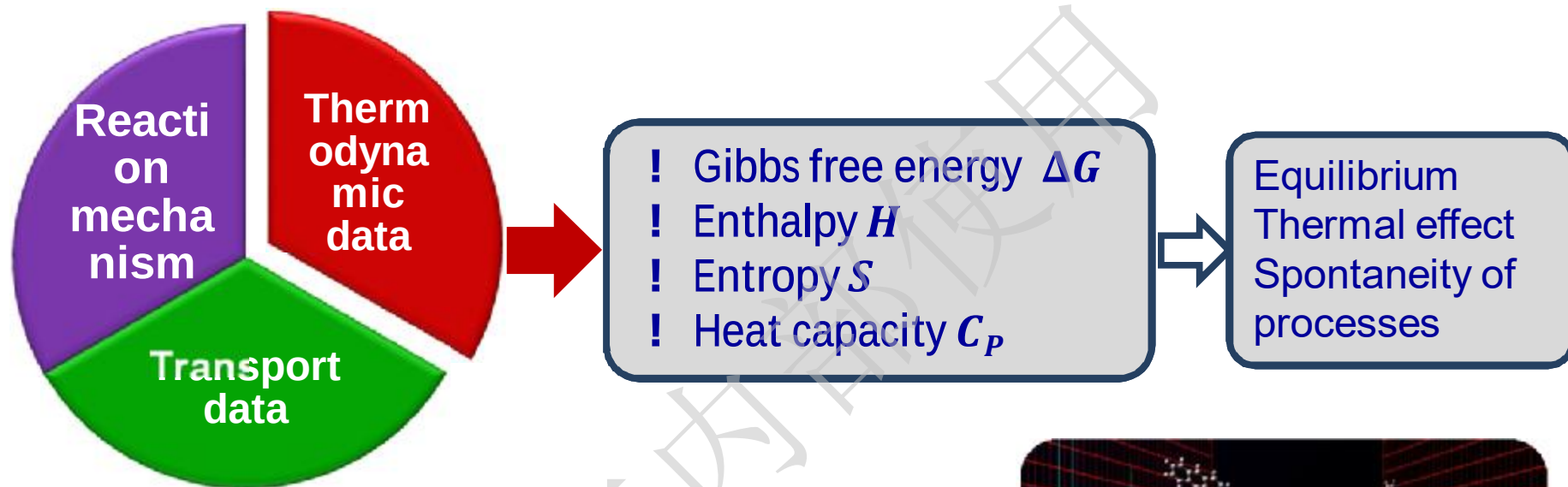
3.1. Theoretical methods

I The Flow Chart of Rate Constant Calculation



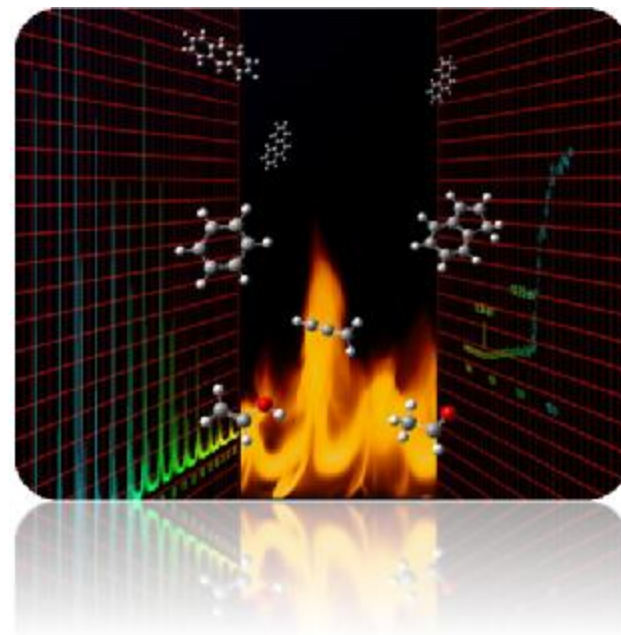
3.1. Theoretical methods

I Thermodynamic Properties Calculation



r Based on Partition Function

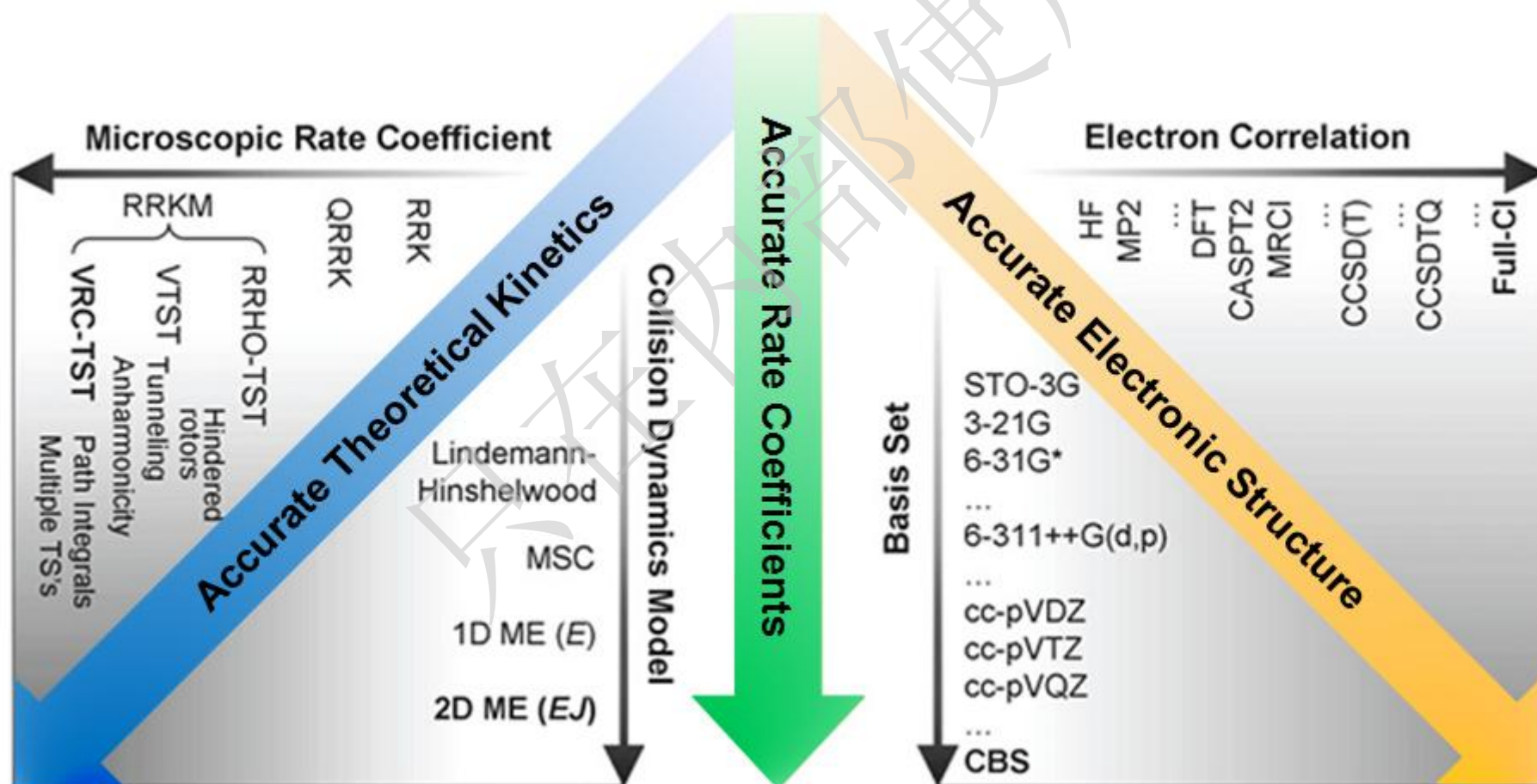
- Compute $\Delta H_{f, 298K}$, S , C_p
- Compute T-dependent thermodynamic parameters



3.1. Theoretical methods

I Theoretical Methods

ab-initio QC + Kinetics theory—more accurate rate parameters



3.2. Experimental Methods

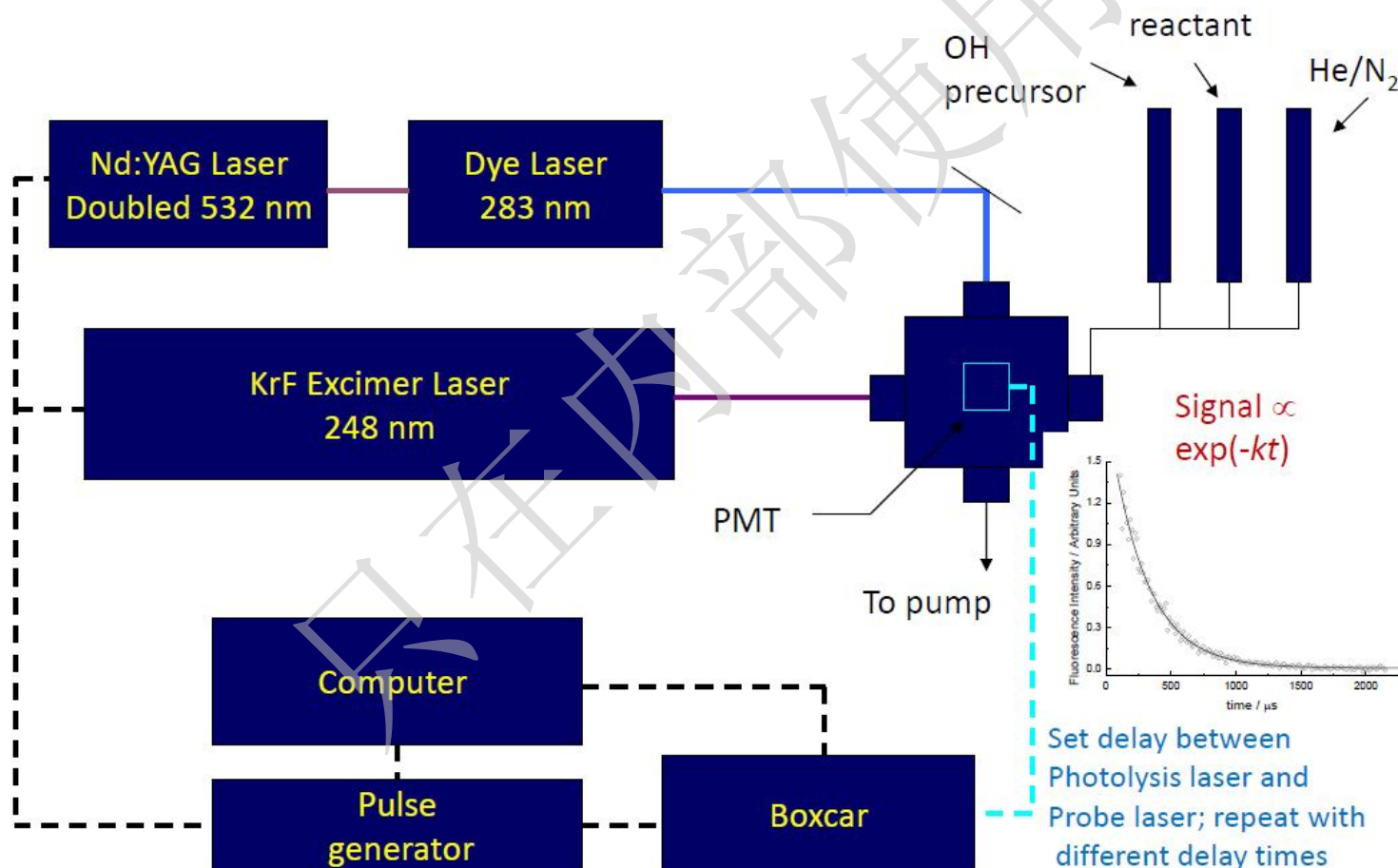
I Measurement of Elementary Reactions

- n Separate experiment for each elementary reaction step
- n The aim is the **determination of the rate coefficient** of a single reaction step at a given temperature, pressure and bath gas
- n **Experimental methods**
 - p Traditional flash photolysis, pulsed laser photolysis (PLP)
 - p Microwave discharge, high temperature (shock tube)
- n **Diagnostic methods for products**
 - p Laser induced fluorescence (LIF)
 - p Resonance fluorescence (RF)
 - p Atom resonantly absorption spectroscopy (ARAS)
 - p Laser absorption spectroscopy (LAS)
 - p Electron-impact ionization mass spectrometry (EIMS)
 - p Synchrotron VUV photoionization mass spectrometry (SVUV-PIMS)
 - p ...

3.2. Experimental Methods

I Measurement of Elementary Reactions

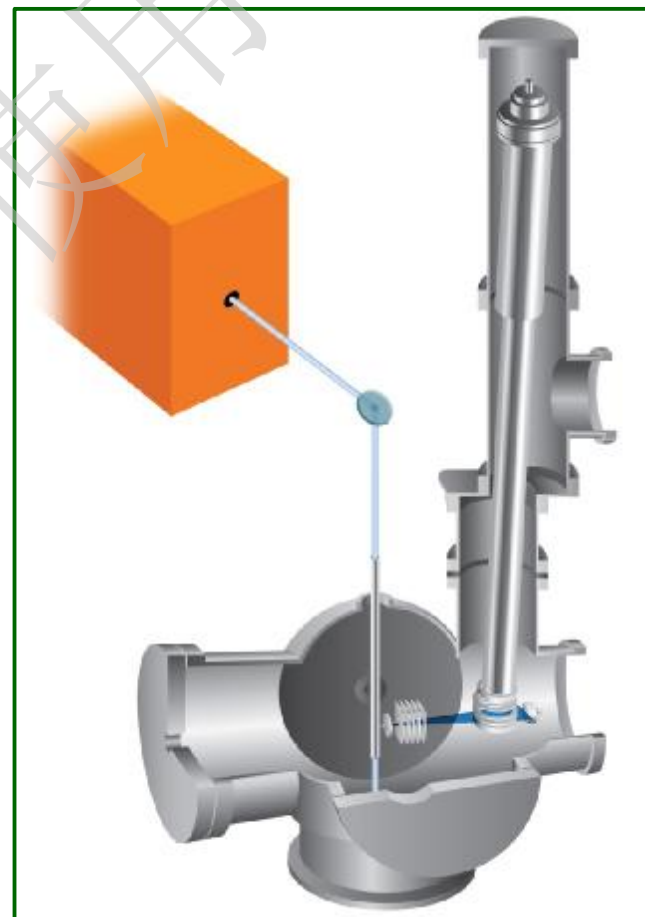
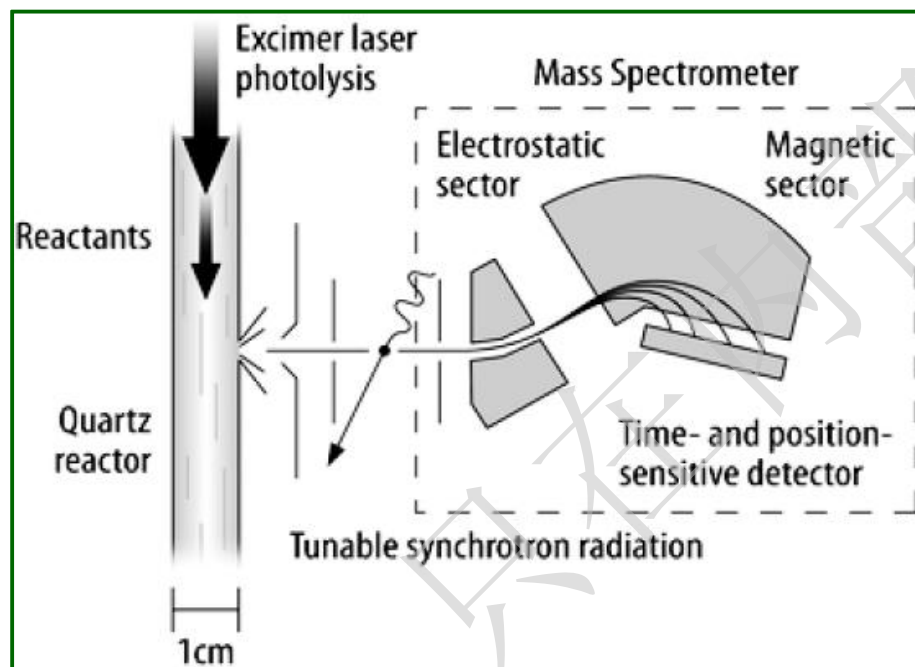
Laser flash photolysis (LFP) / laser induced fluorescence (LIF)



3.2. Experimental Methods

I Measurement of Elementary Reactions

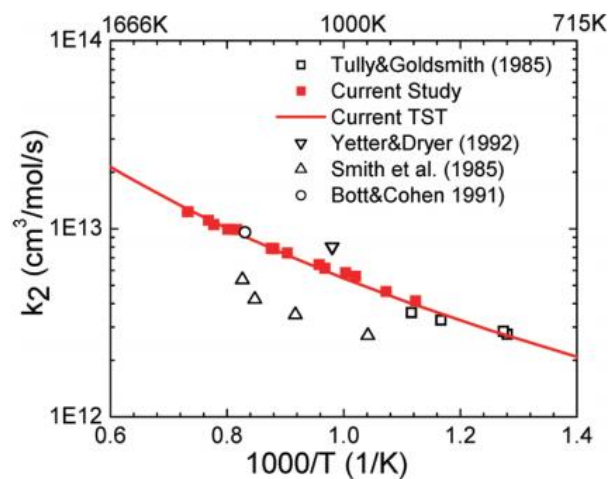
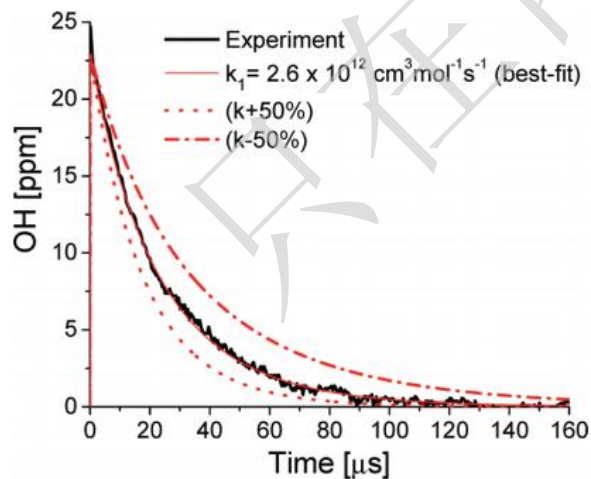
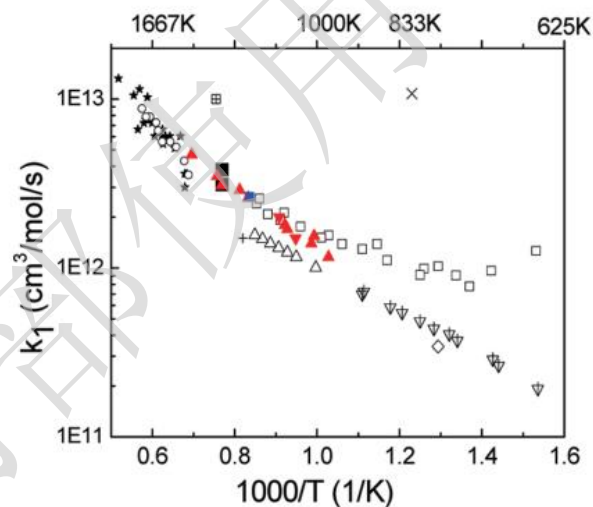
Laser flash photolysis (LFP) + SVUV-PIMS



3.2. Experimental Methods

I Measurement of Elementary Reactions

Shock tube + laser absorption



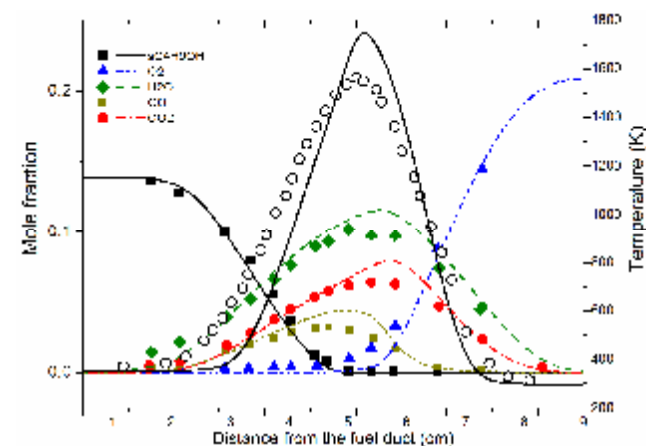
3.2. Experimental Methods

I Speciation and Global Parameters

- n** Can be interpreted only with the simulation of a detailed reaction mechanism
- n** Speciation for pyrolysis, oxidation and flames using laser spectroscopy, gas chromatography, and mass spectrometry
 - p** Shock tube/flow reactor
 - p** Jet stirred reactor
 - p** Laminar premixed flame
 - p** Coflow/counterflow diffusion flame
 - p** ...
- n** Global combustion parameters
 - p** Ignition delay times
 - p** Laminar flame speeds
 - p** Flame extinction measurements
 - p** ...

I Speciation and Global Parameters

- [illegible]



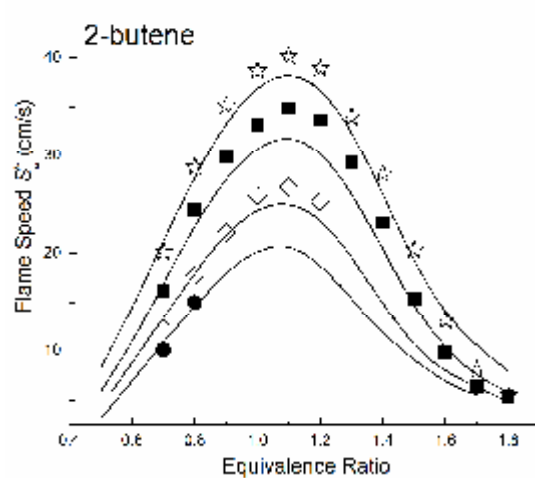
Y. Y. Li et al. Proc. Combust. Inst. 34 (2013) 1739
J. H. Cai et al., Combust. Flame 160 (2013) 1939

3.2. Experimental Methods

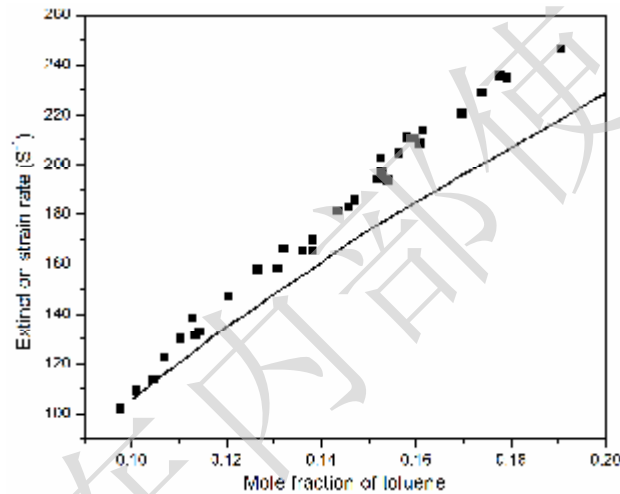
I Speciation and Global Parameters

n Global parameters have specific engineering meanings.

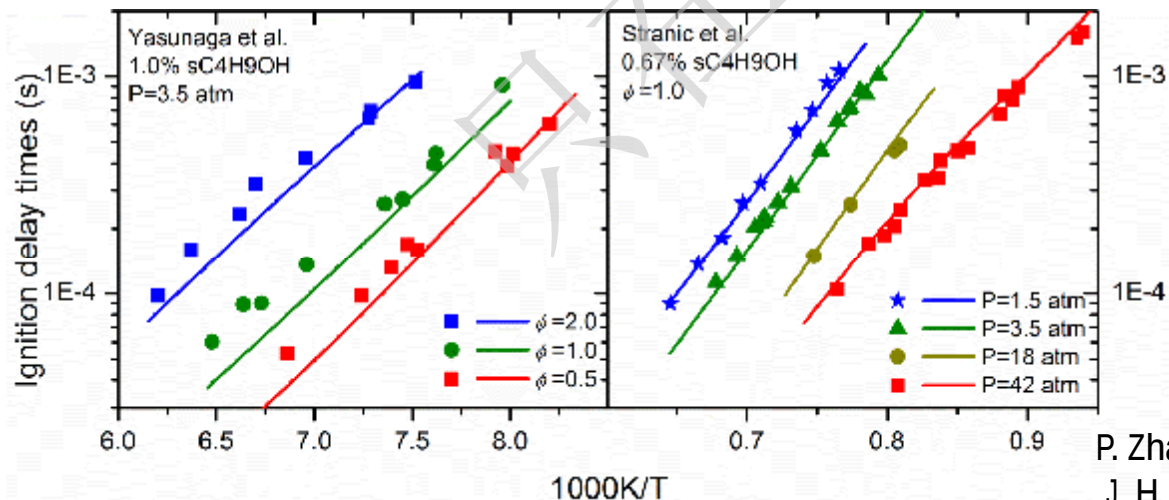
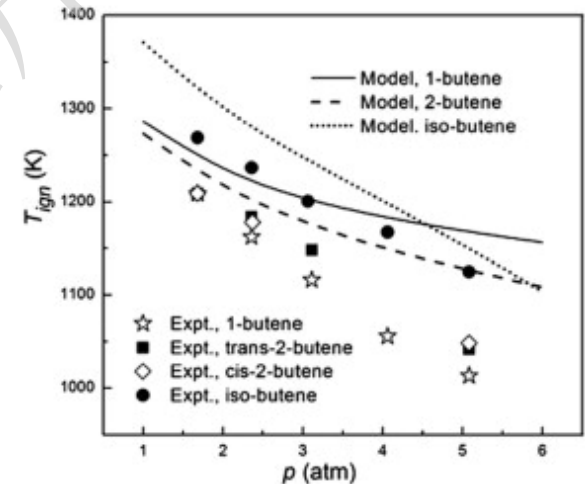
Laminar flame speeds



Extinction strain rate



Ignition temperature



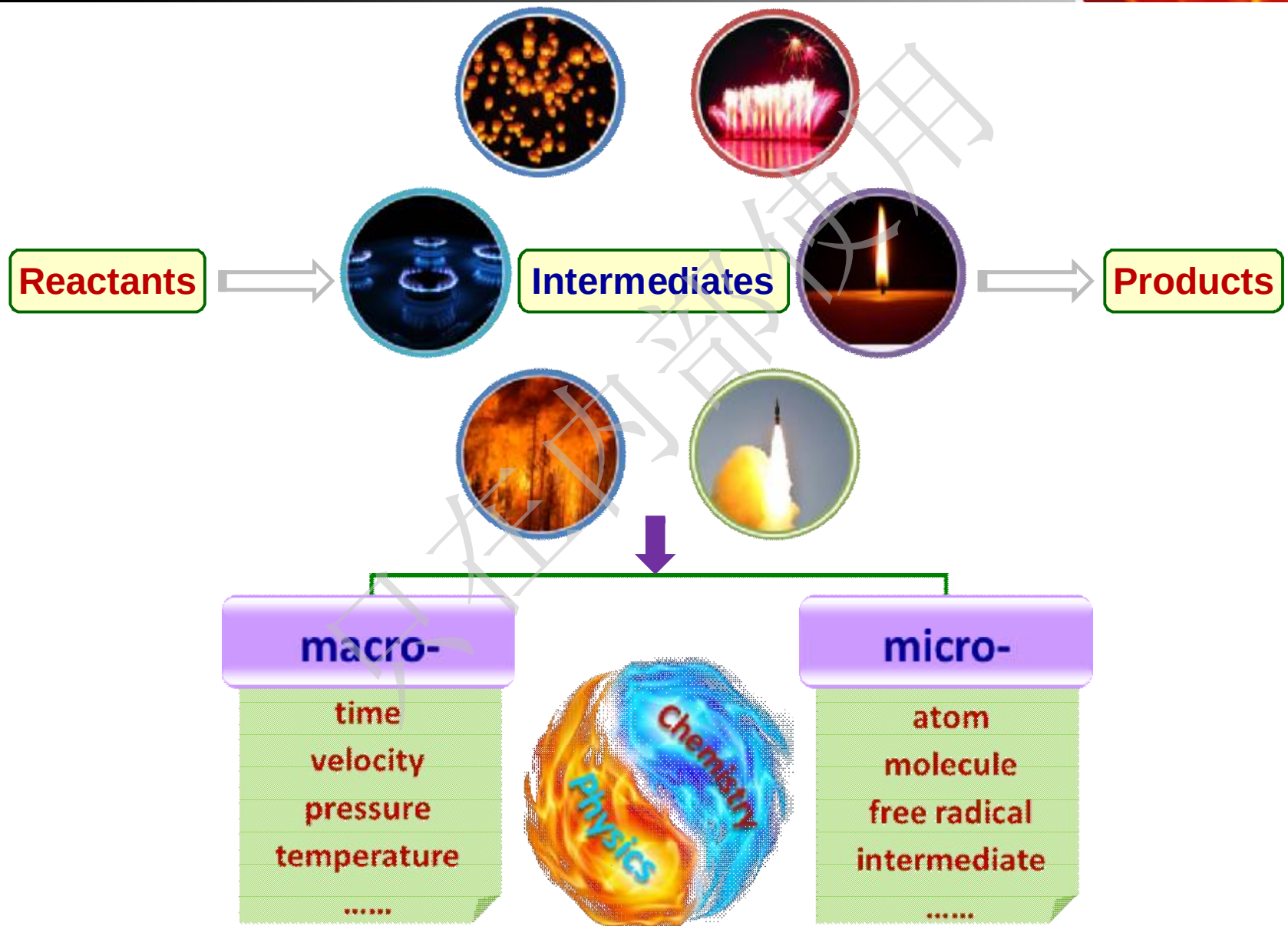
Ignition delay times

P. Zhao et al., Proc. Combust. Inst. 35 (2015) 309

J. H. Cai et al., Combust. Flame 160 (2013) 1939

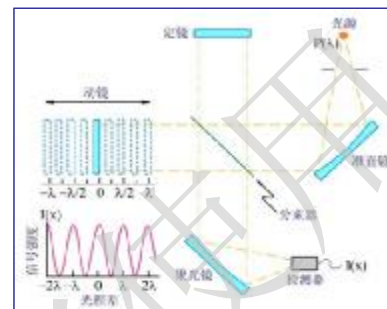
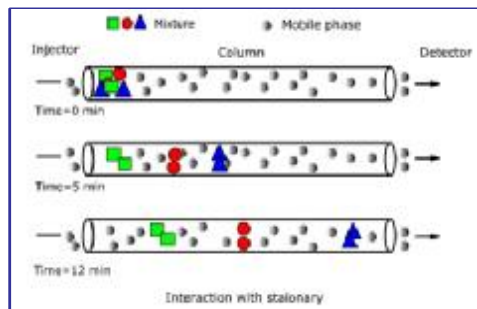
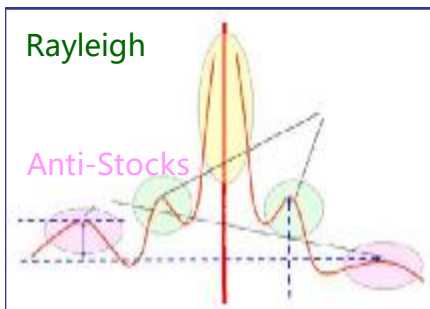
3.2. Experimental Methods

I Combustion Diagnostics



3.2. Experimental Methods

I Combustion Diagnostic Methods



'20

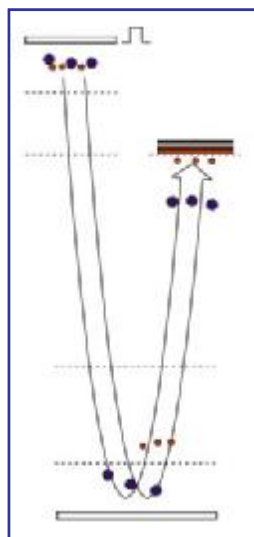
'50

'60

'70

'80

'90



Quadrupole MS

Chromatography

TOF MS

Abs. spect. (AS)

Raman scattering

Rayleigh scattering

CARS

SRS

REMPI

LIF

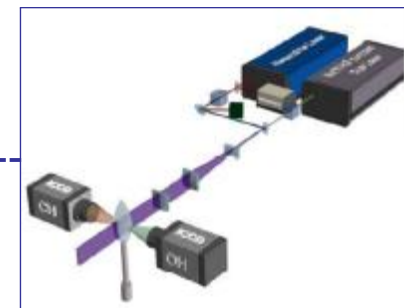
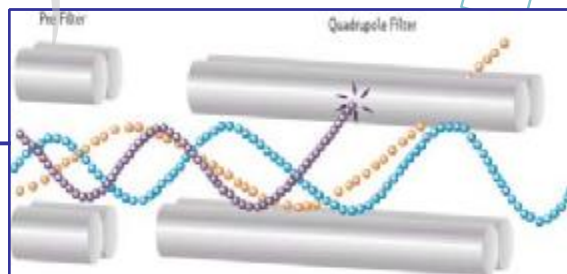
FTIR

NMR

LII

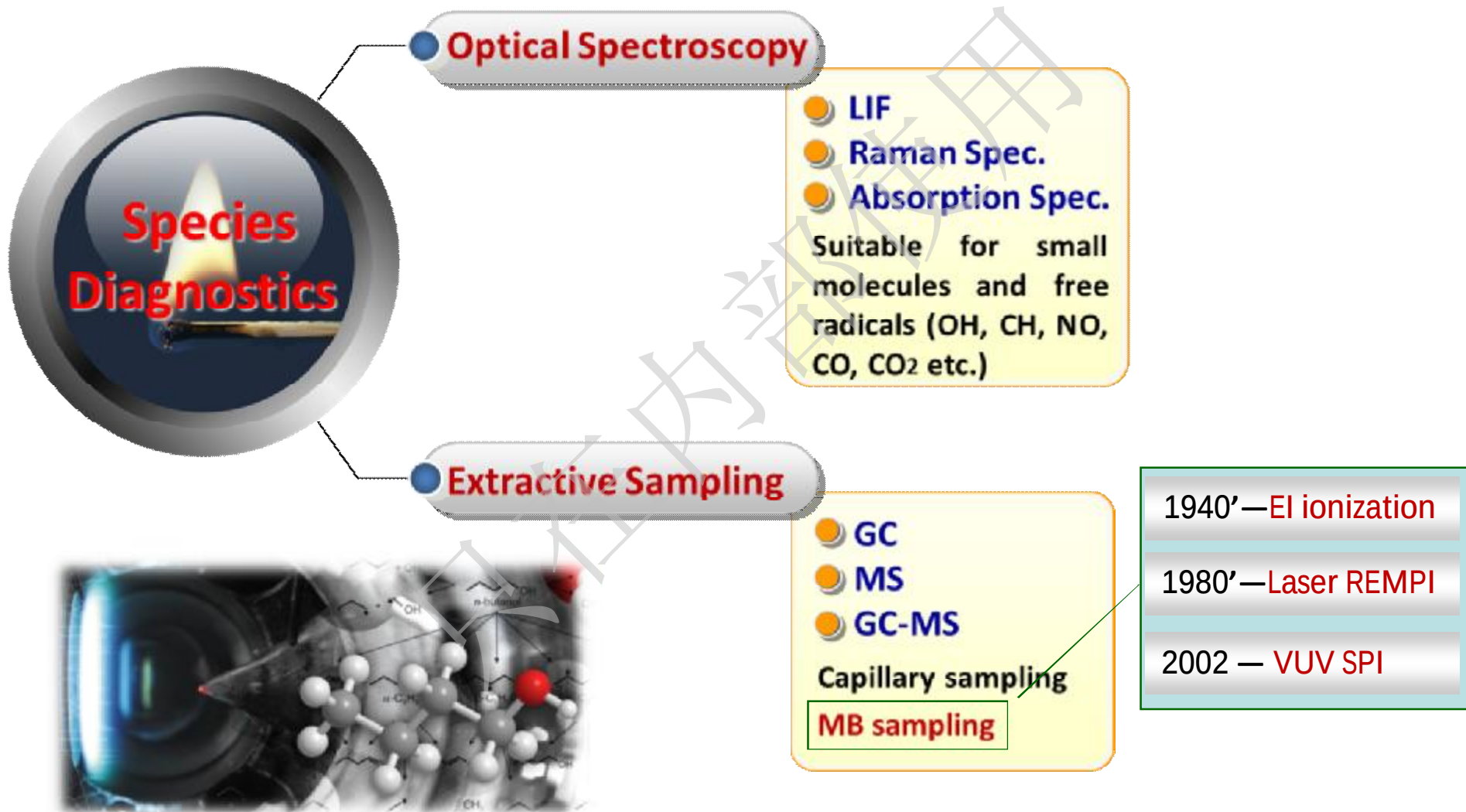
CRDS

SVUV-PIMS



3.2. Experimental Methods

I Combustion Diagnostic Methods



3.2. Experimental Methods

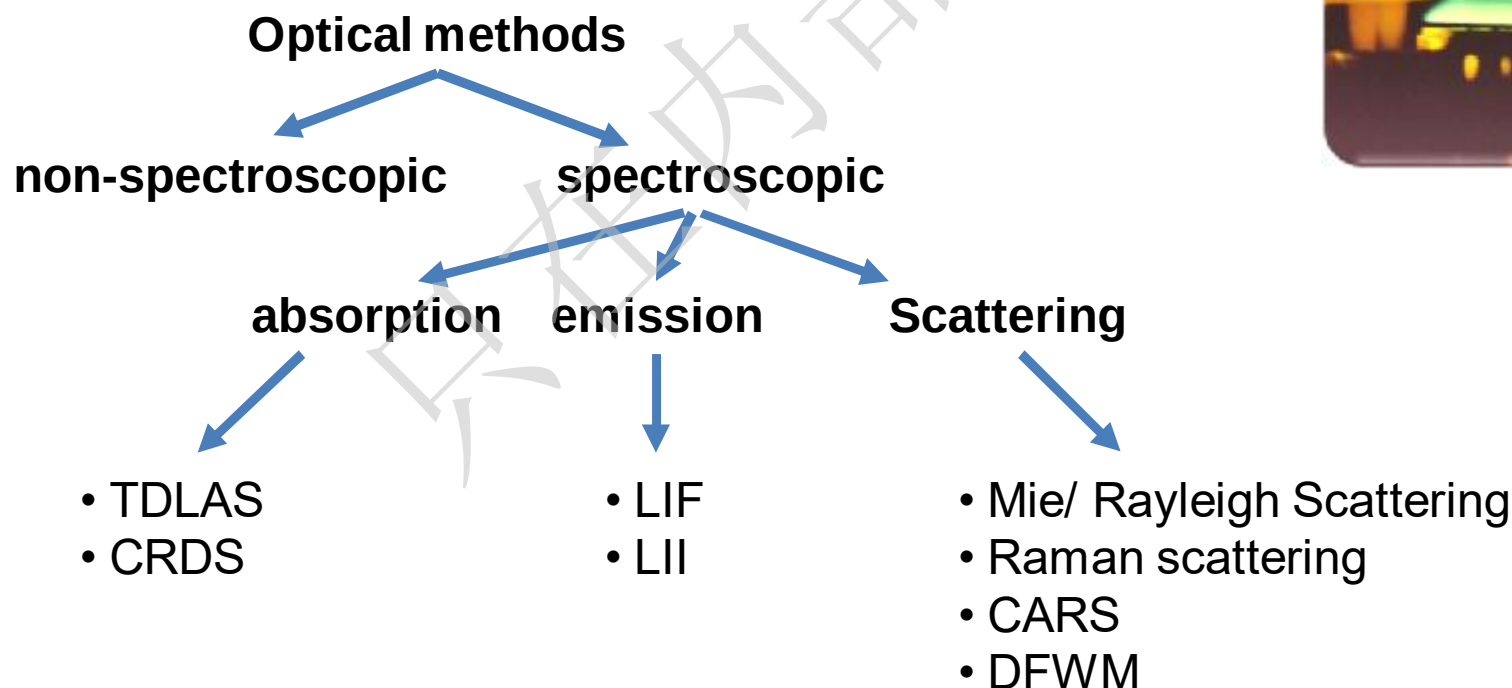
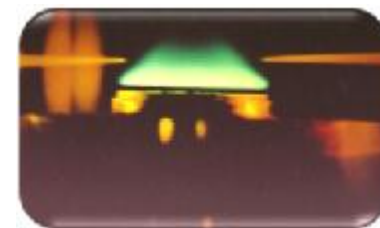
I Optical Spectroscopy

Advantage:

- Non-invasive
- High spatial resolution ($<0.001 \text{ mm}^3$)
- High temporal resolution ($<10 \text{ ns}$)
- High spectral resolution ($\sim \text{MHz}$)
- Multiplex (multi-species, multi-point)

Disadvantage:

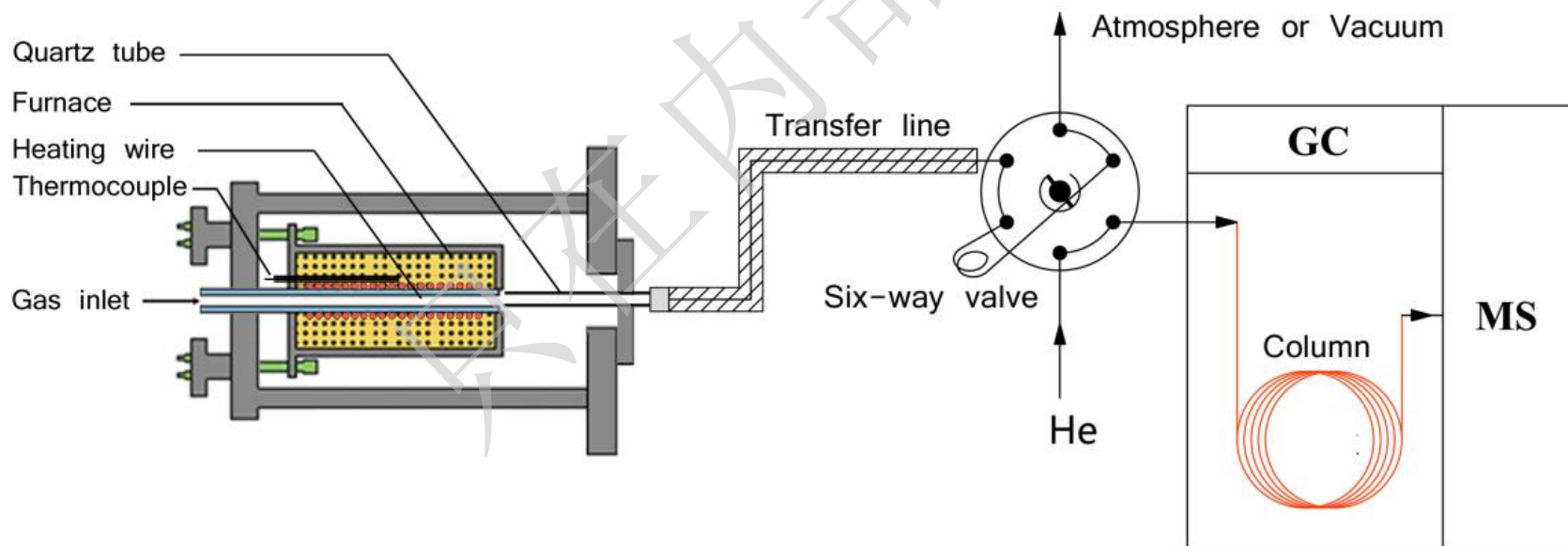
- Mostly expensive and complicated equipment
- May have difficulties to give quantitative data
- Hard to detect large species, not "universal"
- Optical access required
- Intrusive?



3.2. Experimental Methods

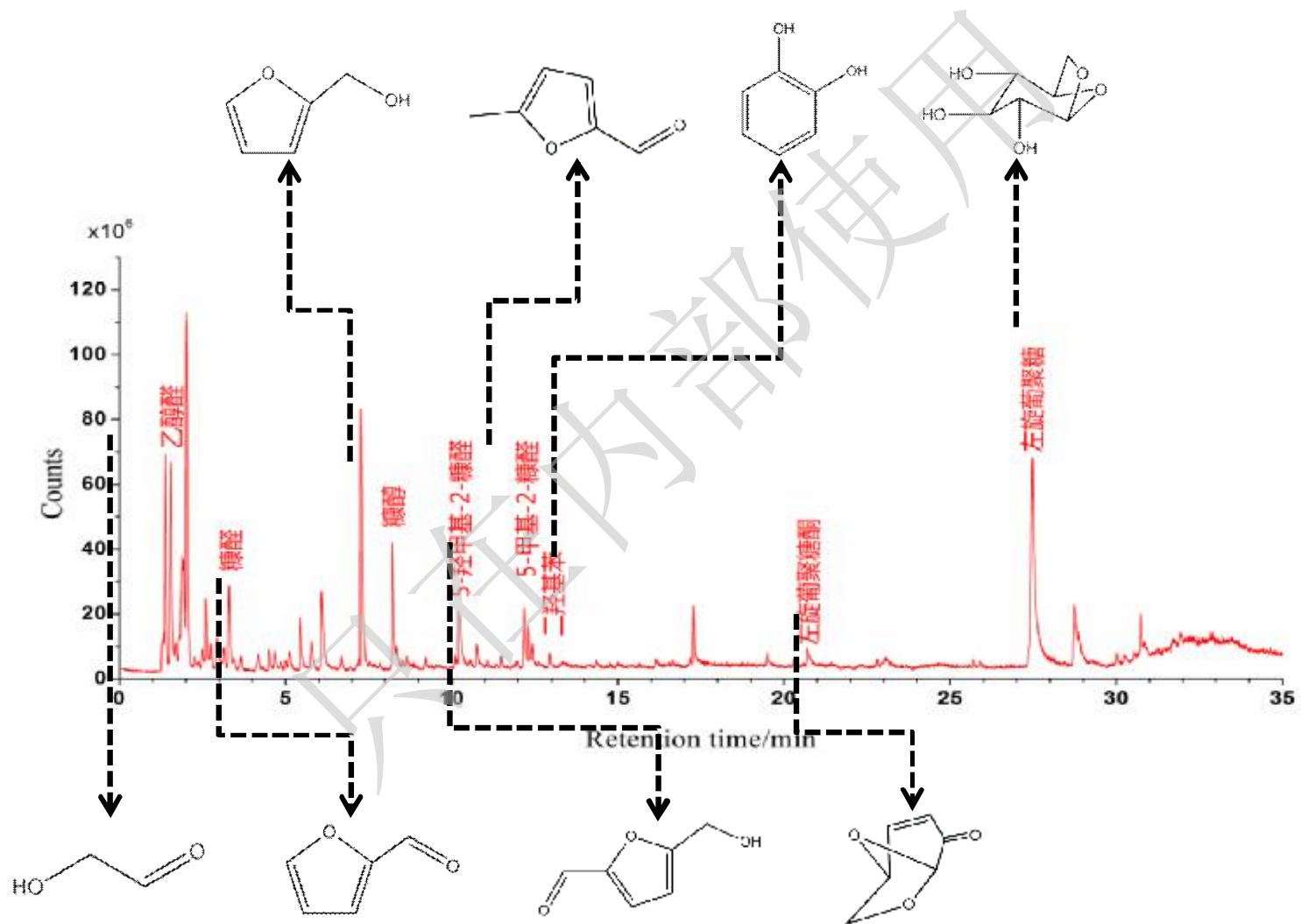
I Gas Chromatography

- n Gas chromatography is usually coupled with mass spectrometry for species measurement
- n Cannot be used to detect reactive species such as radicals and peroxides



3.2. Experimental Methods

I Gas Chromatography



3.2. Experimental Methods

I Mass Spectrometry

(c)

Ar/PI (~ 1 meV)

El ionization *Universal, not selective*

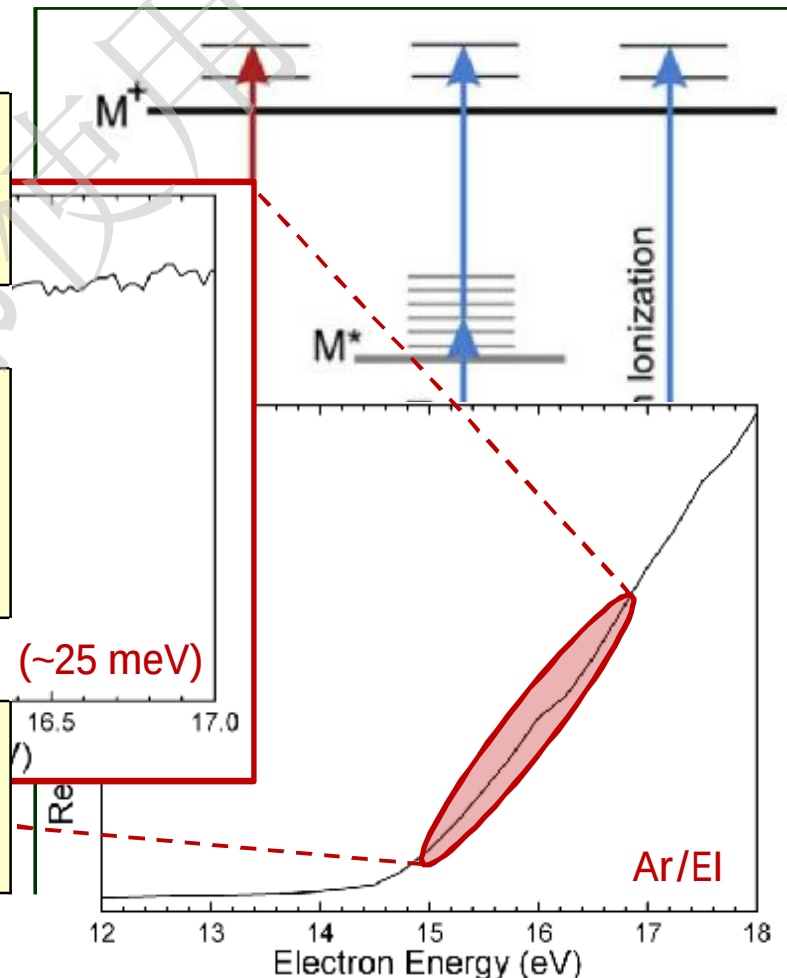
An ionization method in which energetic electrons interact with gas phase atoms or molecules to produce ions.

REMPI *Selective, not universal*

REMPI involves a resonant single or multiple photon absorption to an electronically excited intermediate state followed by another photon which ionizes the atom or molecule.

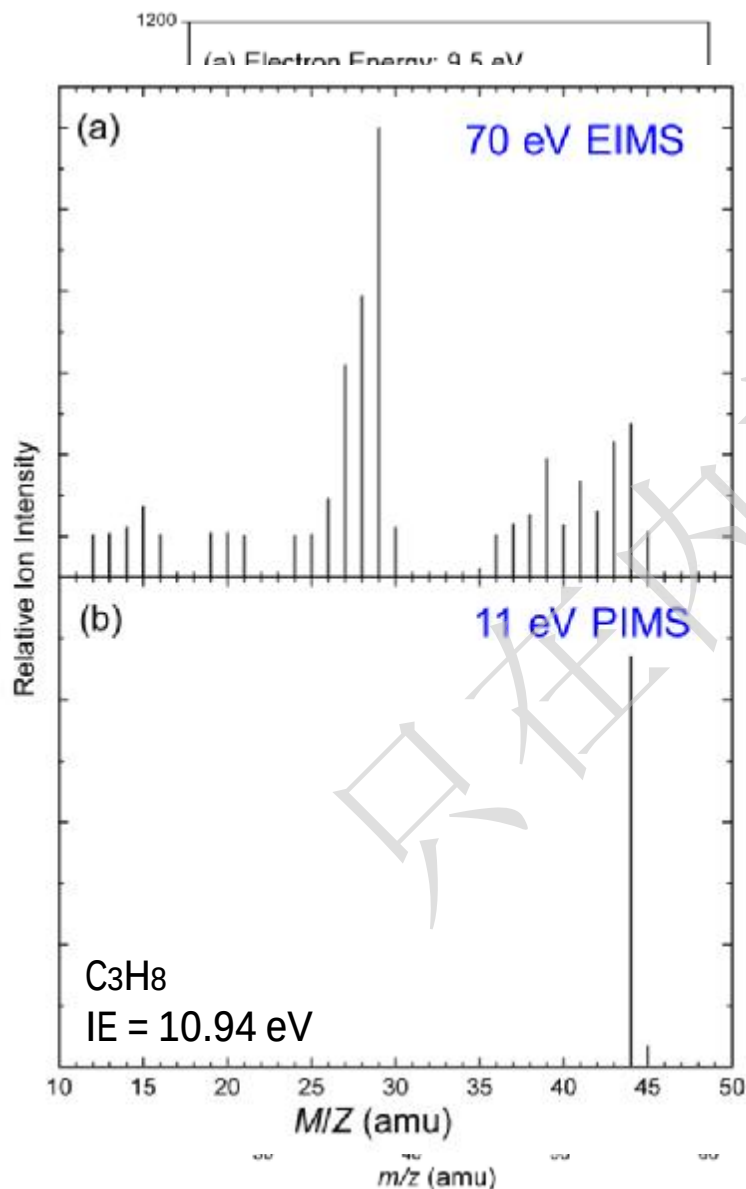
VUV PI *Universal, also selective*

Photoionization is the physical process in which an incident photon ejects one or more electrons from an atom or molecule.



3.2. Experimental Methods

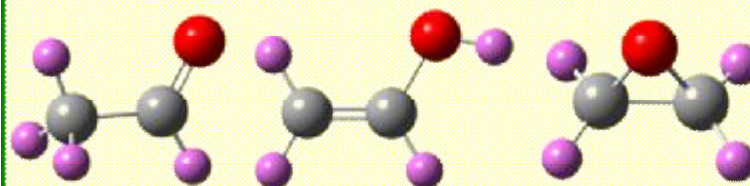
I Photoionization Mass Spectrometry



PIMS can resolve different mass, and gives a single identifiable peak for each compound.

Question: how to distinguish isomers?

C_2H_4O ($m/z = 44$)



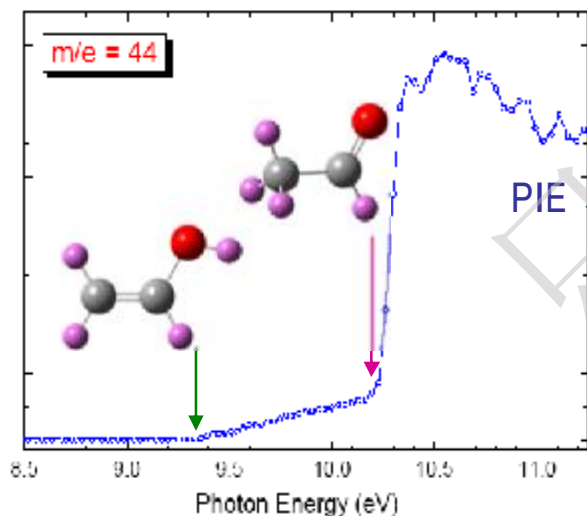
3.2. Experimental Methods

I Synchrotron VUV Photoionization Mass Spectrometry

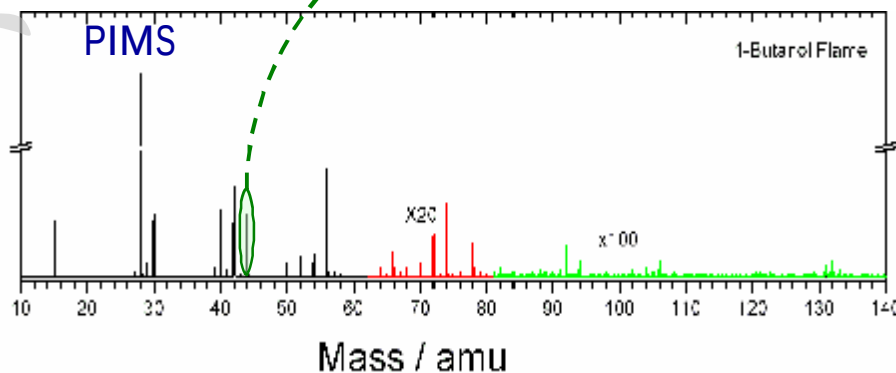
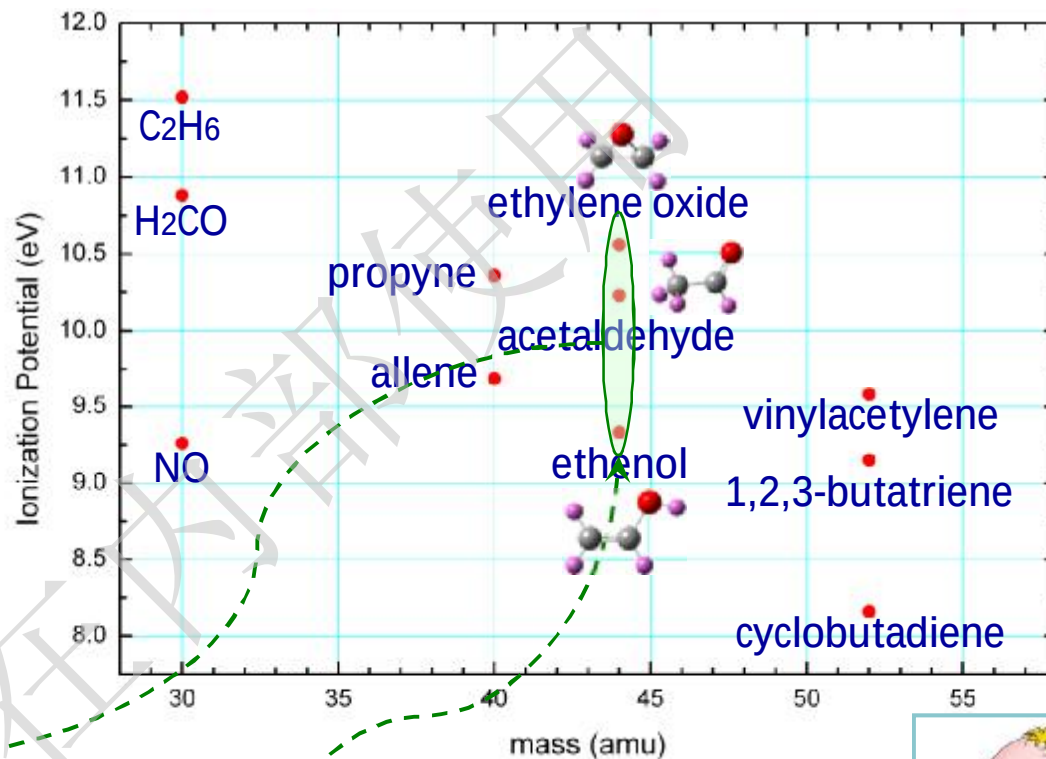


Height Scale

II



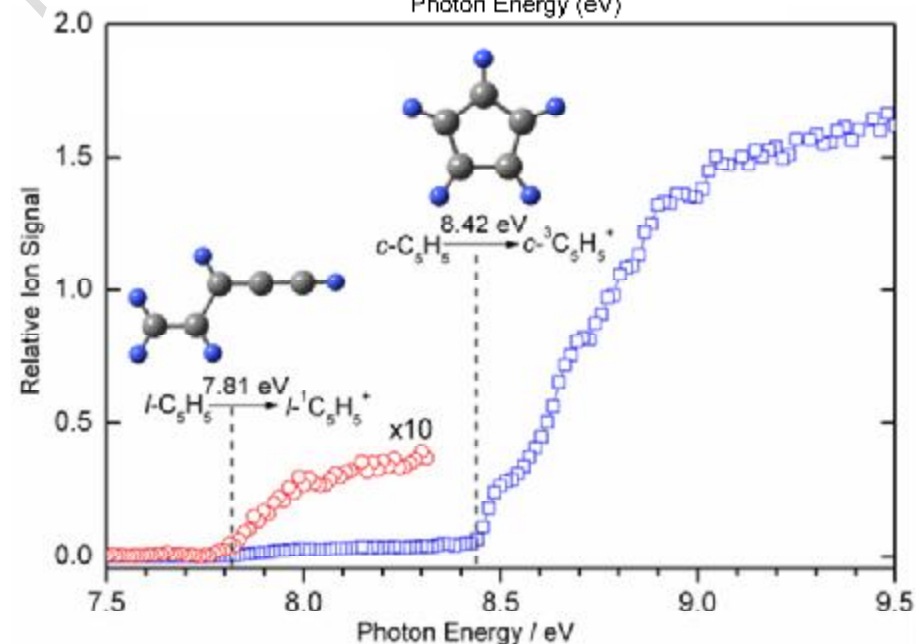
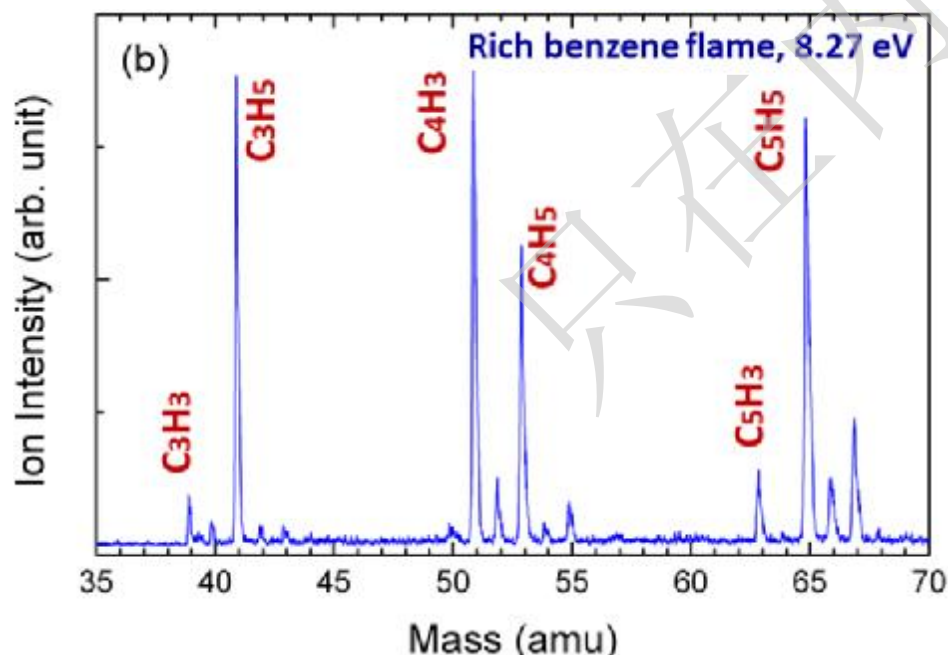
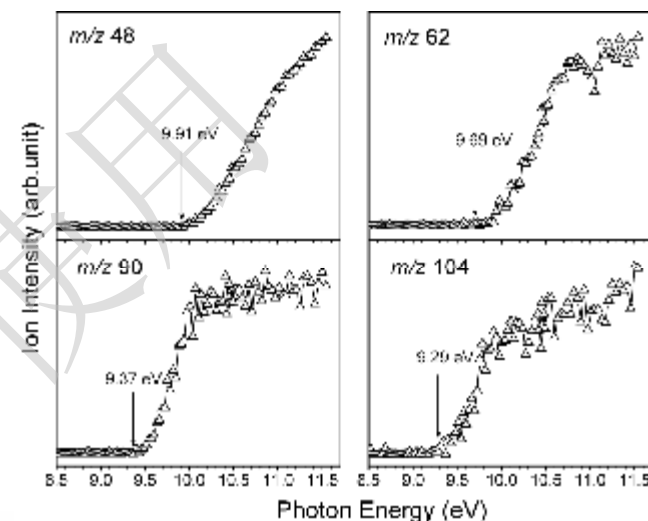
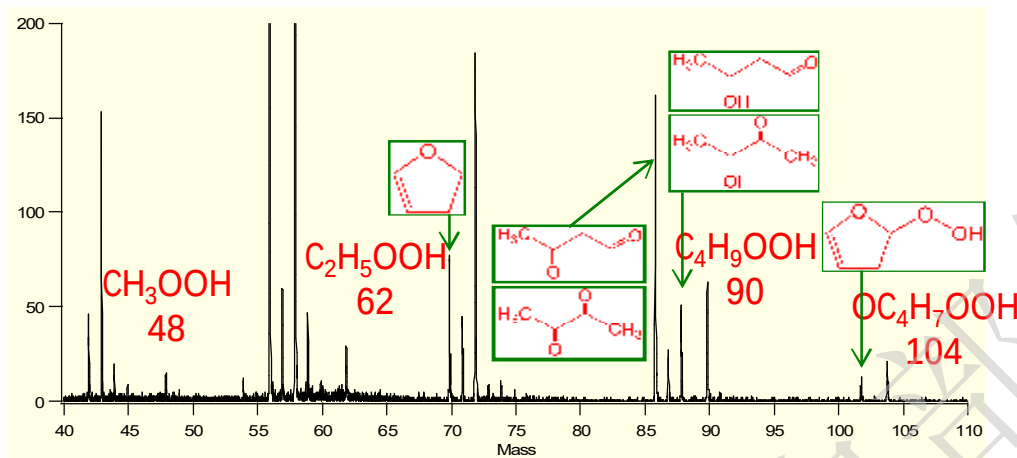
PIE



Weight Scale

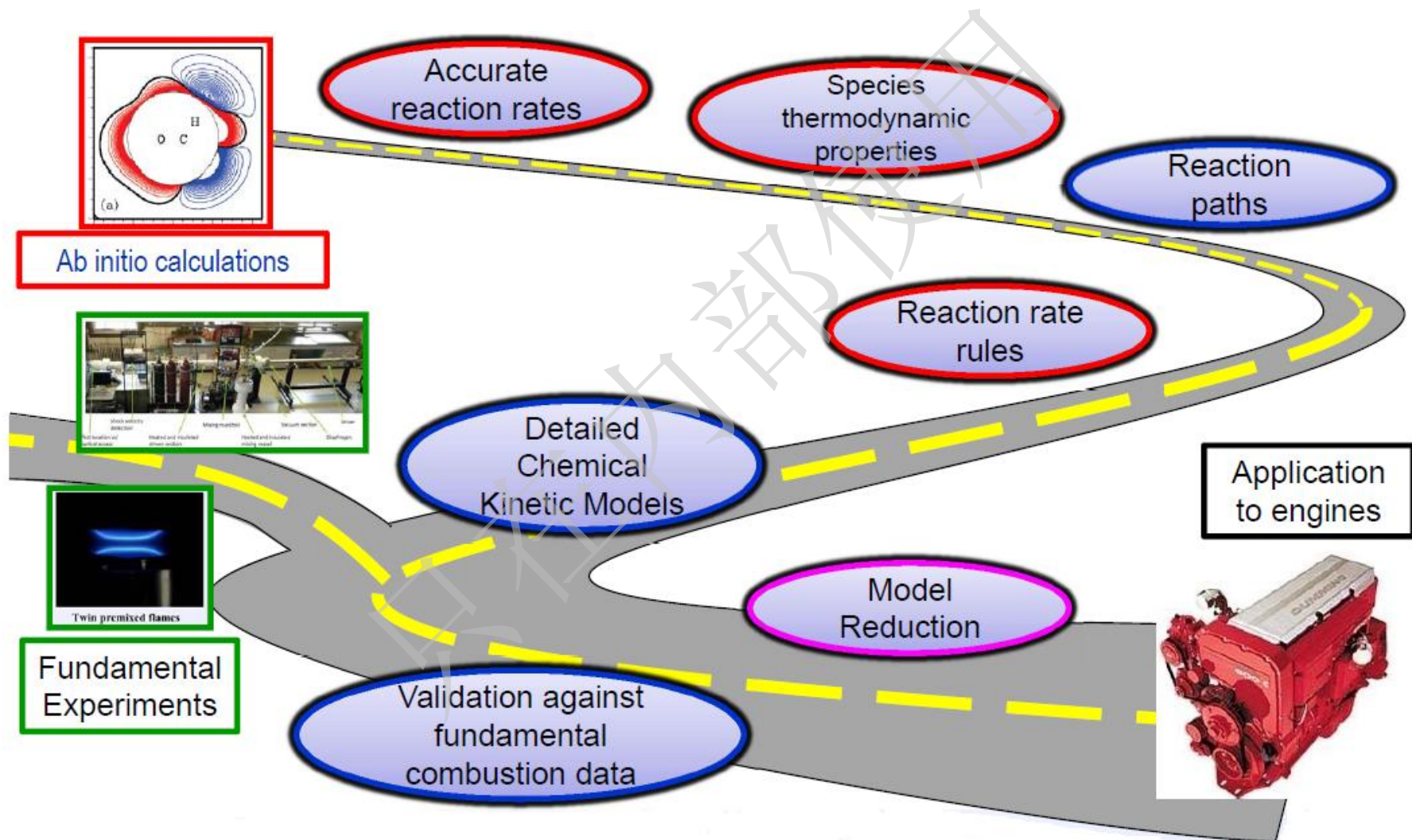
3.2. Experimental Methods

I Synchrotron VUV Photoionization Mass Spectrometry



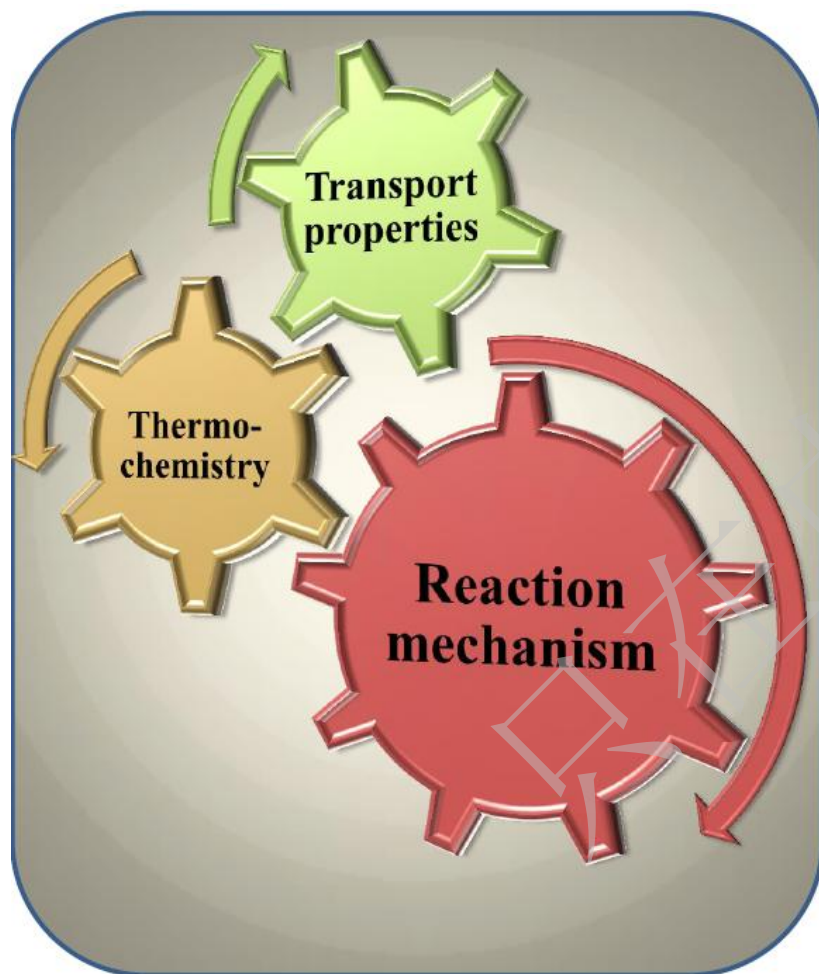
3.3. Kinetic Modeling Methods

I Roadmap for Model Development



3.3. Kinetic Modeling Methods

I Key Components of Kinetic Model



Literature research

Google
scholar

NIST



ELSEVIER

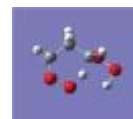
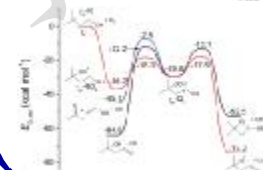


GRI-MECH 3.0

Combustion Kinetics Laboratory
Aerospace and Mechanical Engineering
Professor Hai Wang

QC calculation/estimation

$$k^{TST}(T) = \kappa \frac{k_B T}{h} \frac{Q^\ddagger(T)}{Q_{\text{rea}}(T)} \exp\left(-\frac{L^\ddagger}{k_B T}\right)$$

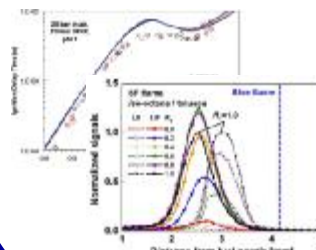


Estimation based on:

- ü Rate rules
- ü TST theory
- ü Structure and BDE
- ü Evans-Polanyi relationship

...

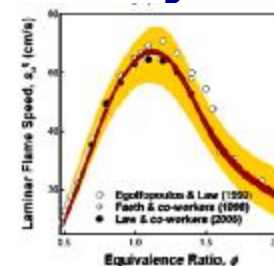
Validation/uncertainty



validation:

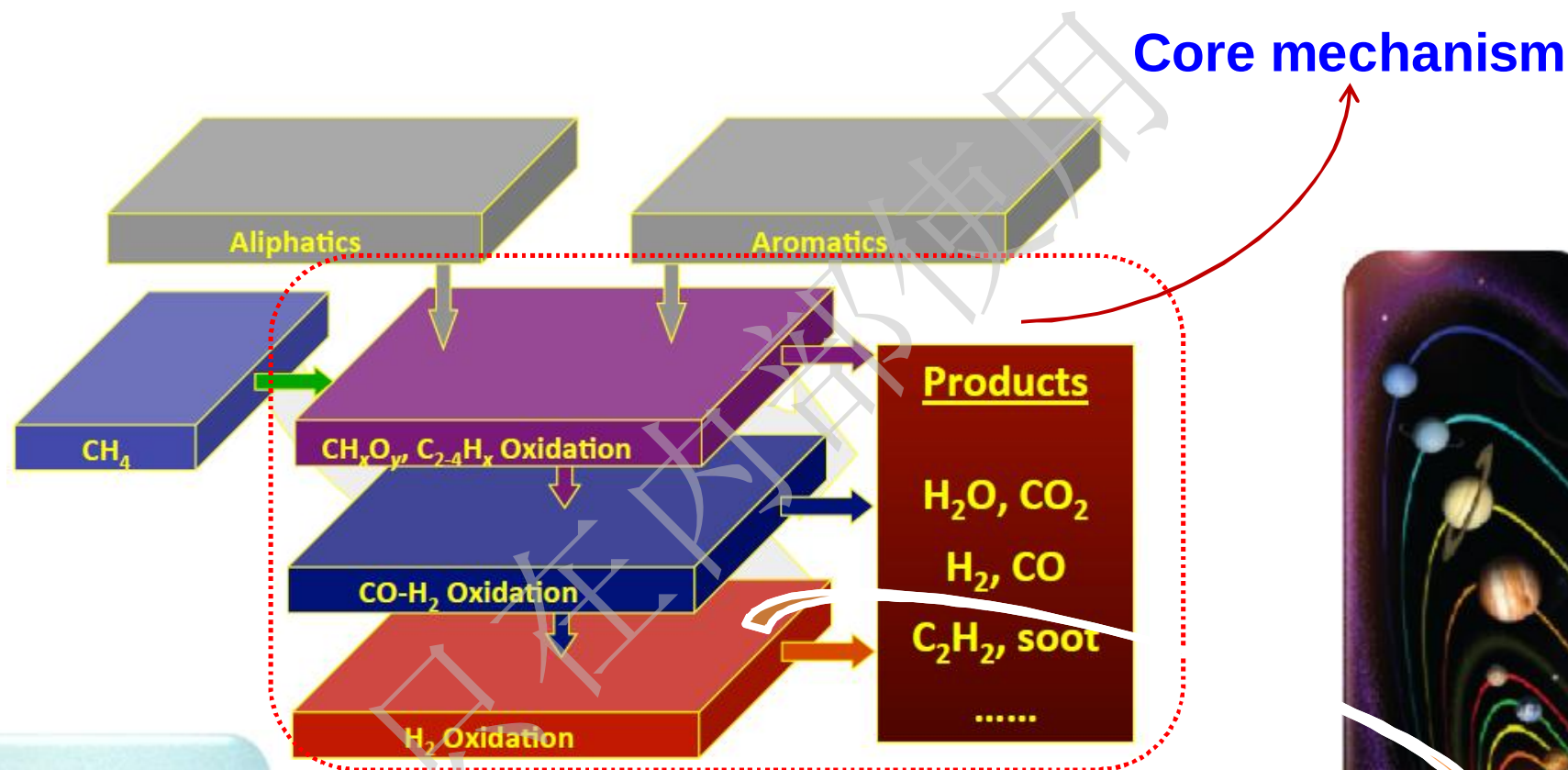
- ü Homogeneous
- ü Premixed flame
- ü Diffusion flame
- ü Soot formation

...



3.3. Kinetic Modeling Methods

I Hierarchical Structure of Kinetic Model



3.3. Kinetic Modeling Methods

I Classification of Reaction Mechanism

Gas C/H/O fuel

Gas N/S/halogen

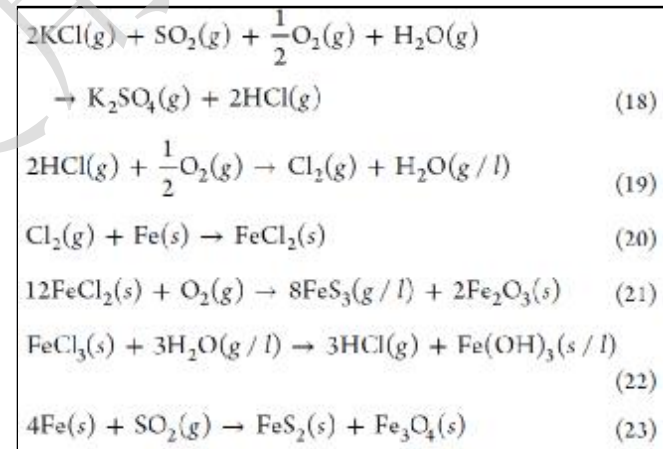
Gas Inorganic

H/O kinetic mechanism: units are $\text{cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ and $K_a, k = A T^n \exp(-E_a/RT)$, UF = uncertainty factor

No.	Reaction	A	n	E_a	Temperatures	UF
1a	$\text{H} + \text{H} + \text{M} = \text{H}_2 + \text{M}^*$ Enhanced third-body efficiencies (relative to Ar): $\text{H}_2 = 0, \text{N}_2 = 0, \text{H} = 0, \text{H}_2\text{O} = 14.3$	$7.00\text{E}+17$	-4.0	0	77-9000	2
1b	$\text{H} + \text{H} + \text{H}_2 = \text{H}_2 + \text{H}_2$	$1.00\text{E}+17$	-0.6	0	50-9000	2.5
1c	$\text{H} + \text{H} + \text{N}_2 = \text{H}_2 + \text{N}_2$	$5.40\text{E}+18$	-1.3	0	77-2000	3.2
1d	$\text{H} + \text{H} + \text{H} = \text{H}_2 + \text{H}$	$3.20\text{E}+15$	0	0	50-9000	3.2
2	$\text{O} + \text{O} + \text{M} = \text{O}_2 + \text{M}^*$ Enhanced third-body efficiencies (relative to Ar): $\text{O} = 28.8, \text{O}_2 = 8, \text{NO} = 2, \text{N} = 2, \text{N}_2 = 2,$ $\text{H}_2\text{O} = 5$	$1.00\text{E}+17$	-1.0	0	300-5000	2
3	$\text{O} + \text{H} + \text{M} = \text{OH} + \text{M}^*$ Enhanced third-body efficiency: $\text{H}_2\text{O} = 5$	$6.75\text{E}+18$	-4.0	0	2950-3700	3
4a	$\text{H}_2\text{O} + \text{M} = \text{H} + \text{OH} + \text{M}^*$ Enhanced third-body efficiencies (relative to Ar): $\text{H}_2\text{O} = 0, \text{H}_2 = 3, \text{N}_2 = 2, \text{O}_2 = 1.5$	$6.06\text{E}+27$	-3.312	120,770	300-3400	2
4b	$\text{H}_2\text{O} + \text{H}_2\text{O} = \text{H} + \text{OH} + \text{H}_2\text{O}$	$1.00\text{E}+26$	-2.44	120,160	300-3400	2
5a	$\text{H} + \text{O}_2 (+\text{M}) = \text{HO}_2 (+\text{M})^{1/2}$ Low-pressure limit: $k_{\text{form}} = 0.5$ Enhanced third-body efficiencies (relative to N_2):	$4.66\text{E}+12$ $5.70\text{E}+19$	0.44 -1.4	0 0	300-2000 300-2000	1.2

R655	$\text{N}_2 + \text{O} = \text{N} + \text{NO}$	$1.00 \cdot 10^{14}$	0.0	75490.0
R701	$\text{N} + \text{OH} = \text{N} + \text{H}$	$3.80 \cdot 10^{13}$	0.0	0.0
R708	$\text{NH} + \text{OH} = \text{NO} + \text{H}_2$	$2.00 \cdot 10^{13}$	0.0	0.0
R710	$\text{NH} + \text{O} = \text{NO} + \text{H}$	$1.84 \cdot 10^{14}$	0.0	0.0
R713	$\text{NH} + \text{NO} = \text{N}_2 + \text{O} + \text{H}$	$4.30 \cdot 10^{14}$	-0.5	0.0
R738	$\text{NNH} + \text{M} = \text{N}_2 + \text{H} + \text{M}$	$1.00 \cdot 10^{15}$	0.0	20000.0
R740	$\text{NNH} + \text{H} = \text{N}_2 + \text{H}_2$	$1.00 \cdot 10^{14}$	0.0	0.0
R744	$\text{NNH} + \text{O} = \text{N}_2 + \text{OH}$	$1.00 \cdot 10^{14}$	0.0	0.0
R745	$\text{HNO} + \text{O} = \text{NO} + \text{OH}$	$1.00 \cdot 10^{12}$	0.0	0.0
R746	$\text{HNO} + \text{OH} = \text{NO} + \text{H}_2\text{O}$	$3.60 \cdot 10^{13}$	0.0	0.0
R747	$\text{HNO} + \text{H} = \text{NO} + \text{H}_2$	$4.40 \cdot 10^{11}$	-0.7	650.0
R762	$\text{NO} + \text{H}_2 = \text{NO}_2 + \text{H}$	$2.10 \cdot 10^{12}$	0.0	-480.0
R764	$\text{NO} + \text{H} + \text{M} = \text{HNO} + \text{M}$	$2.34 \cdot 10^{16}$	-0.9	0.0

$\text{H} + \text{Cl} + \text{M} = \text{HCl} + \text{M}^*$	$2.0\text{E}23$	-2.450	0	-103.1
$\text{Cl} + \text{H}_2 = \text{HCl} + \text{H}$	$9.5\text{E}07$	1.720	3060	1.1
$\text{HCl} + \text{O} = \text{Cl} + \text{OH}$	$5.9\text{E}05$	2.114	4024	0.4
$\text{HCl} + \text{OH} = \text{Cl} + \text{H}_2\text{O}$	$4.1\text{E}05$	2.120	-1284	-15.7
$\text{Cl} + \text{H}_2\text{O}_2 = \text{HCl} + \text{HO}_2$	$6.6\text{E}12$	0.000	1950	-15.5
$\text{Cl} + \text{HO}_2 = \text{HCl} + \text{O}_2$	$7.5\text{E}14$	-0.630	0	-54.0



I Transportation fuels

I Biofuels

I Nitrogen fuels

I Coal combustion

I NOx/SOx formation

I SNCR

I Flame inhibition

I Coal combustion

I Biomass pyrolysis

I Metal fuels

3.3. Kinetic Modeling Methods

I Classification of Reaction Mechanism

Soot formation

1. HACA mechanism				
1a. H-abstraction				
$H + BIN_i \rightarrow H_2 + BIN_i^*$	$5.4 \cdot 10^5$	2	10.5	
1b. Acetylene addition				
$C_2H_2 + BIN_i^* \rightarrow products$	$1.0 \cdot 10^9$	0	5	
2. Soot inception ($i, n < 5$)				
$BIN_i^* + BIN_n^* \rightarrow products$	$1.0 \cdot 10^9$	0	6*	
$BIN_i + BIN_n^* \rightarrow products$	$1.0 \cdot 10^9$	0	8*	
$BIN_i + BIN_n \rightarrow products$	$1.0 \cdot 10^6$	0.5	0	
3. Surface growth				
3a. Small RR* addition				
$RR^* + BIN_i \rightarrow products$	$2.0 \cdot 10^9$	0	19*	
$RR^* + BIN_i^* \rightarrow products$	$2.5 \cdot 10^9$	0	3*	
3b. PAH condensation				
$i \geq 5$				
$PAH^* + BIN_i^* \rightarrow products$	$5.0 \cdot 10^{11}$	0.5	0	
$PAH^* + BIN_i \rightarrow products$				
$i < 5$ and $n \geq 5$				
$BIN_i^* + BIN_n^* \rightarrow products$	$3.0 \cdot 10^{11}$	0.5	0	
$BIN_i + BIN_n^* \rightarrow products$				

Catalytic combustion

Reaction	A* (cm, mol, s)	E _a (kJ/mol)	β (-)	References
R1 $H_2 + 2Ni(s) \rightarrow 2H(s)$	1.000×10^{-22}	0.0	0.0	[66], s.c. ^c
R2 $O_2 + 2Ni(s) \rightarrow 2O(s)$	1.000×10^{-22}	0.0	0.0	[18], s.c. ^c
R3 $CH_4 + Ni(s) \rightarrow CH_4(s)$	8.000×10^{-3}	0.0	0.0	s.c. ^{c,e}
R4 $H_2O + Ni(s) \rightarrow H_2O(s)$	1.000×10^{-14}	0.0	0.0	[18], s.c. ^c
R5 $CO_2 + Ni(s) \rightarrow CO_2(s)$	1.000×10^{-5}	0.0	0.0	s.c. ^{c,e}
R6 $CO + Ni(s) \rightarrow CO(s)$	5.000×10^{-14}	0.0	0.0	[67, 68], s.c. ^c
R7 $2H(s) \rightarrow 2Ni(s) + H_2$	$3.000 \times 10^{+21}$	98.0	0.0	[31, 69]
R8 $2O(s) \rightarrow 2Ni(s) + O_2$	$1.300 \times 10^{+22}$	464.0	0.0	[45]
R9 $H_2O(s) \rightarrow H_2O + Ni(s)$	$6.000 \times 10^{+13}$	68.9	0.0	[45]
R10 $CO(s) \rightarrow CO + Ni(s)$	$1.000 \times 10^{+13}$	122.4 - $50\theta_{CO(s)}$	0.0	[66]
R11 $CO_2(s) \rightarrow CO_2 + Ni(s)$	$1.000 \times 10^{+10}$	27.3	0.0	[45, 68]
R12 $CH_4(s) \rightarrow CH_4 + Ni(s)$	$2.000 \times 10^{+14}$	25.1	0.0	[47, 67]
R13 $CH_4(s) + Ni(s) \rightarrow CH_4(s) + H(s)$	$3.700 \times 10^{+21}$	57.7	0.0	[47]
R14 $CH_4(s) + H(s) \rightarrow CH_4(s) + Ni(s)$	$3.700 \times 10^{+21}$	56.1	0.0	[47]
R15 $CH_4(s) + Ni(s) \rightarrow CH_4(s) + H(s)$	$3.700 \times 10^{+24}$	100.0	0.0	[47]
R16 $CH_4(s) + H(s) \rightarrow CH_4(s) + Ni(s)$	$3.700 \times 10^{+21}$	49.8	0.0	[47]
R17 $CH_4(s) + Ni(s) \rightarrow CH_4(s) + H(s)$	$3.700 \times 10^{+24}$	97.1	0.0	[47]
R18 $CH(s) + H(s) \rightarrow CH_2(s) + Ni(s)$	$3.700 \times 10^{+24}$	73.6	0.0	[47]
R19 $CH(s) + Ni(s) \rightarrow C(s) + H(s)$	$3.700 \times 10^{+21}$	18.8	0.0	[47]
R20 $C(s) + H(s) \rightarrow CH(s) + Ni(s)$	$3.700 \times 10^{+21}$	173.6	0.0	[47]
R21 $CH_4(s) + O(s) \rightarrow CH_4(s) + OH(s)$	$1.700 \times 10^{+24}$	88.3	0.0	This work
R22 $CH_4(s) + OH(s) \rightarrow CH_4(s) + O(s)$	$3.700 \times 10^{+21}$	23.4	0.0	This work

CVD mechanism

$SiH_4 (+M) = SiH_2 + H_2 (+M)$	3.2×10^9	1.7	54.7
Low-pressure parameters	4.0×10^{12}	0	45.1
$Si_2H_6 (+M) = SiH_4 + SiH_2 (+M)$	1.8×10^{10}	1.7	50.2
Low-pressure parameters	1.8×10^{35}	-10.4	56.0
$Si_2H_6 (+M) = Si_2H_4B + H_2 (+M)$	9.1×10^9	1.8	50.2
Low-pressure parameters	7.8×10^{40}	-7.8	59.0
$Si_3H_8 (+M) = SiH_2 + Si_2H_6 (+M)$	7.0×10^{12}	1.0	52.7
Low-pressure parameters	8.4×10^{55}	-15.1	60.5
$Si_3H_8 (+M) = Si_2H_4B + SiH_4 (+M)$	3.7×10^{12}	1.0	50.9
Low-pressure parameters	4.4×10^{66}	-17.3	59.3
$Si_2H_4B (+M) = Si_2H_4A (+M)$	2.5×10^{13}	0.2	5.4
Low-pressure parameters	1.1×10^{33}	-5.8	9.2
$SiH_4 + p_X \rightarrow p_X(\eta_{Si} + 1, \eta_H + 2) + H_2$	S.R.	$3.0 \times 10^{+3}$	37.5
$SiH_2 + p_X \rightarrow p_X(\eta_{Si} + 1, \eta_H + 2)$	Cond.	1.0	-
$Si_2H_6 + p_X \rightarrow p_X(\eta_{Si} + 2, \eta_H + 2) + 2H_2$	S.R.	$3.0 \times 10^{+4}$	37.5
$Si_2H_4A + p_X \rightarrow p_X(\eta_{Si} + 2, \eta_H + 4)$	Cond.	1.0	-
$Si_2H_4B + p_X \rightarrow p_X(\eta_{Si} + 2, \eta_H + 4)$	Cond.	1.0	-
$Si_3H_8 + p_X \rightarrow p_X(\eta_{Si} + 3, \eta_H + 2) + 3H_2$	S.R.	$3.0 \times 10^{+4}$	37.5
$p_X \rightarrow p_X(\eta_{Si}, \eta_H - 2) + H_2$	H ₂	$1.2 \times 10^{+9}$	47.0

I Soot and PAH formation

I Coking mechanism

I Catalytic combustion

I Surface effect on combustion

I Chemical Vapor

Deposition

I Flame synthesis

3.3. Kinetic Modeling Methods

I Classification of Reaction Mechanism

Plasma mechanism

$e + CH_4 = e + CH_3 + H$	σ^e
$e + CH_4 = e + CH_3^+ + H$	σ
$e + CH_4(+) = CH_2 + 2H$	$1.7 \times 10^{-7} (300/T)^{0.5}$
$e + CH_4(+) = CH_3 + H$	$1.7 \times 10^{-7} (300/T)^{0.5}$
$CH_4(+) + O_2 = CH_4 + O_2^+(+)$	5×10^{-10}
$Ar + e = Ar^+ + e$	Σ
$Ar + e = Ar^+(+) + 2e$	Σ
$Ar^+ + O_2 = Ar + 2O$	2×10^{-10}
$Ar^+(+) + O_2 = Ar + O_2^+(+)$	1×10^{-10}
$Ar^+ + CH_4 = Ar + CH_2 + 2H$	3.3×10^{-10}
$Ar^+ + CH_4 = Ar + CH + H_2 + H$	5.8×10^{-10}
$Ar^+ + CH_4 = Ar + CH_3 + H$	5.8×10^{-10}
$Ar^+ + CH_4 = Ar + CH_2 + H_2$	5.8×10^{-10}
$Ar^+(+) + CH_4 = Ar + CH_3^+(+) + H$	6.5×10^{-10}
$Ar^+(+) + CH_4 = Ar + CH_2^+(+) + H_2$	1.4×10^{-10}
$He + e = He^+ + e$	Σ
$He + e = He^+(+) + 2e$	Σ
$He^+ + O_2 = He + O_2^+(+) + e$	$1.5 \times 10^{-11} T^{0.5}$
$He^+ + O = He + O^+(+) + e$	$1.5 \times 10^{-11} T^{0.5}$
$He^+(+) + O_2 = He + O^+(+) + O$	$0.6 \times 10^{-11} T^{0.5}$
$He^+(+) + O_3 = He + O_2 + O^+(+)$	$0.6 \times 10^{-11} T^{0.5}$
$He^+(+) + O_2(^1\Delta_g) = He + O^+(+) + O$	$0.6 \times 10^{-11} T^{0.5}$
$He^+(+) + O(^1D) = He + O^+(+)$	$2.9 \times 10^{-12} T^{0.5}$
$He + 2O = He^+ + O_2$	1×10^{-33}
$He + O(^1D) = He + O$	1×10^{-10}
$He^+ + CH_4 = He + CH + H_2 + H$	5.6×10^{-13}

Excited state species

$H_2 + O_2(^1\Delta) \Rightarrow H_2O + O(^1D)$	3.5E+13	0	39,750
$H_2 + O_2(^1\Delta) \Rightarrow OH + OH$	1.7E+15	0	33,780
$O_2(^1\Delta) + O + M \Rightarrow O_3 + M$	6.9E+12	0	-2090
$HO_2 + O_2(^1\Delta) \Rightarrow O_2 + HO_2^+$	1.0E+13	0	0
$O_2(^1\Delta) + HO_2^+ \Rightarrow H + O_2 + O_2$			
$HO_2^+ + O_2 \Rightarrow HO_2 + O_2$	3.6E+12	0	0
$O + OH + M \Rightarrow HO_2 + M$	1.0E+15	0	0
$O_3 + H \Rightarrow OH + O_2(^1\Delta)$			
$O_3 + H \Rightarrow O + HO_2$			
$O_3 + OH \Rightarrow HO_2 + O_2(^1\Delta)$	4.8E+11	0	2000
$O_3 + HO_2 \Rightarrow OH + O_2 + O_2(^1\Delta)$	1.0E+10	0	2000
$O_3 + H_2 \Rightarrow OH + HO_2$	6.0E+10	0	20,000
$H + OH + OH \Rightarrow OH^+ + H_2O$	1.1E+16	0	0
$O(^1D) + H + M \Rightarrow OH^+ + M$	1.5E+18	-1	0
$H_2 + HO_2 \Rightarrow H_2O + OH^+$	4.8E+19	-1.7	38,000
$H_2O + OH \Rightarrow H_2 + HO_2$	7.9E+09	0.43	71,700

$OH^+ + O_2 \Rightarrow OH + O_2$	8.40E+11	0.5	-482
$OH^+ + N_2 \Rightarrow OH + N_2$	1.08E+11	0.5	-1238
$OH^+ + H_2O \Rightarrow OH + H_2O$	2.96E+12	0.5	-861
$OH^+ + H_2 \Rightarrow OH + H_2$	3.54E+11	0.5	-444
$OH^+ + OH \Rightarrow OH + OH$	1.50E+12	0.5	0
$OH^+ + H \Rightarrow OH + H$	1.50E+12	0.5	0
$OH^+ + O \Rightarrow OH + O$	1.50E+12	0.5	0
$OH^+ + Ar \Rightarrow OH + Ar$	2.17E+10	0.5	2060
$OH^+ + H_2 \Rightarrow H_2O + H$	2.60E+12	0.5	-444
$OH^+ + O_2 \Rightarrow O_3 + H$	2.52E+11	0.5	-482
$OH^+ + O_2 \Rightarrow HO_2 + O$	1.008E+12	0.5	-482
$OH^+ + H_2O \Rightarrow H_2O_2 + H$	2.96E+12	0.5	-861
$OH^+ \Rightarrow OH + h\nu$	1.40E+06	0	0

- I Plasma assisted combustion
- I Electrically assisted combustion
- I Flame ionization mechanism

- I Flame radiation
- I Chemiluminescence

3.3. Kinetic Modeling Methods

I Reaction Types

Reactant number/P,T dependence

- n Unimolecular reaction: $k(P,T)$
- n Chemically-activated reaction: $k(P,T)$
- n Bimolecular reaction: $k(T)$

Trends of complexity

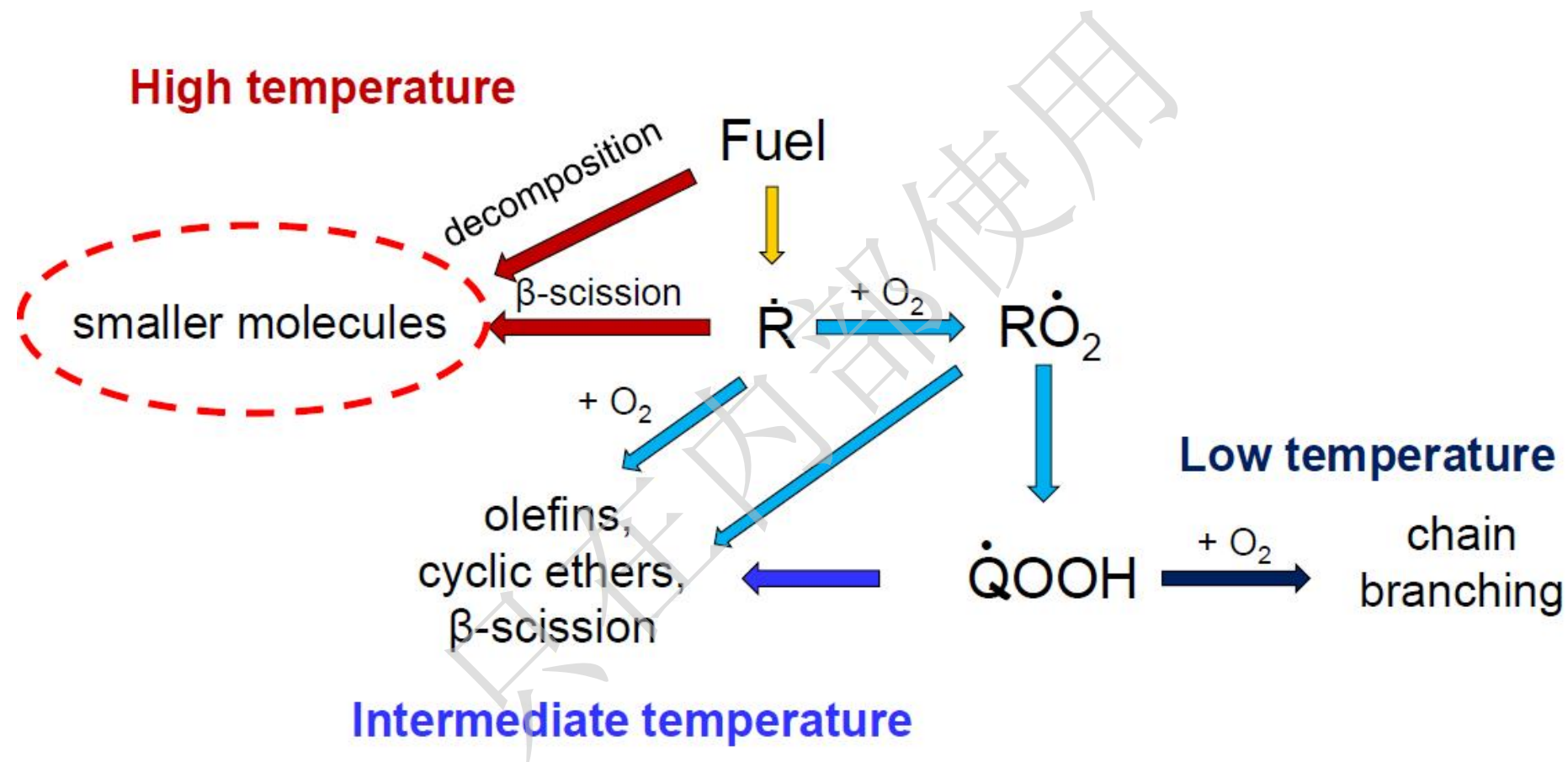
- n Elementary reaction
- n Formally direct reaction
- n Vertical lumped reaction
- n Lumped species reaction
- n Semi-global reaction
- n Global/lumped reaction

Typical bimolecular reaction type

Reaction type	Concentrations		Rate Coefficient	Rate
	Reactant 1	Reactant 2		
Molecule + molecule	<i>large</i>	<i>large</i>	<i>very small</i>	<i>small</i>
Molecule + radical	<i>large</i>	<i>small</i>	<i>intermediate</i>	<i>large</i>
Radical + radical	<i>small</i>	<i>small</i>	<i>large</i>	<i>small</i>

3.3. Kinetic Modeling Methods

I Classification of Reaction



I Classification of Reaction

[illegible]

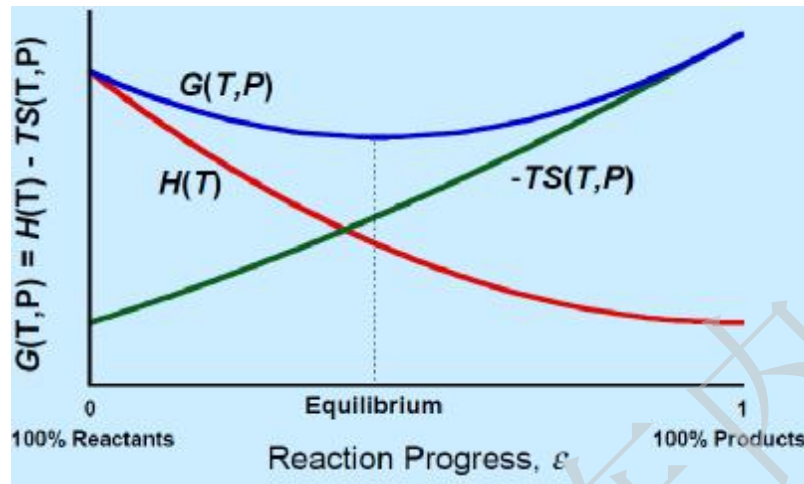
Reaction class 1: Unimolecular fuel decomposition
Reaction class 2: H-atom abstractions
Reaction class 3: Alkyl radical decomposition
Reaction class 4: Alkyl radical + O₂ = olefin + HO₂
Reaction class 5: Alkyl radical isomerization
Reaction class 6: H atom abstraction from olefins
Reaction class 7: Addition of radical species to olefins
Reaction class 8: Alkenyl radical decomposition
Reaction class 9: Olefin decomposition

- 10: Alkyl radical addition to O_2 ($R + O_2 = RO_2$)
- 12, 13, and 14: $R + R'O_2 = RO + R'O$
- 15: Alkylperoxy radical isomerization ($RO_2 = QOOH$)
- 16: Concerted eliminations ($RO_2 = \text{alkene} + HO_2$)
- 17: $RO_2 + HO_2 = ROOH + O_2$
- 18: $RO_2 + H_2O_2 = ROOH + HO_2$
- 19: $RO_2 + CH_3O_2 = RO + CH_3O + O_2$
- 20: $RO_2 + R'O_2 = RO + R'O + O_2$
- 21: $ROOH = RO + OH$
- 22: RO decomposition
- 23: $QOOH = \text{cyclic ether} + OH$ (cyclic ether formation)
- 24: $QOOH = \text{alkene} + HO_2$ (radical beta to OOH)
- 25: $QOOH = \text{alkene} + \text{carbonyl} + OH$ (radical gamma to OOH)
- 26: Addition of QOOH to molecular oxygen O_2 ($QOOH + O_2 = O_2QOOH$)
- 27: O_2QOOH isomerization to carbonylhydroperoxide + OH
- 28: Carbonylhydroperoxide decomposition
- 29: Reactions of cyclic ethers with OH and HO_2

3.3. Kinetic Modeling Methods

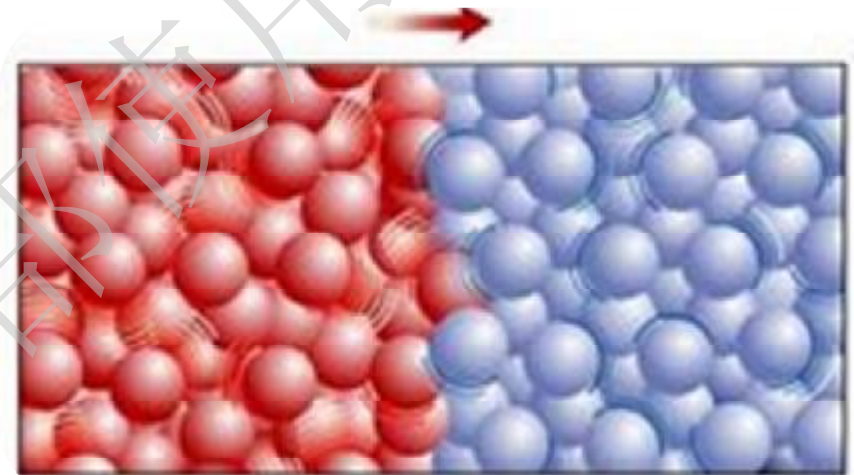
I Thermodynamic and Transport Data

Enthalpy, entropy, heat capacity



- n Chemical equilibria
- n Chemical insight: endo vs. exothermic, spontaneous or not
- n Heat release, adiabatic flame temperature
- n ...

Heat and mass transfer



- n Diffusion coefficients
- n Viscosities
- n Thermal conductivities
- n Thermal diffusion coefficients

3.3. Kinetic Modeling Methods

I Numerical Model

n Conservation equations

Mass $\frac{\partial \rho}{\partial t} + \nabla(\rho \mathbf{v}) = 0$

Momentum $\frac{\partial}{\partial t}(\rho \mathbf{v}) + \nabla(\rho \mathbf{v} \mathbf{v} + p \mathbf{I}) = \nabla \boldsymbol{\tau} + \rho \mathbf{g}$

Energy $\rho C_p \frac{\partial T}{\partial t} + \rho C_p \mathbf{v} \nabla T = -\nabla \mathbf{q} - \nabla \mathbf{q}_{rad} - \rho \sum_{k=1}^{NC} C_{p,k} \omega_k \mathbf{v}_k - \sum_{k=1}^{NC} h_k \dot{\Omega}_k$

Species $\frac{\partial}{\partial t}(\rho \omega_k) + \nabla(\rho \omega_k \mathbf{v}) = -\nabla(\rho \omega_k \mathbf{V}_k) + \dot{\Omega}_k \quad k = 1 \dots NC$

n Boundary conditions

Inlet $(x = 0) \begin{cases} T = T_0 \\ \rho u \omega_k + \rho \omega_k V_k = (\rho u \omega_k)_0 \end{cases} \quad k = 1 \dots N_s$

Outlet $(x = L) \begin{cases} \frac{dT}{dx} = 0 \\ \frac{d\omega_k}{dx} = 0 \end{cases} \quad k = 1 \dots N_s$

n Numerical algorithms

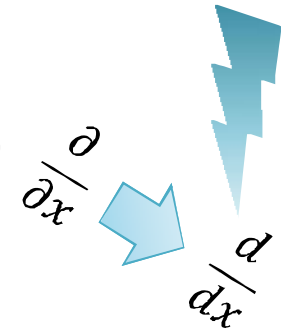
p Finite difference method

p Solve ODE

p Damping Newton iteration

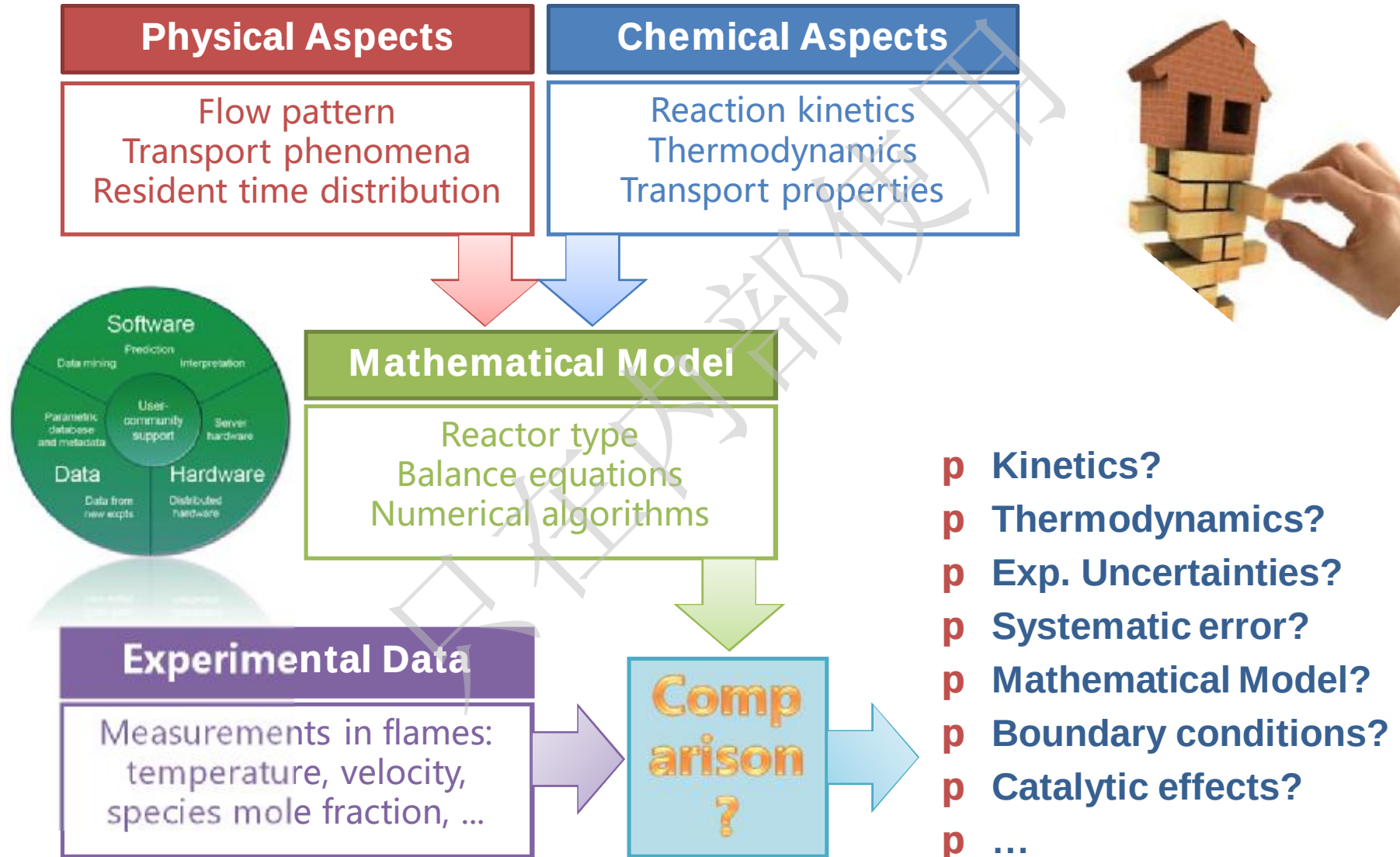
p Time step integration

p Operator splitting.....



3.3. Kinetic Modeling Methods

I Combustion Kinetic Modeling



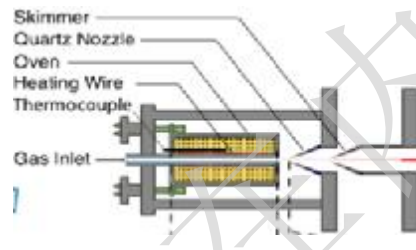
3.3. Kinetic Modeling Methods

I Validation Experiments – Homogeneous Reactors

Jet Stirred Reactor



Flow Reactors



Shock Tube Rapid Compression Machine



Homogeneous

Homogeneous

Homogeneous

Non
Homogeneous

Non
Homogeneous

Non
Homogeneous

Homogeneous

Shock Tube

Turbulent (Plug)
Flow Reactor

Rapid
Compression

Laminar Flow
Reactor

Laminar Premixed
Flame

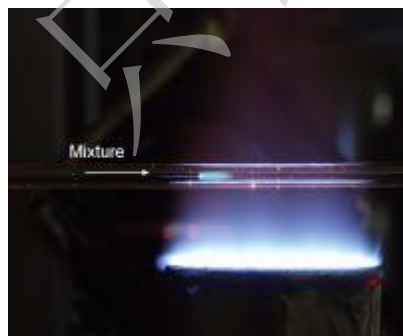
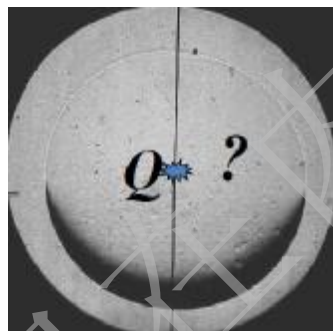
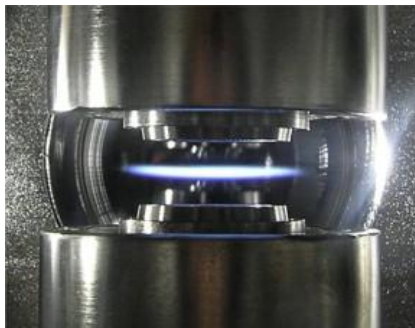
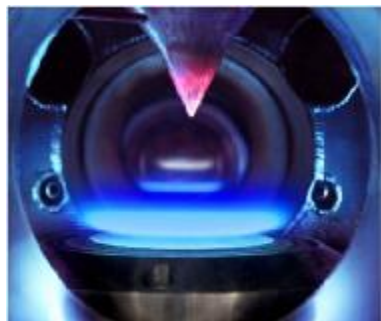
Laminar Diffusion
Flame

Perfectly Stirred
Reactor

Degree of Mixing

3.3. Kinetic Modeling Methods

I Validation Experiments – Flames



Homogeneous

Homogeneous

Homogeneous

Non
Homogeneous

Non
Homogeneous

Non
Homogeneous

Homogeneous

Shock Tube

Turbulent (Plug)
Flow Reactor

Rapid
Compression

Laminar Flow
Reactor

Laminar Premixed
Flame

Laminar Diffusion
Flame

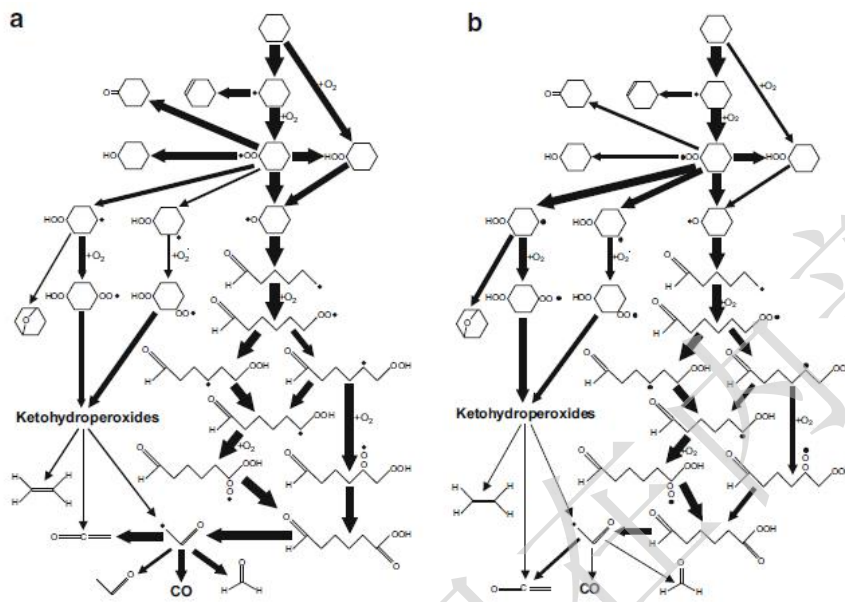
Perfectly Stirred
Reactor

Degree of Mixing

3.3. Kinetic Modeling Methods

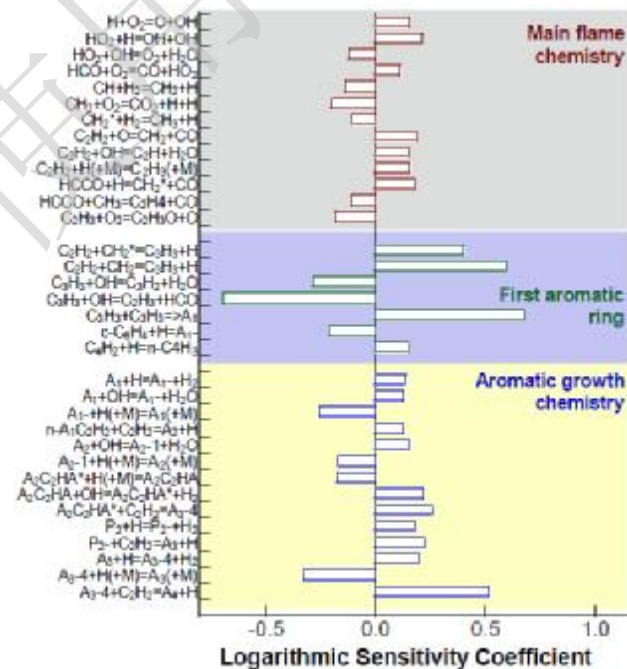
I Modeling Analysis

Rate of production (ROP) analysis



- n Species conversion pathways
- n Pathways leading to the consumption or production of a species
- n Flux balance analysis

Sensitivity analysis

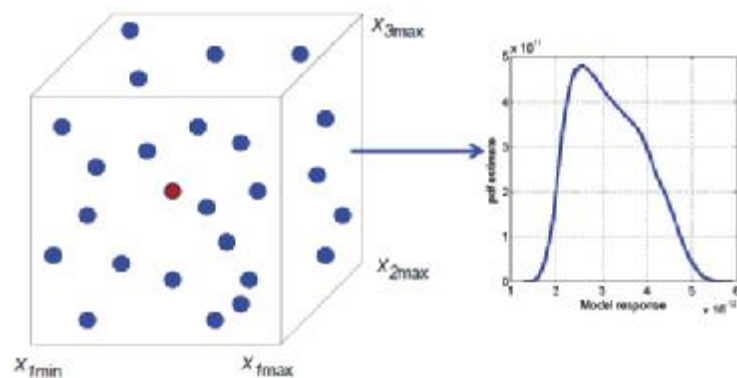


- n Determine how great each input parameter contributes to the output uncertainty

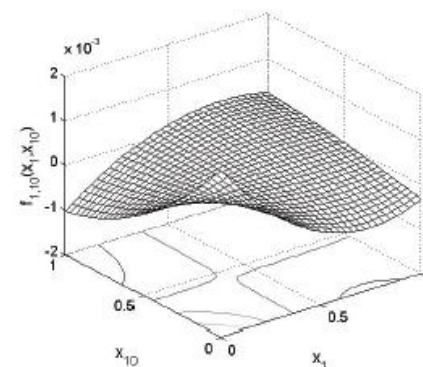
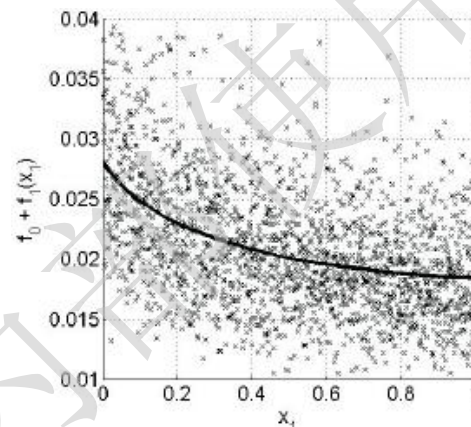
3.3. Kinetic Modeling Methods

I Uncertainty/Global sensitivity analysis

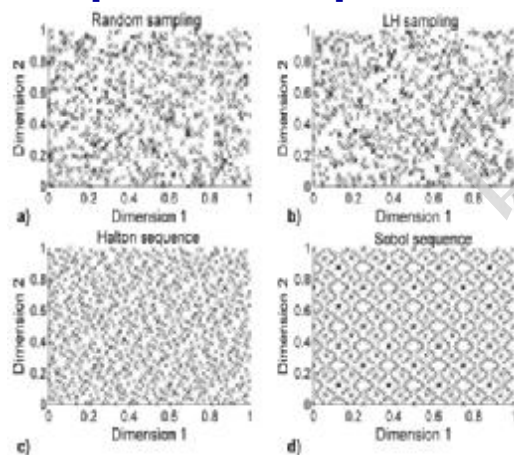
Attempts to cover whole input space using a sampling method



Perform many simulations (N) until output mean/variances converge



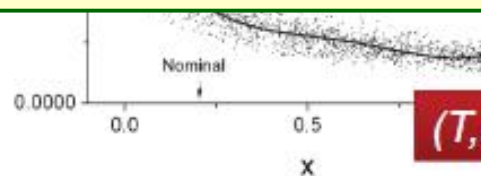
Sample from input distributions



Explore which parameters contribute most to the width of the predicted output distribution

0.0006
CH3OH+HO2

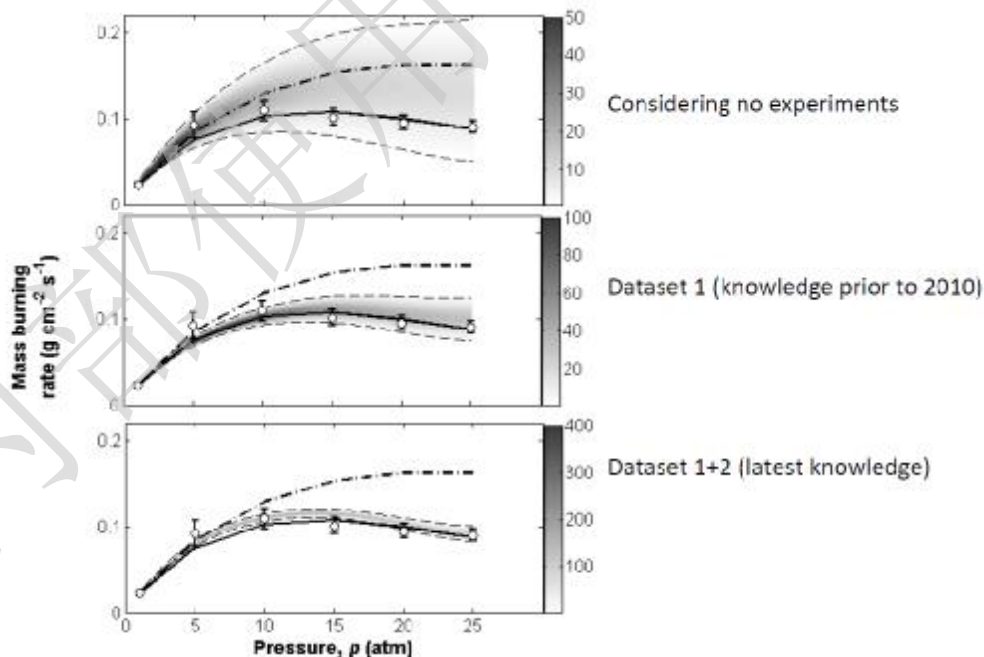
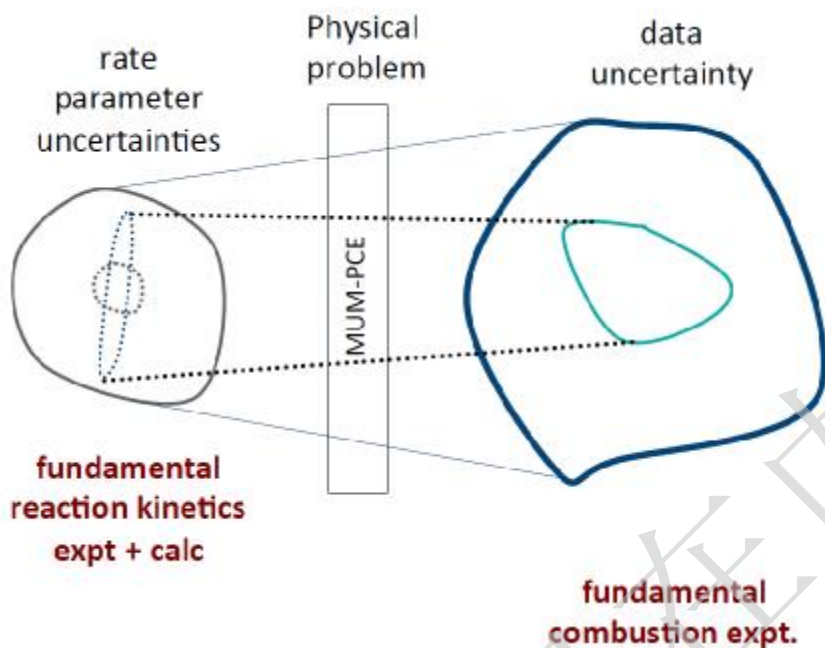
Low scatter indicates low



$(T, P, \phi) = (1150K, 5bar, 1)$

3.3. Kinetic Modeling Methods

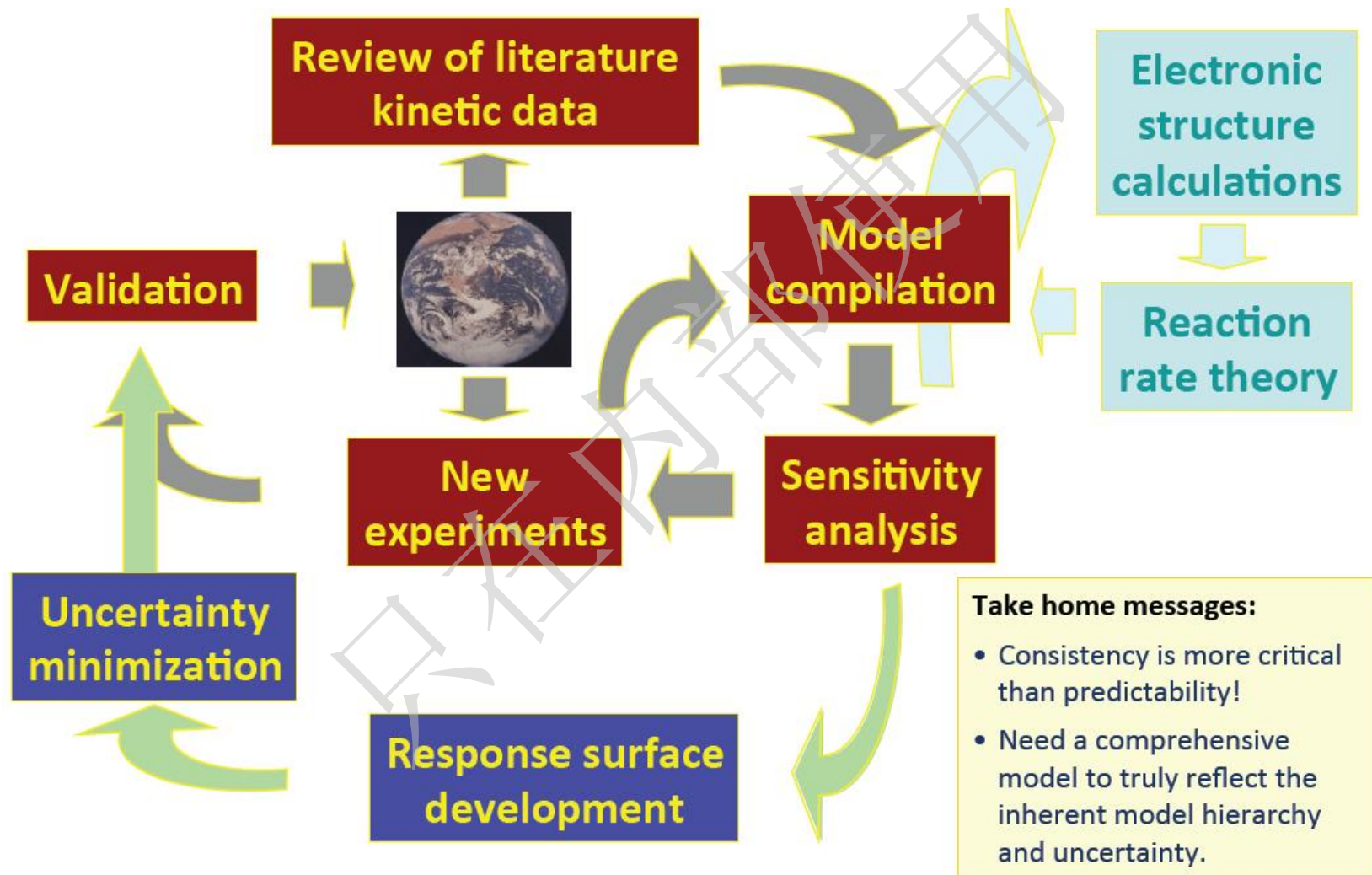
I Uncertainty analysis based model optimization



- n Chemical model + associated uncertainty
- n Physical model
- n Predictions + associated uncertainty
- n Model uncertainty constraining

3.3. Kinetic Modeling Methods

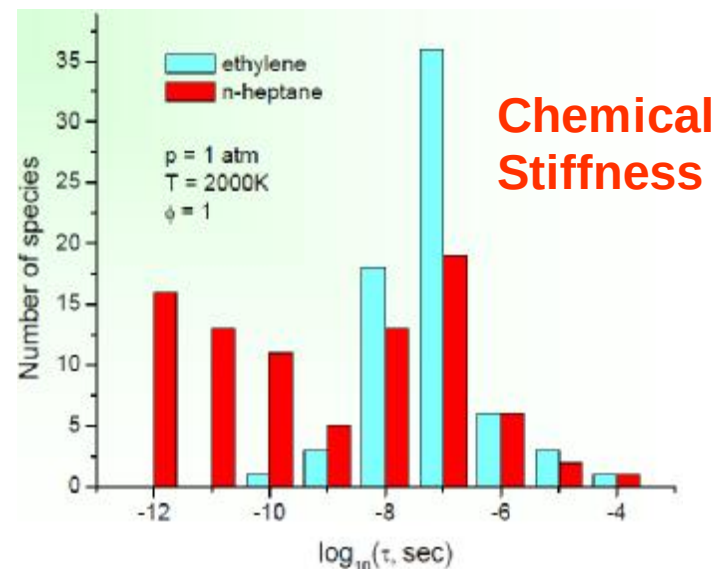
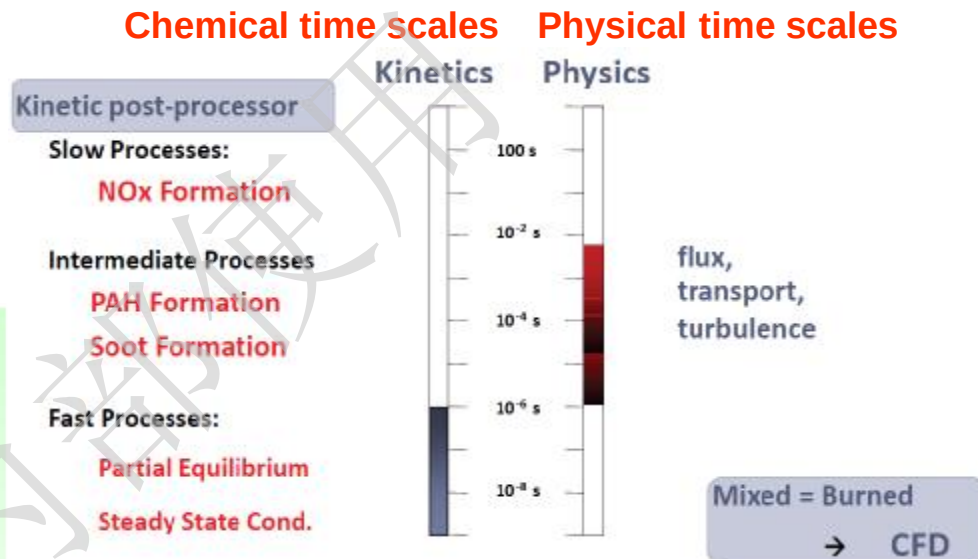
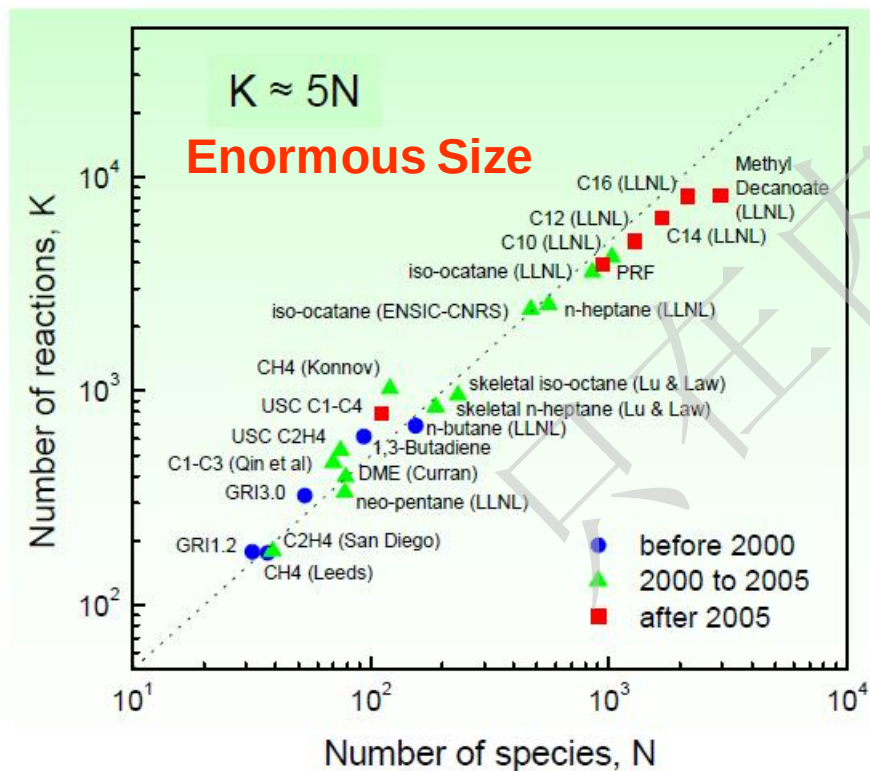
I Uncertainty analysis based model optimization



3.3. Kinetic Modeling Methods

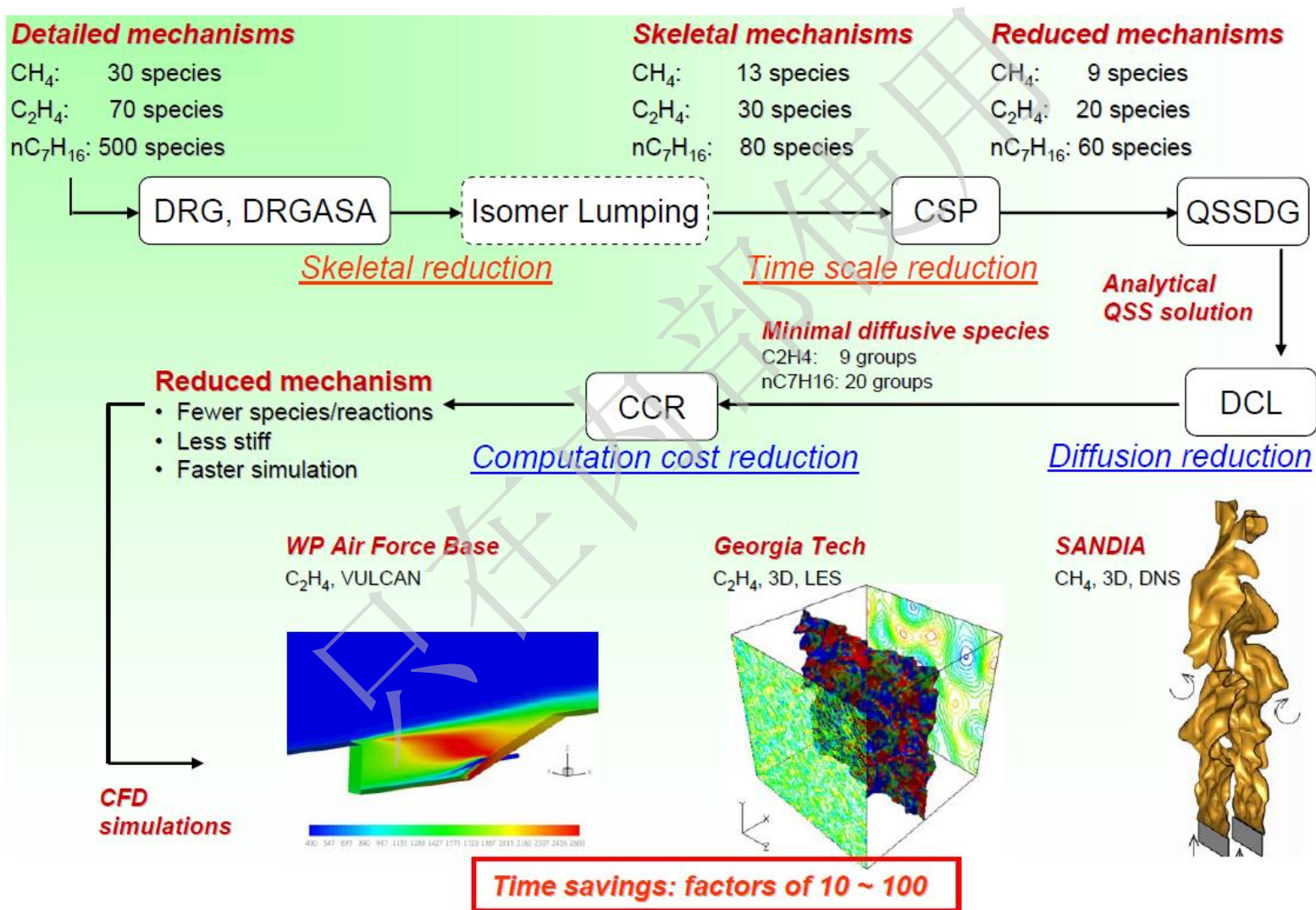
I Model Reduction – Necessity

n Problem: Detailed mechanism cannot always be used in CFD simulations; computation time $\sim N^2$



3.3. Kinetic Modeling Methods

I Model Reduction – Flow Chart



3.3. Kinetic Modeling Methods

I Model Reduction – Methods

Skeletal reduction (Removal of redundant species)

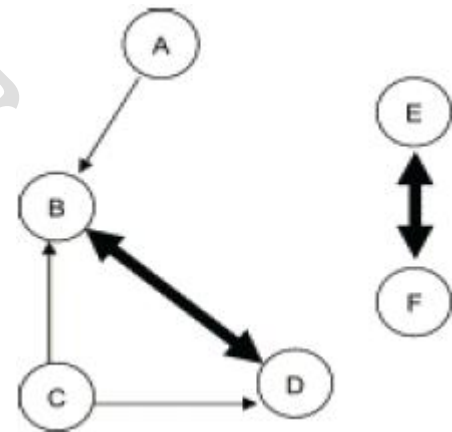
- n Trial & error with validation
- n Sensitivity analysis
- n Principal component analysis (PCA)
- n Graph based methods: DRG, DRGEP, PFA, DRGEPSA...

Timescale based reduction

- n Quasi steady state approximations (QSSA)
- n Partial equilibrium approximations (PEA)
- n Rate controlled constrained equilibrium (RCCE)
- n Computational singular perturbation (CSP)
- n Intrinsic low dimensional manifold (ILDM)

Other methods

- n In-situ adaptive tabulation (ISAT)
- n Mathematically based lumping
- n Dynamic adaptive chemistry (DAC)
- n



Kinetic post-processor

Slow Processes:

NO_x Formation

Intermediate Processes

PAH Formation

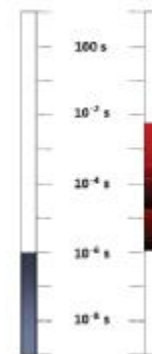
Soot Formation

Fast Processes:

Partial Equilibrium

Steady State Cond.

Kinetics Physics



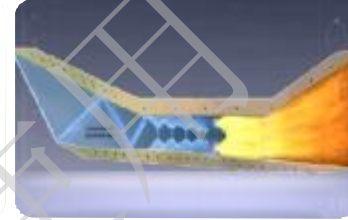
flux,
transport,
turbulence

Mixed = Burned

→ CFD

4. Applications

Engine combustion



Combustors design



New fuels



New combustion technology

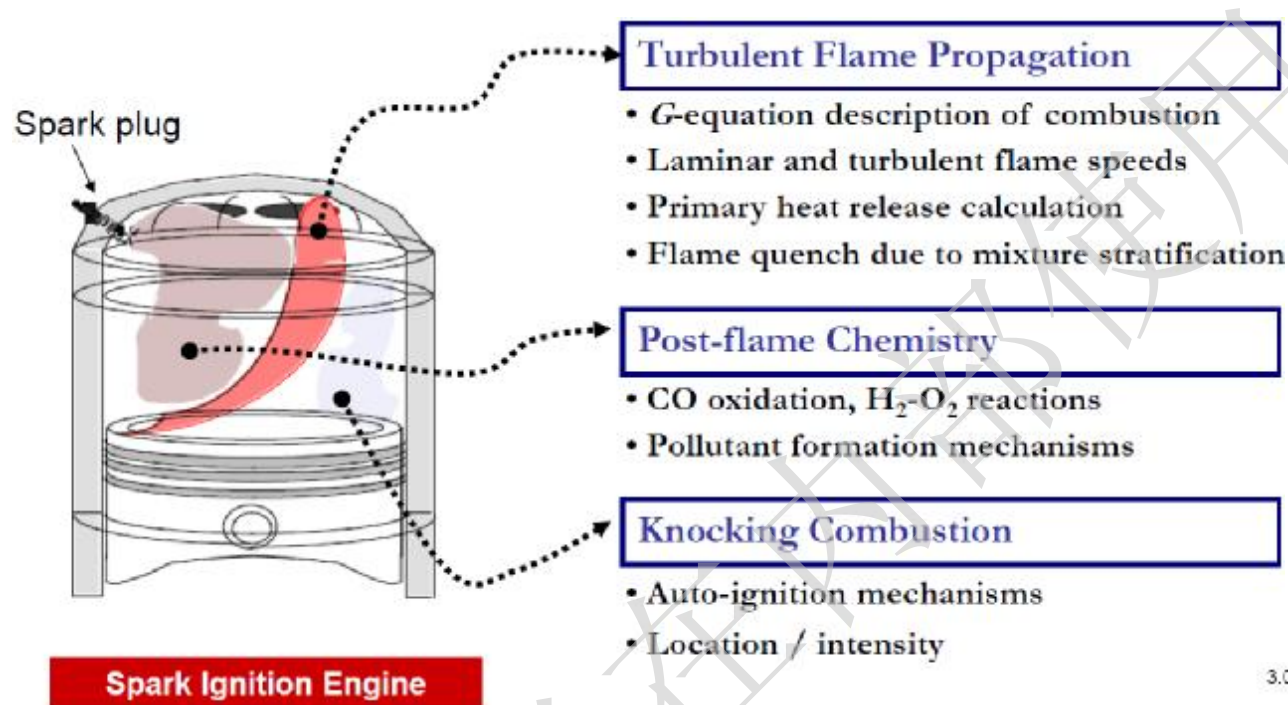


Polluants control

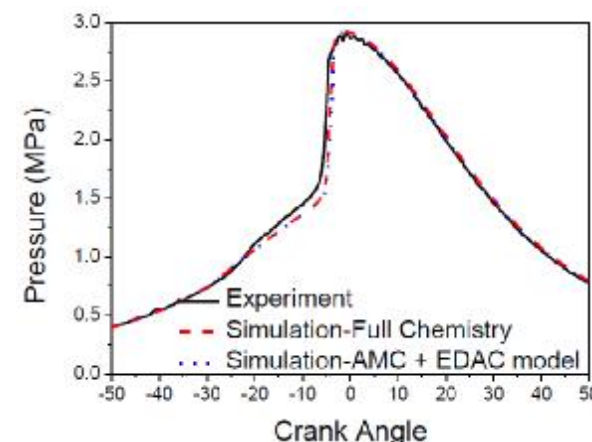
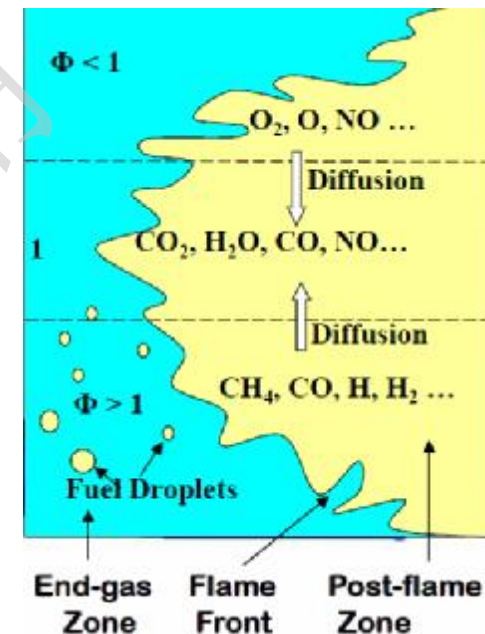


4.1. Gasoline Engine Applications

I Role of Chemistry in SI Engine



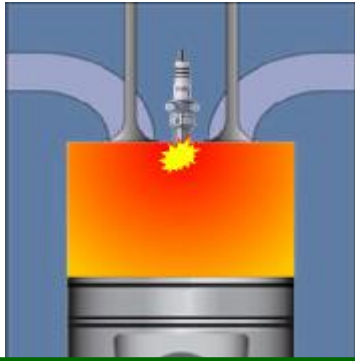
- n Improve efficiency
- n Predict ignition and knock
- n Control pollutant emission
- n ...



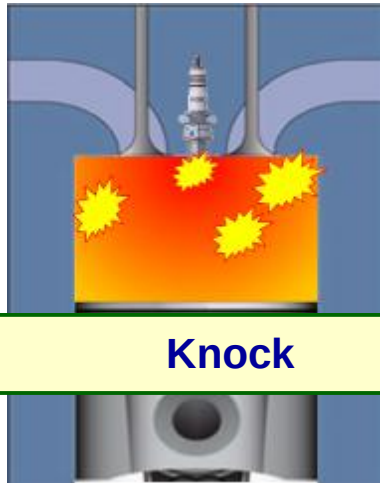
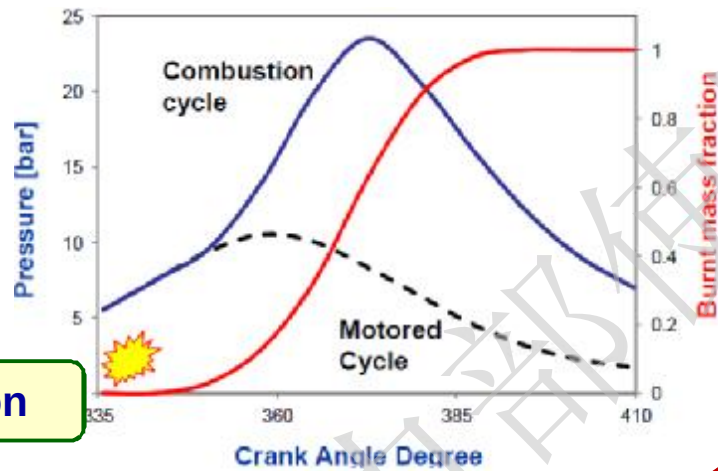
SOURCE: R. D. Reitz, 2014

4.1. Gasoline Engine Applications

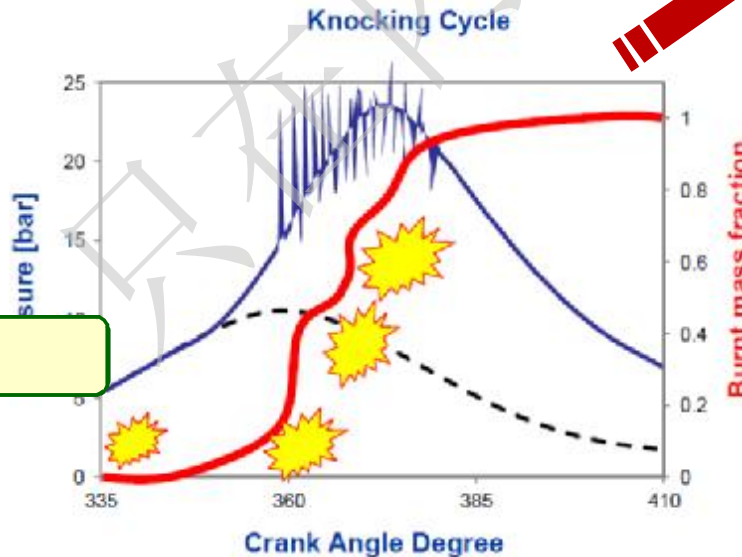
I Knock in SI Engine



Normal spark ignition



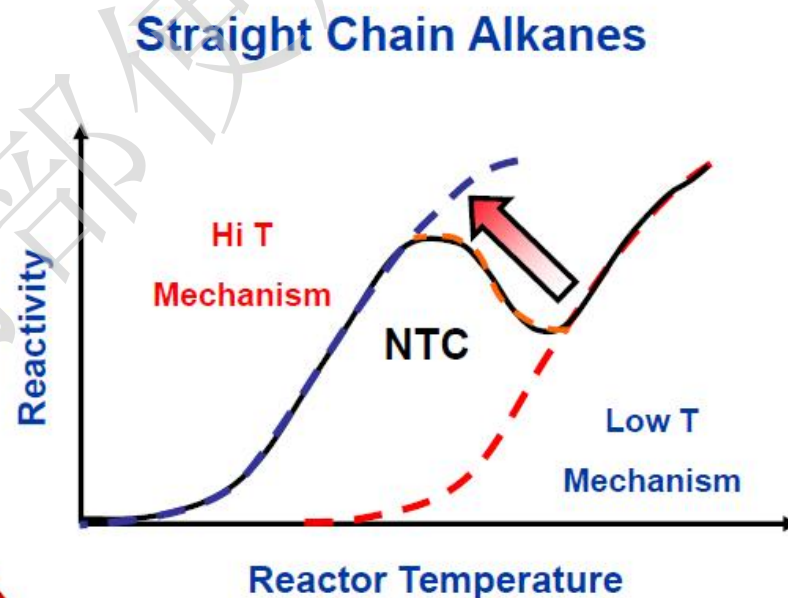
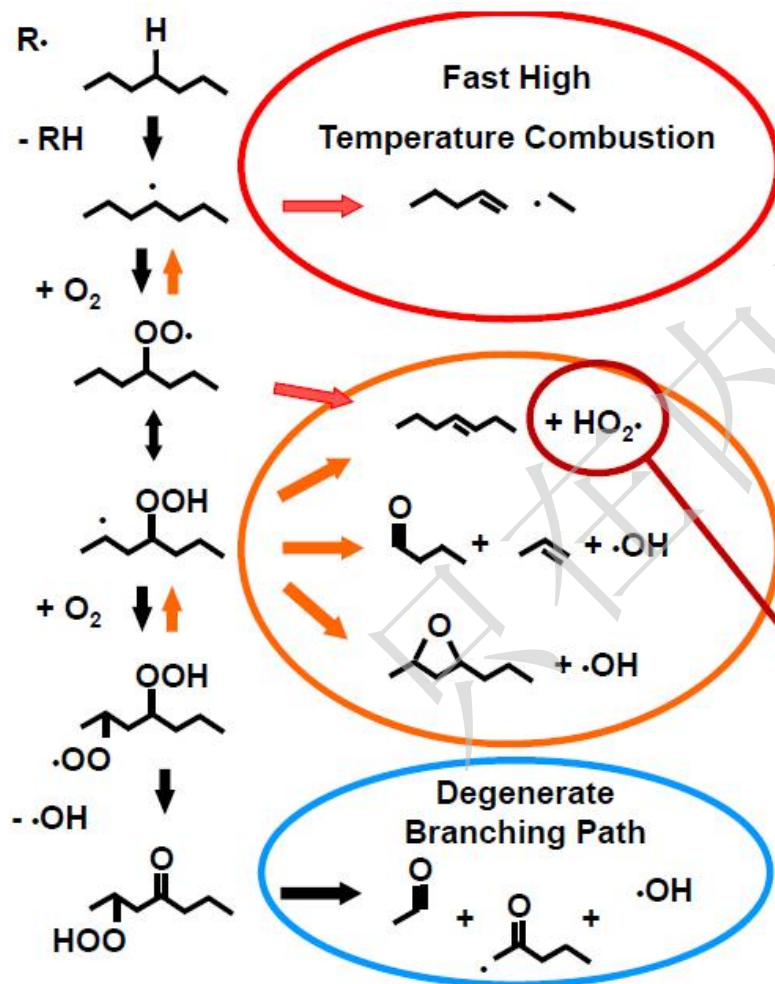
Knock



4.1. Gasoline Engine Applications

I Knock & Fuel – Linear Alkane Chemistry

- n Early gasoline engines uses reactive *n*-alkanes as fuel, thus easy to cause knock

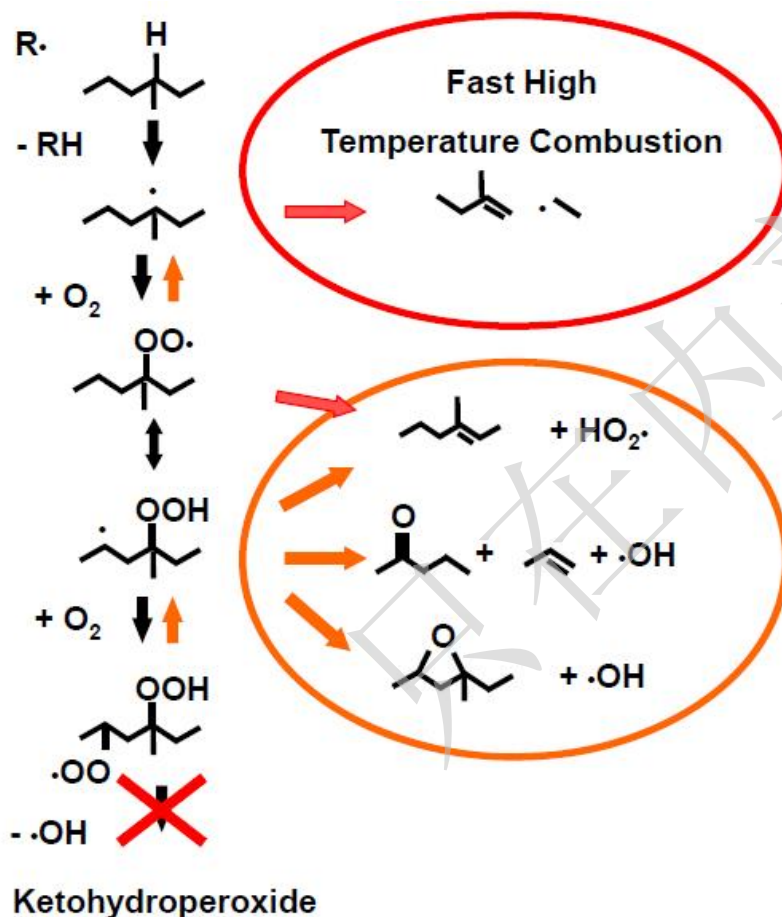


HO₂ Radicals lead to the formation of H₂O₂, when the temperature increases H₂O₂ → 2OH triggering the transition to high temperature ignition

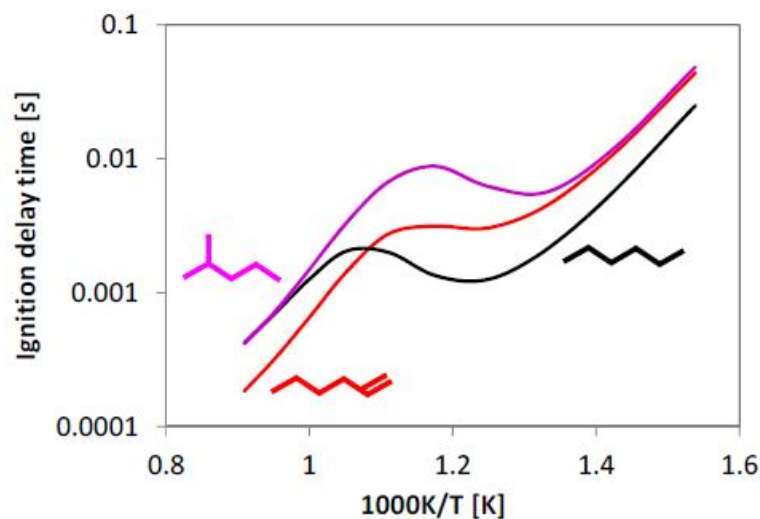
4.1. Gasoline Engine Applications

I Knock & Fuel – Branched Alkane Chemistry

- n If the abstraction reaction happens at a tertiary site of an iso-alkane, the formation of ketohydroperoxides is not possible



The internal isomerization steps required by the low temperature degenerate branching mechanism are slower because of the higher activation energy required for primary hydrogens



4.1. Gasoline Engine Applications

I Octane Number

N-heptane (PRF0)

Iso-octane (PRF100)

Research Octane Number (RON)

- n Speed 600 rpm

- n Inlet Temperature 325K

- n Spark Advance 13°

- n Representative of idling/low load conditions

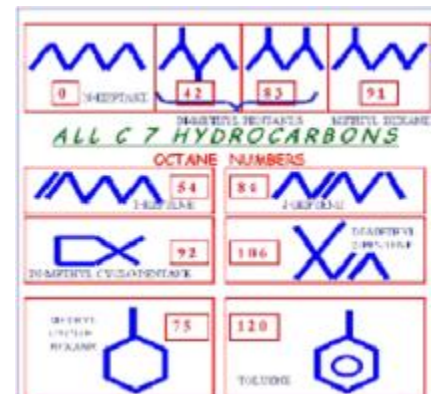
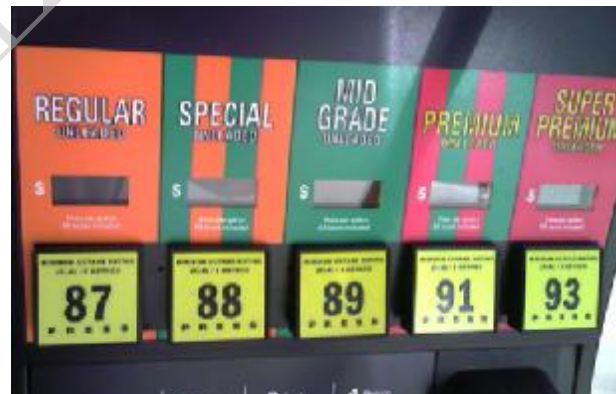
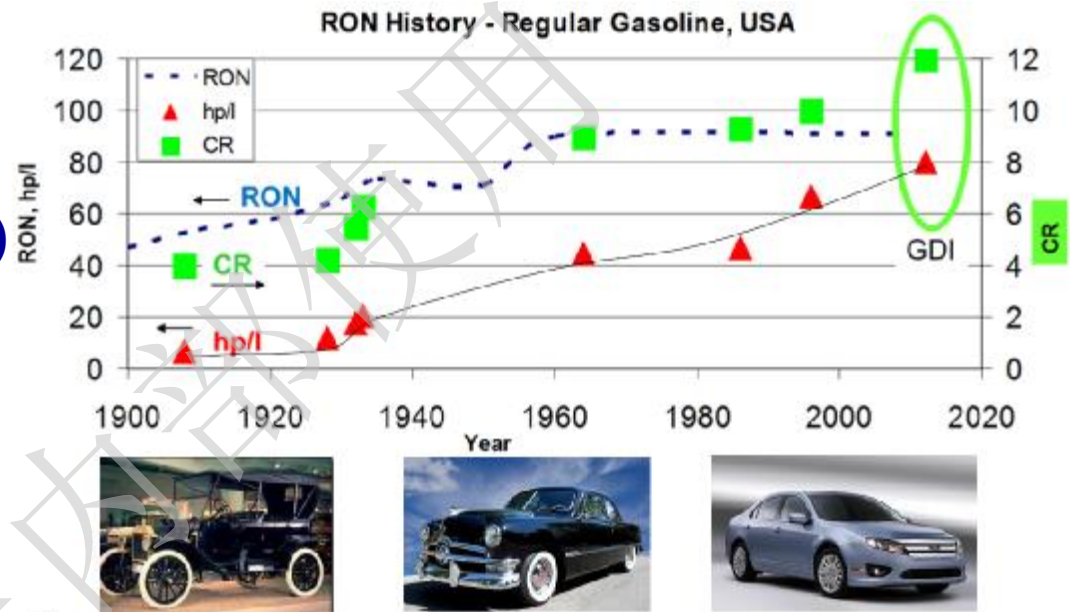
Motor Octane Number (MON)

- n Speed 900 rpm

- n Inlet Temperature 422K

- n Spark Advance 19-26°

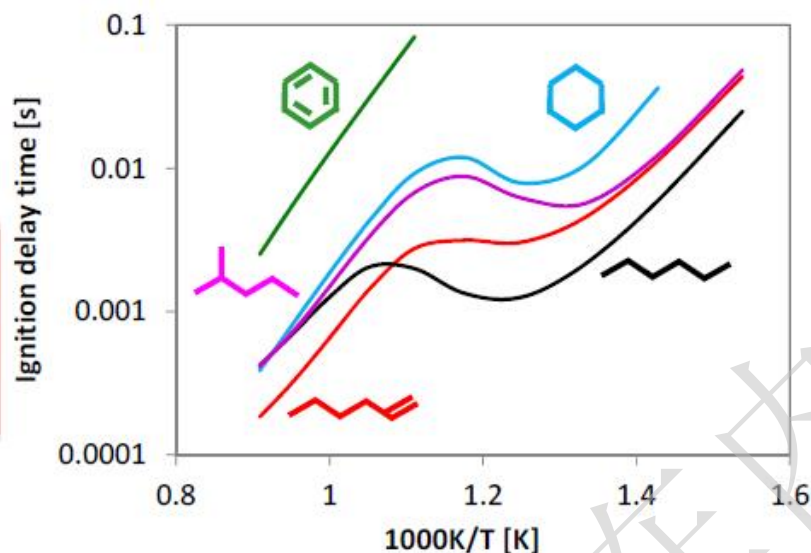
- n Representative of road/high load conditions



4.1. Gasoline Engine Applications

I Hydrocarbons with High Octane Number

Naphthenes

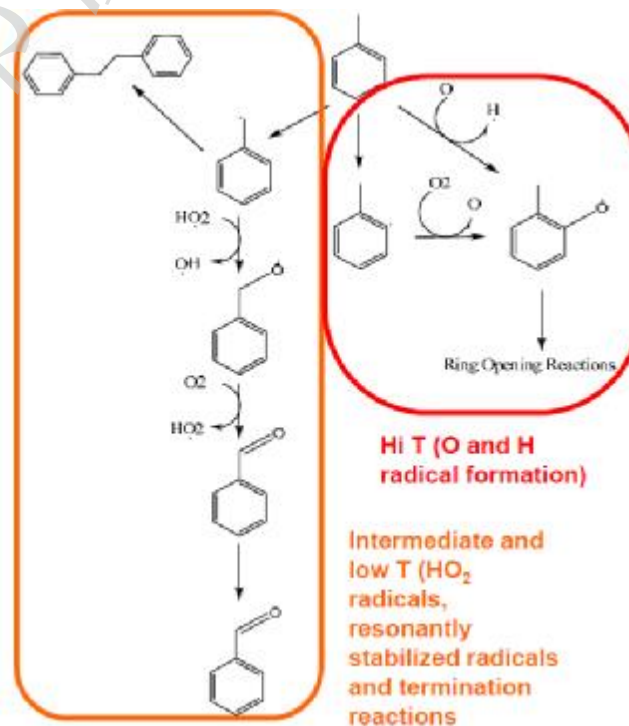


25 atm, $\Phi=1$, constant volume

The 3D structure of cycloalkane makes H atom unavailable for the internal abstraction reactions, reducing the reactivity of the fuel at low temperature

Aromatics

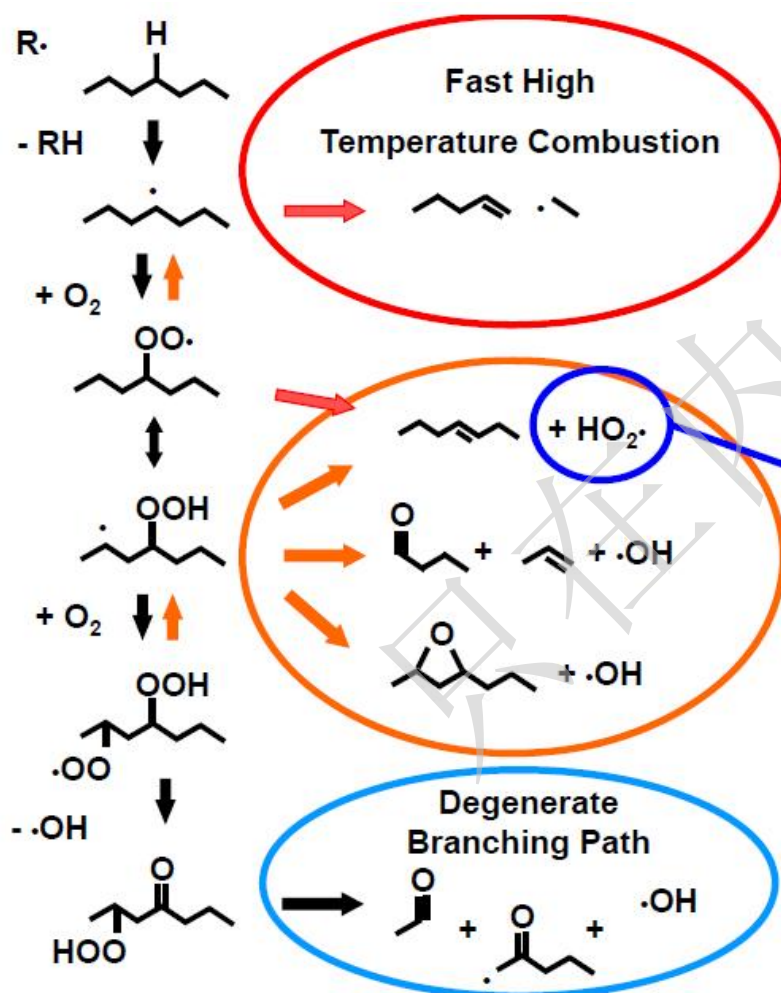
Aromatics do not have Low-T degenerate branching pathway and act as radical scavengers in mixtures (e.g. resonantly stabilized benzyl radical)



4.1. Gasoline Engine Applications

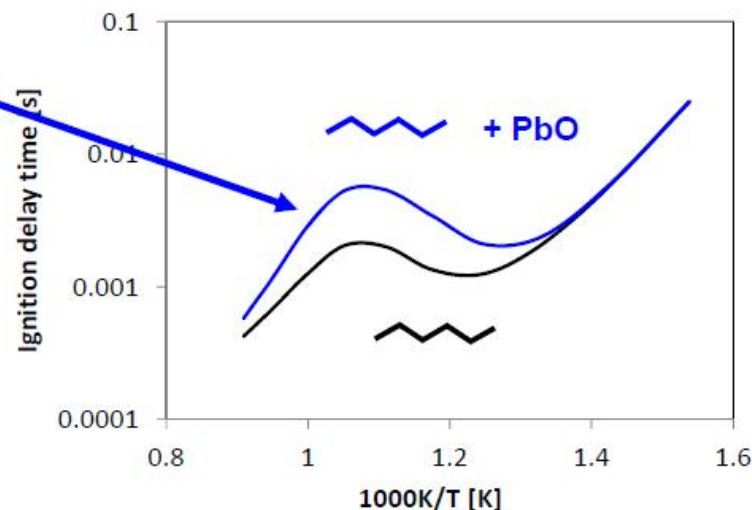
I Gasoline Additives

- Thomas Midgley (1924) demonstrated Tetraethyl Lead to be an effective anti-knock agent (despite its toxicity)



It is believed that TEL forms a "fog" of PbO which scavenges HO_2 radicals and prevents the build up of H_2O_2 inhibiting the transition to high temperature ignition

Pitz & Westbrook, Combustion and Flame 63: 113-133 (1986)

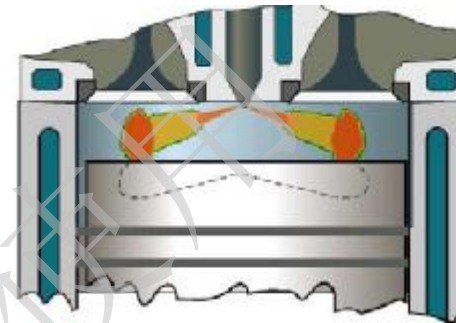


25 atm, $\Phi=1$, constant volume

4.2. Diesel Engine Applications

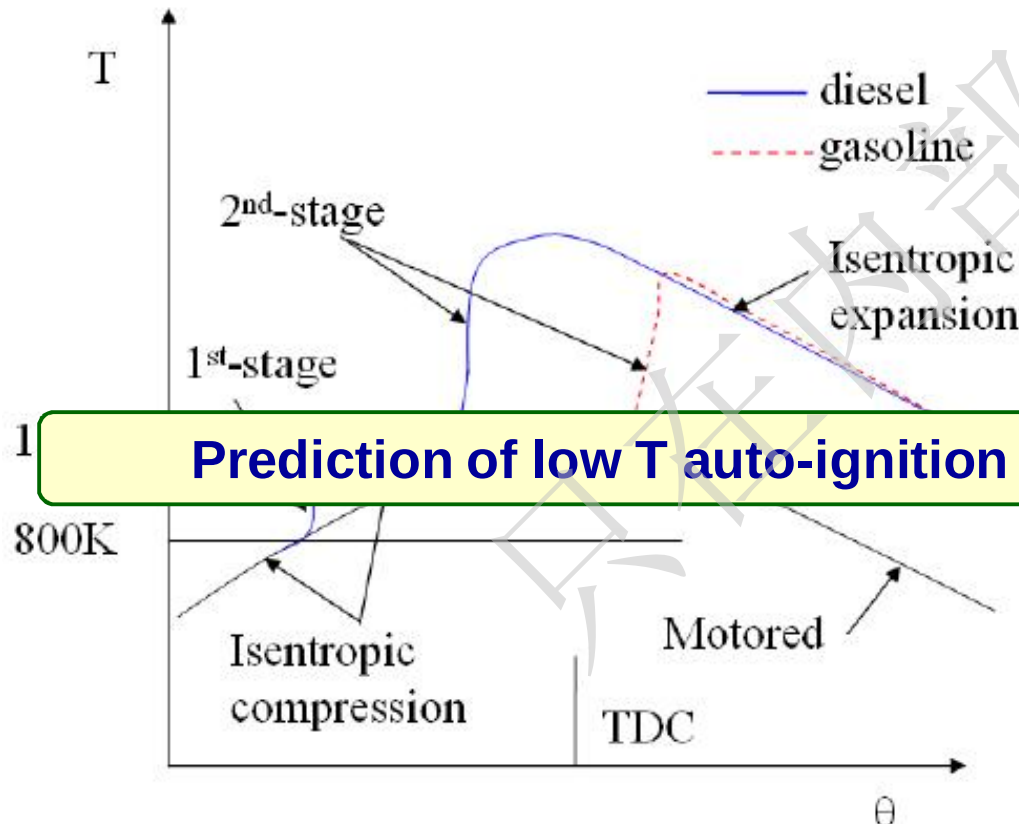
I Diesel Engine Ignition

Ignition is controlled by chemistry

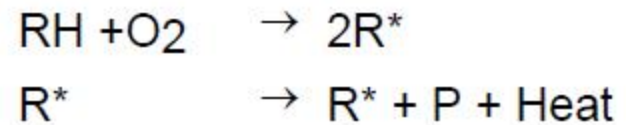


Compression
ignition

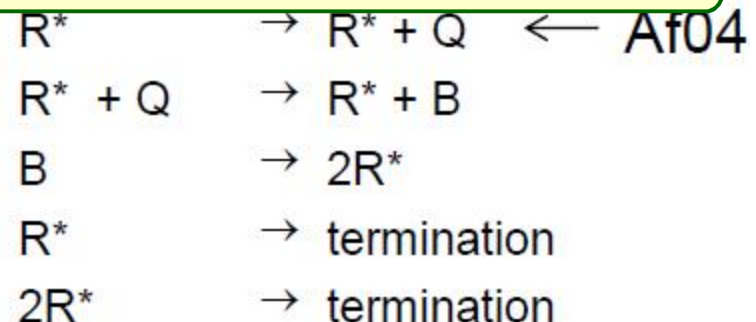
Spray
combustion



Shell Ignition Model

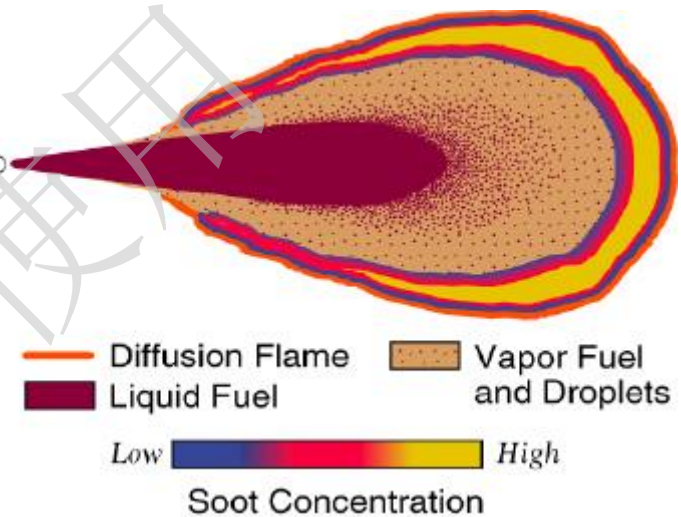
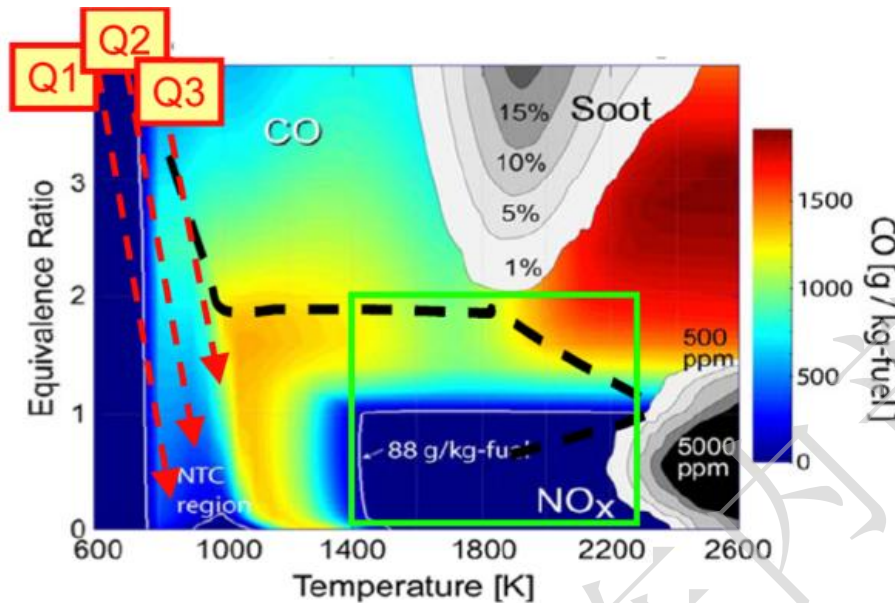


Prediction of low T auto-ignition needs detailed chemistry



4.2. Diesel Engine Applications

I Pollutant Formation in Diesel Engine



- n High compression ratio: high efficiency
- n Rich combustion: soot production
- n High temperature flame front: NO_x production



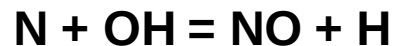
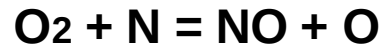
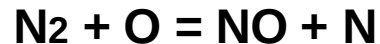
4.2. Diesel Engine Applications

I NOx Formation Mechanism

Formation of NO and the combustion kinetics

Formation mechanism

u Thermal NO

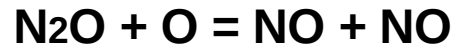


u Prompt NO

Prediction of NOx formation needs detailed chemistry



u N₂O mechanism

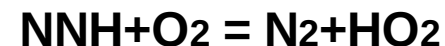
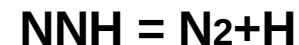
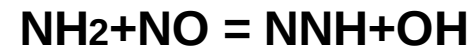


Control methods

u Water injection

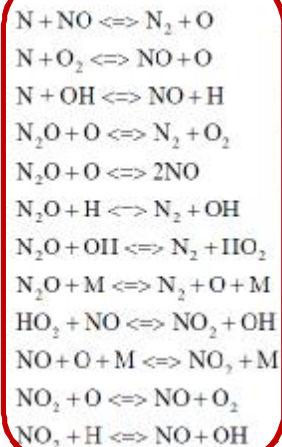
Decrease the temperature

Decrease the concentration of O

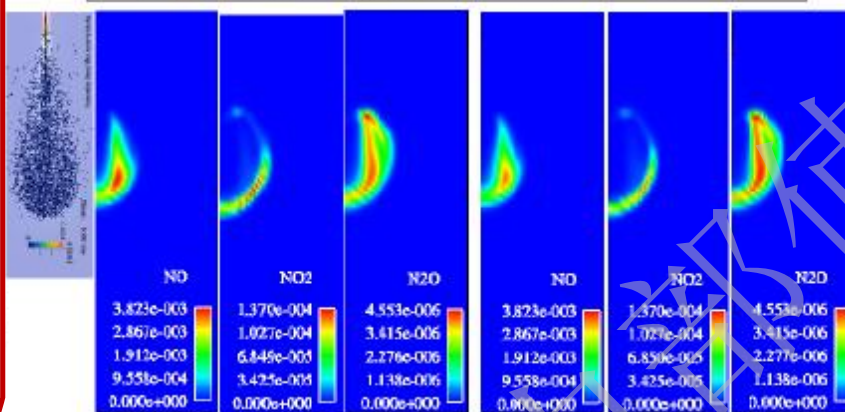


4.2. Diesel Engine Applications

I NOx Prediction in Diesel Engine



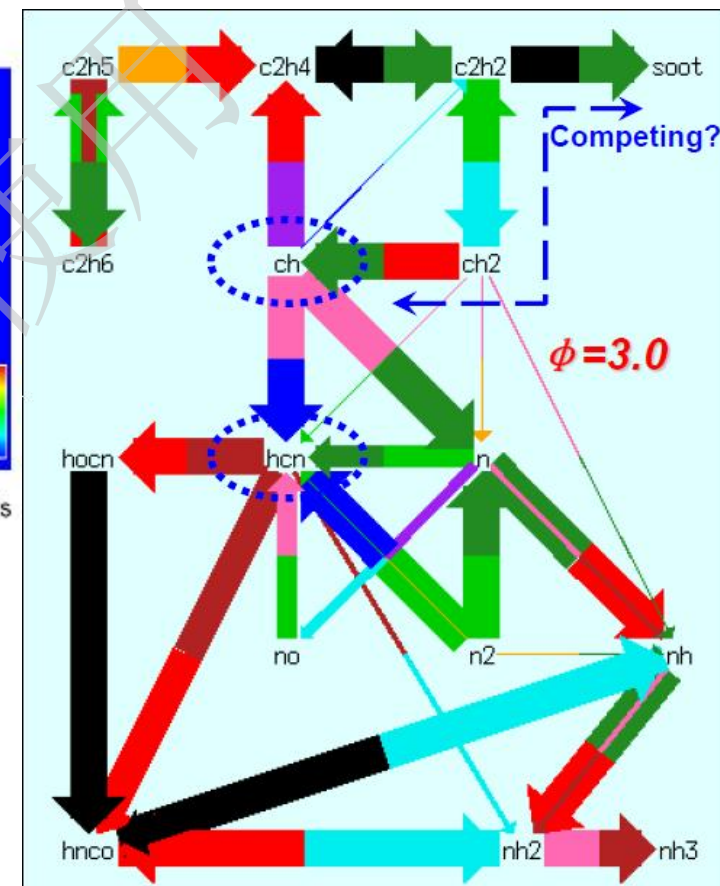
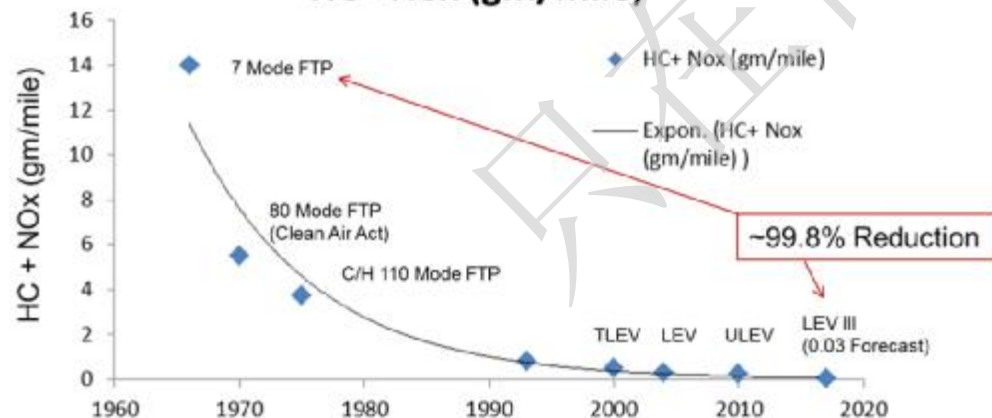
Comparison of NOx predictions (T=900K, P=3.7MPa)



GRI mechanism results

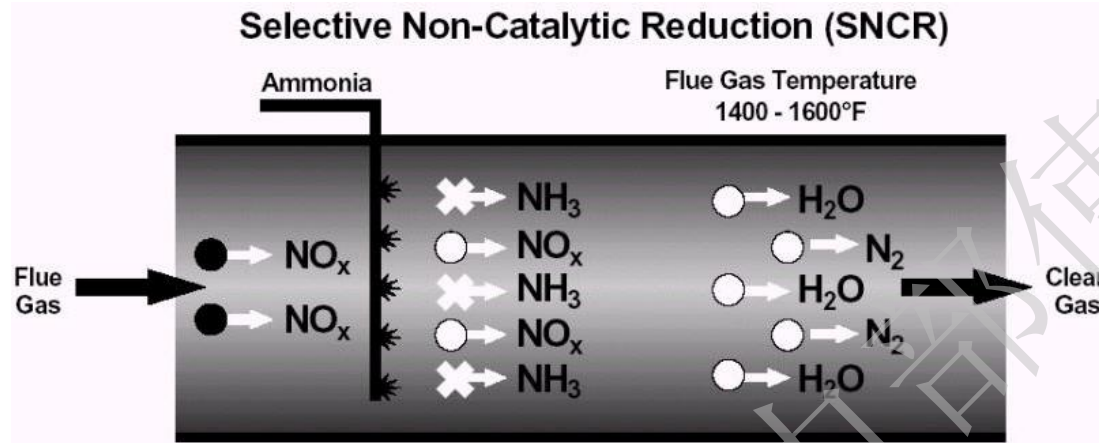
Reduced mechanism results

HC + NOx (gm/mile)

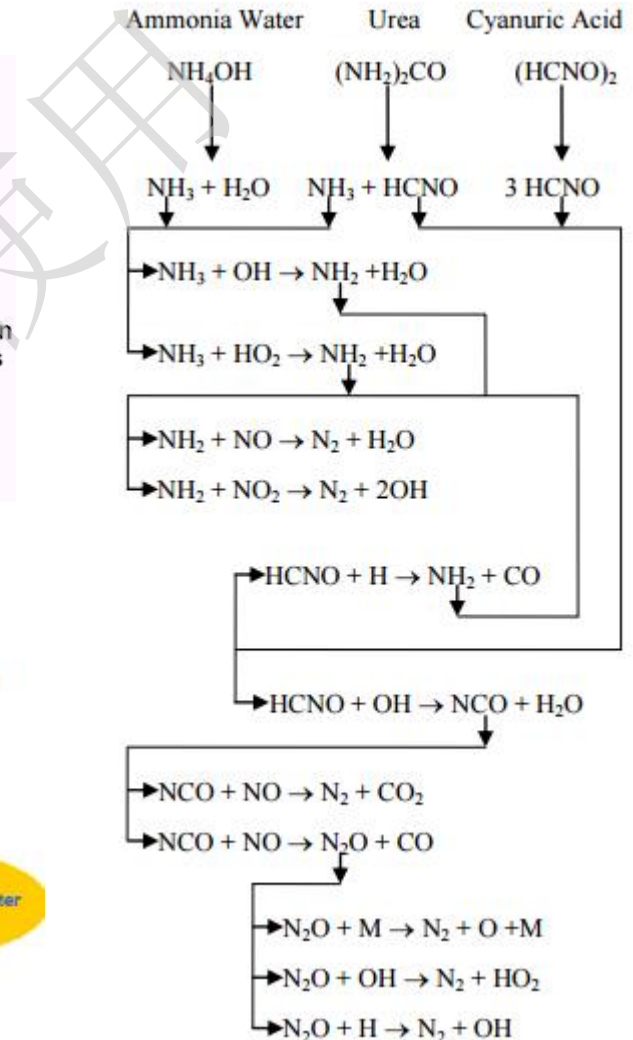
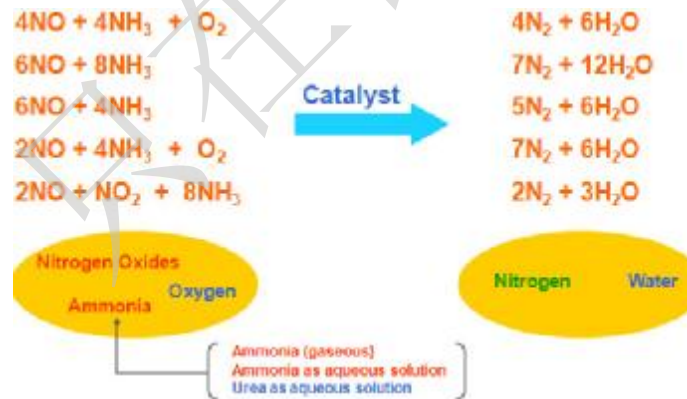
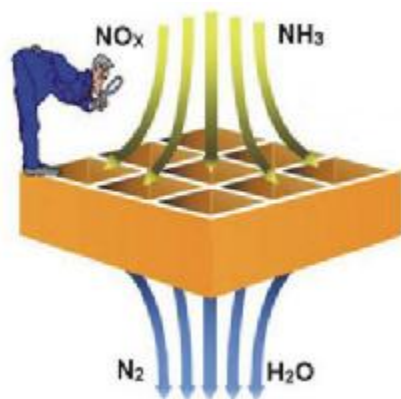


4.2. Diesel Engine Applications

I Control Strategy of NOx Emission



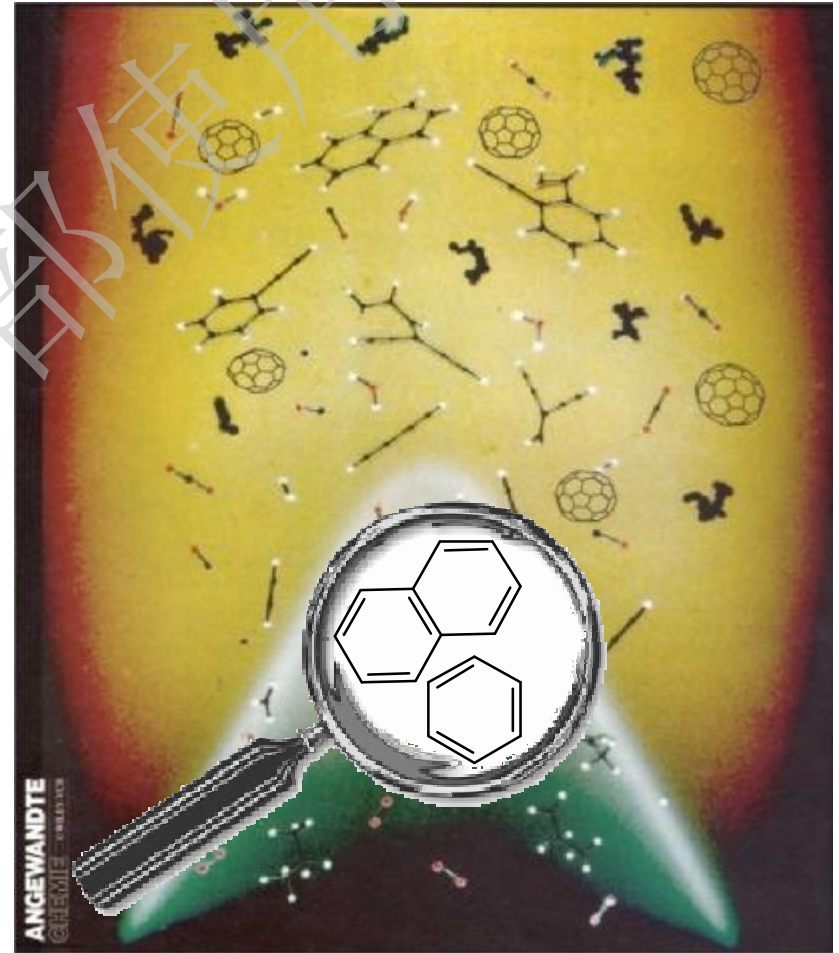
Selective catalytic reduction



4.2. Diesel Engine Applications

I Soot Formation Mechanism

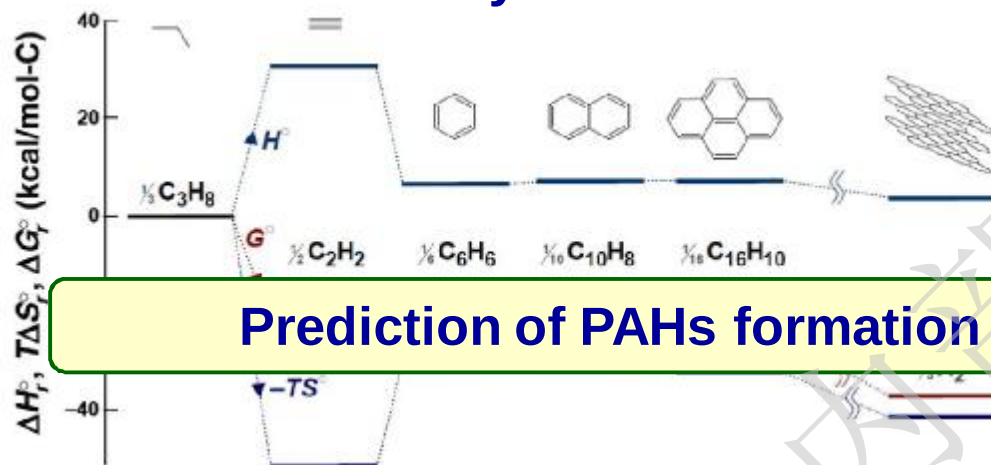
Soot formation in combustion



4.2. Diesel Engine Applications

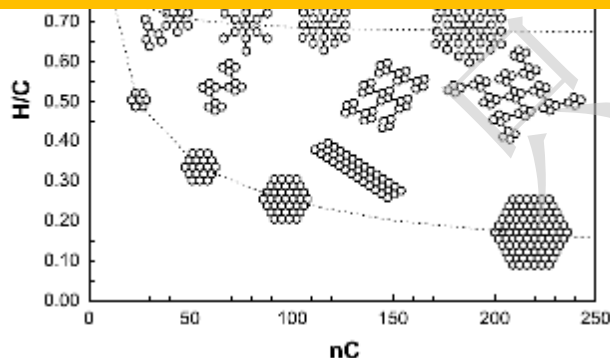
I Soot Formation Mechanism

Thermodynamics

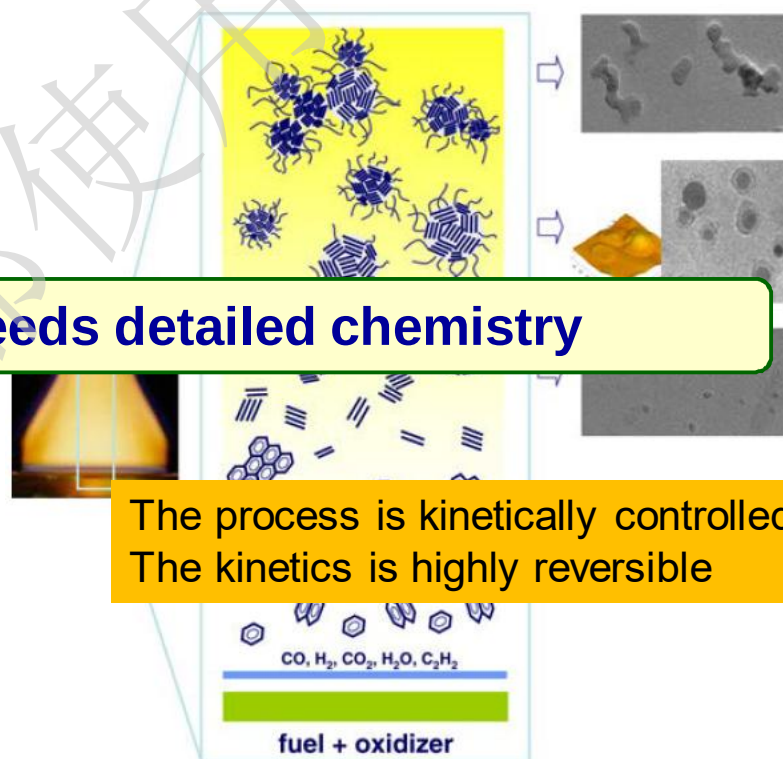


Prediction of PAHs formation needs detailed chemistry

The process of soot formation is entropy-driven
The overall process is slightly endothermic



Kinetics

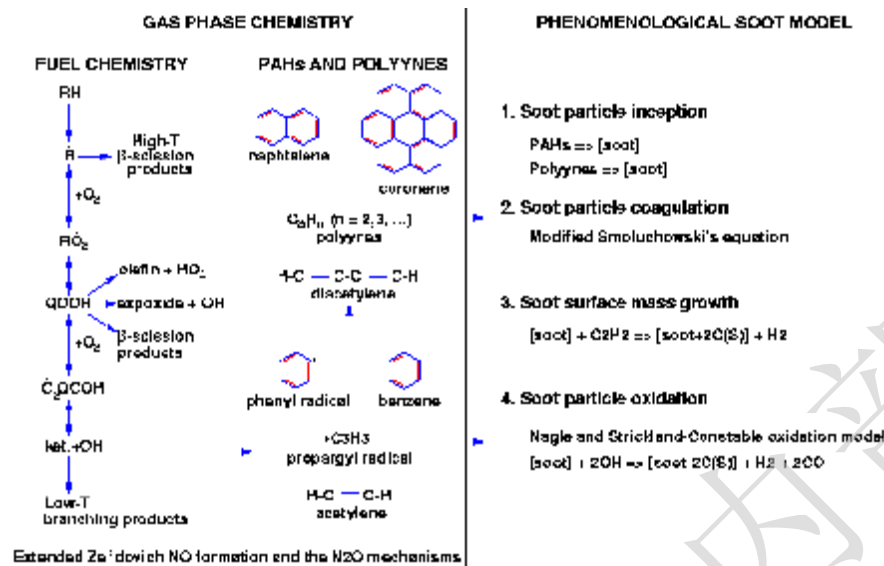


The process is kinetically controlled
The kinetics is highly reversible

1. PAH formation and growth
2. Soot nucleation, inception, surface growth
3. Dehydrogenation, particle coalescence, aggregation
4. Oxidation

4.2. Diesel Engine Applications

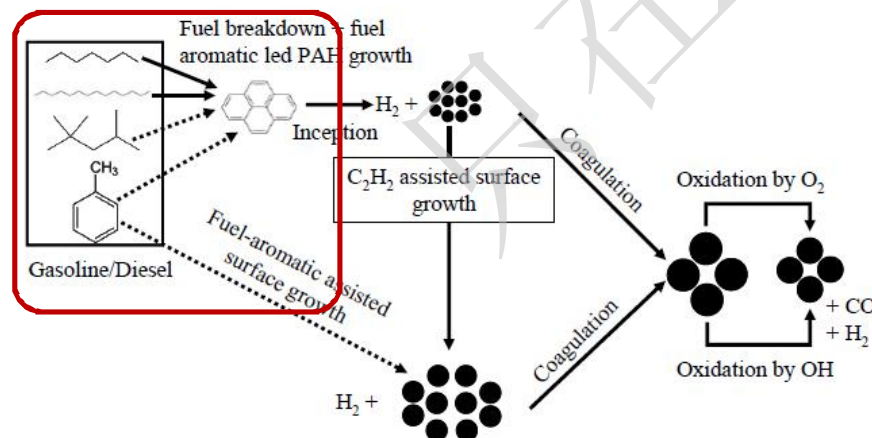
I Soot Prediction in Diesel Engine



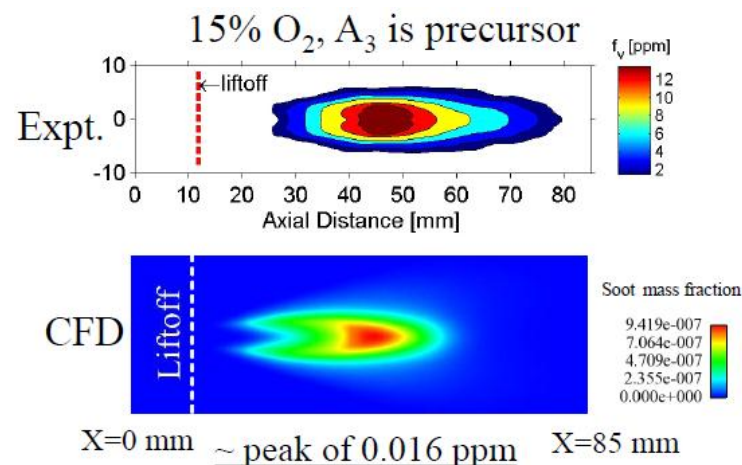
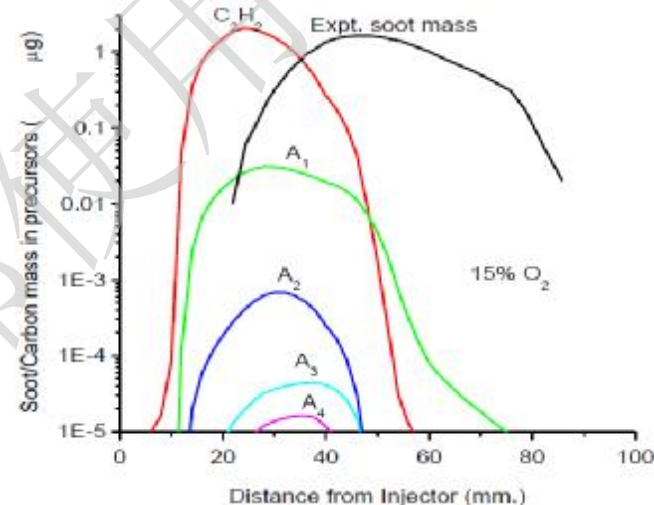
PHENOMENOLOGICAL SOOT MODEL

- Soot particle inception**
PAHs ⇒ [soot]
Polynyne ⇒ [soot]
- Soot particle coagulation**
Modified Smoluchowski's equation
- Soot surface mass growth**
[soot] + C₂H₂ ⇒ [soot + 2C(S)] + H₂
- Soot particle oxidation**
Nagle and Strickland-Constable oxidation model
[soot] + 2OH ⇒ [soot - 2C(S)] + H₂ + 2CO

CO₂, H₂O, SOOT, NO, AND OTHER PRODUCTS



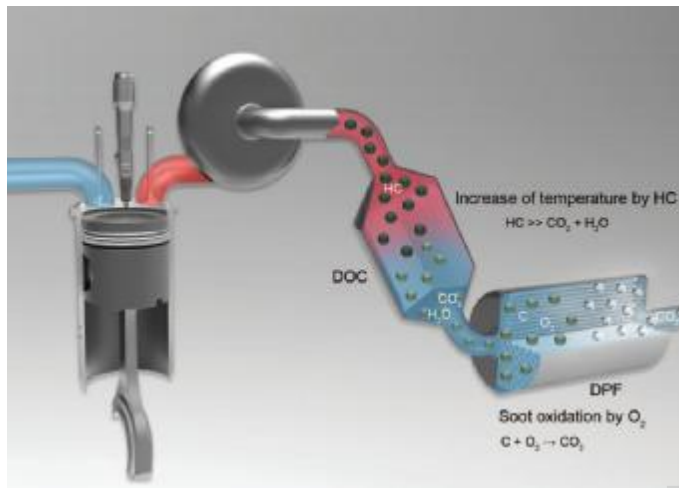
Soot emission



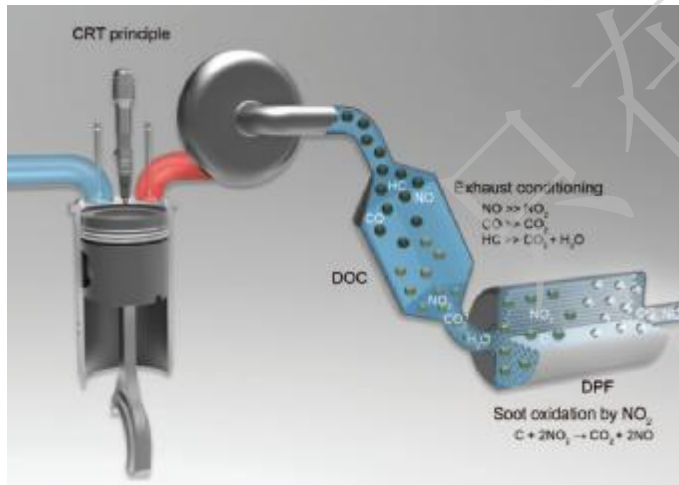
4.2. Diesel Engine Applications

I Control Strategy of Soot Emission

Diesel Particulate Filter

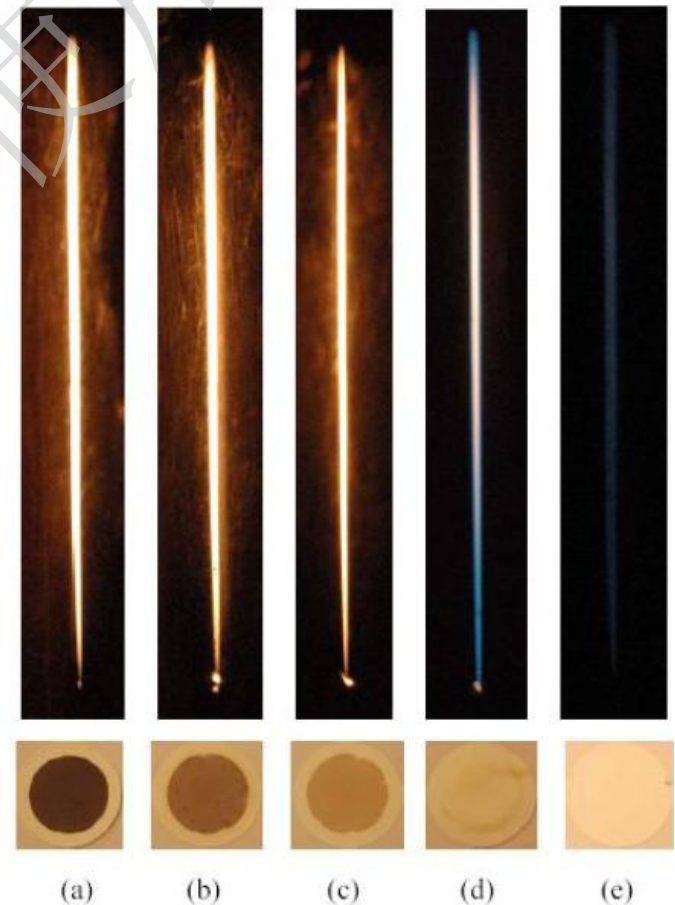


**Soot
catalytic
oxidation**



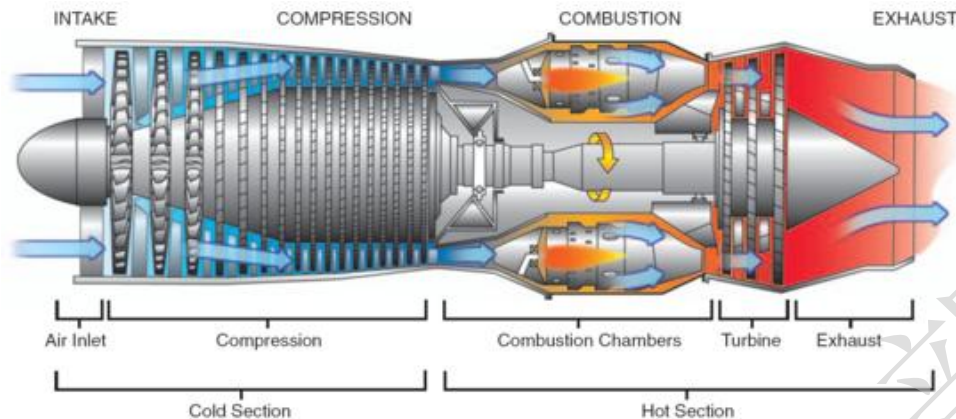
Oxygenated fuels blend

Biodiesel Blending: Soot Reduced



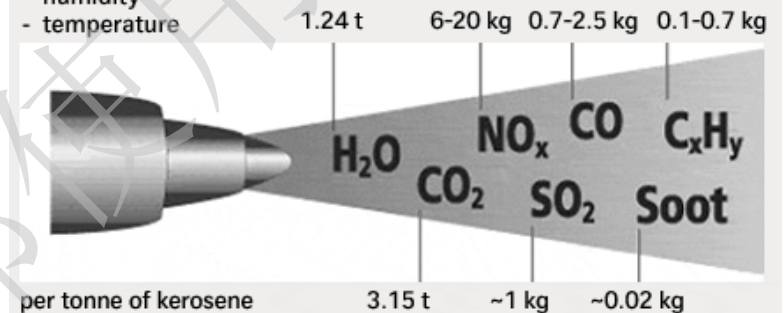
4.3. Jet Engine Applications

Jet engine and gas turbine design

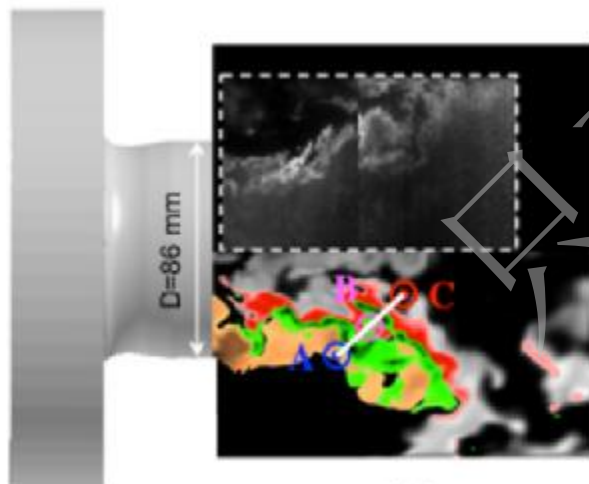


Kerosene combustion by-products depending on

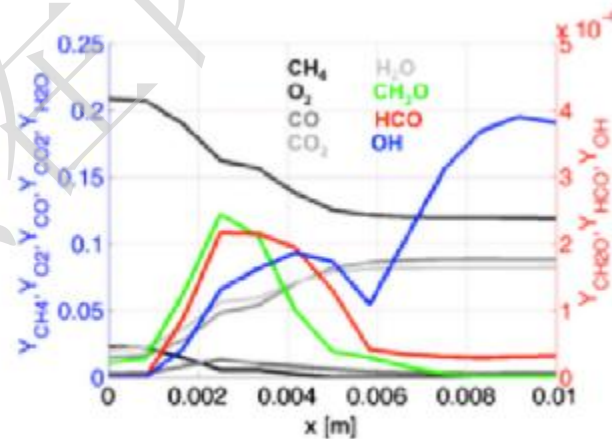
- operating conditions
- altitude
- humidity
- temperature



CFD based design

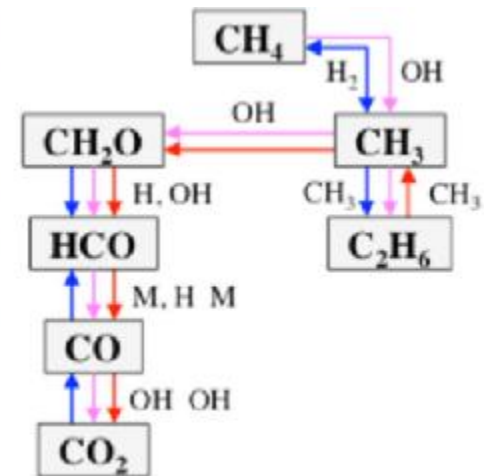


(a)



(b)

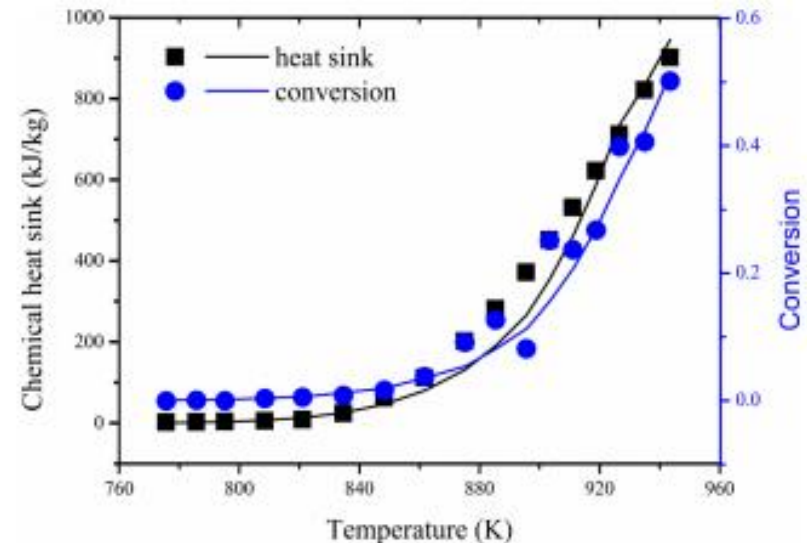
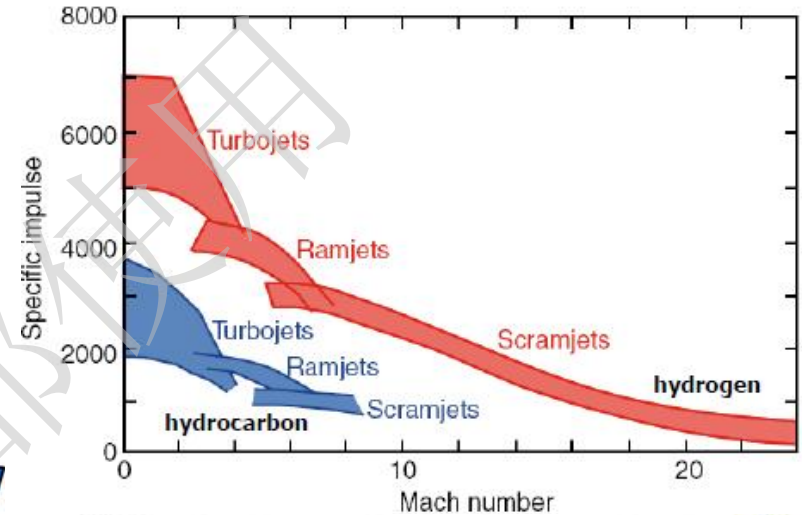
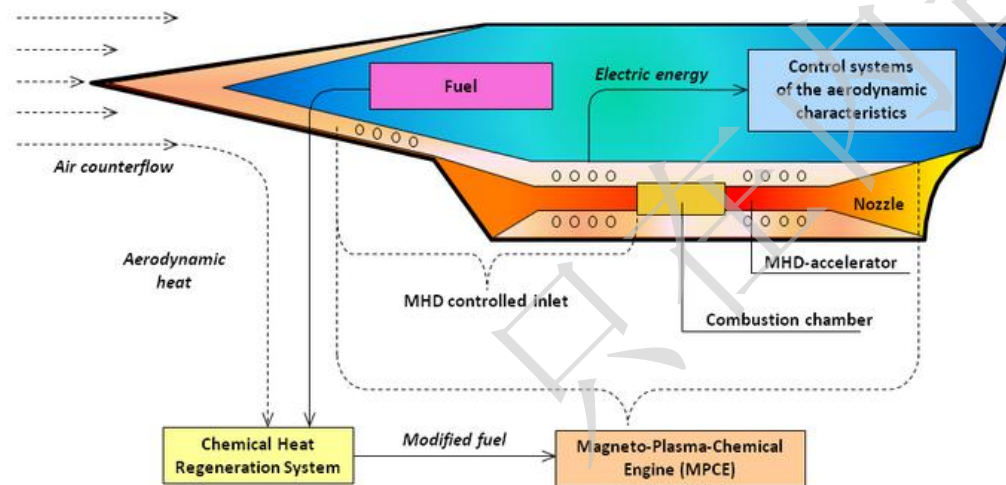
Pollutant formation



(c)

4.3. Jet Engine Applications

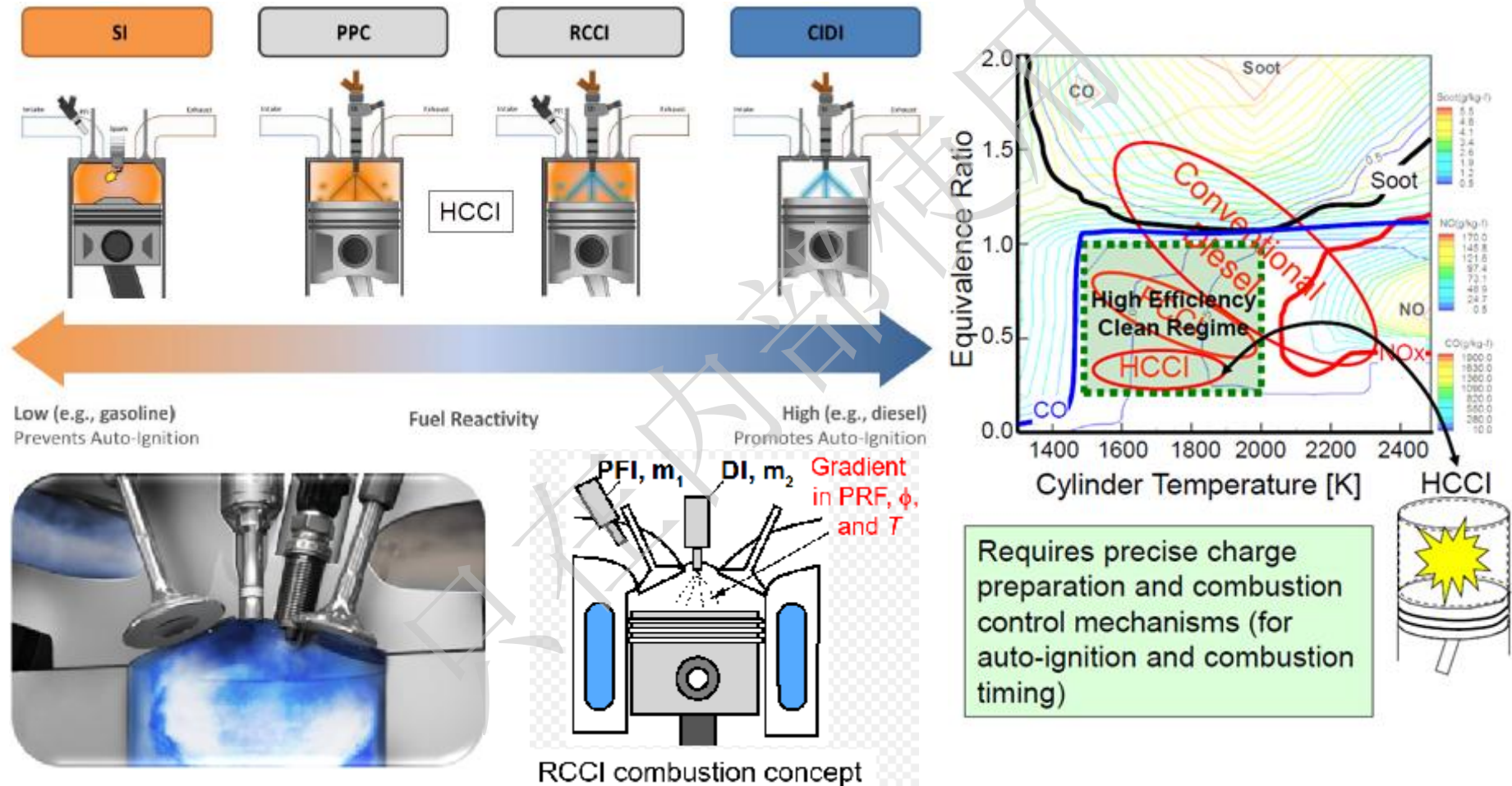
I Hypersonic engine and active cooling



Pyrolysis/catalytic pyrolysis of fuel

4.4. New Technology Applications

I Low Temperature Engine Combustion



n Simultaneous reduction of NOx and soot

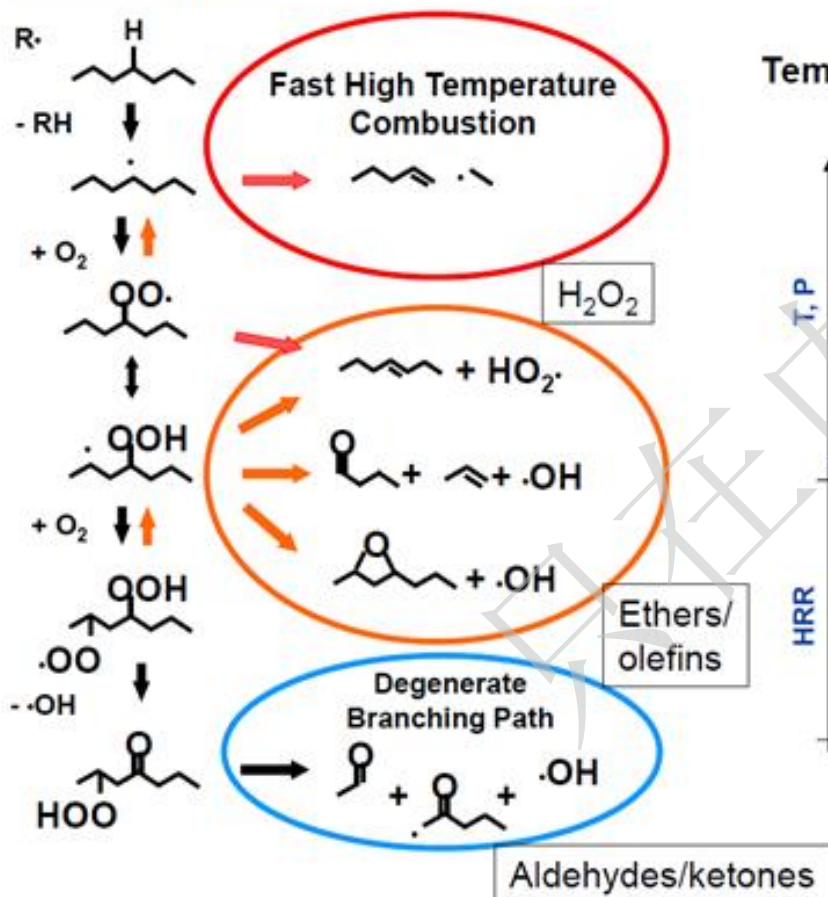
n Increase the formation of CO and HC

4.4. New Technology Applications

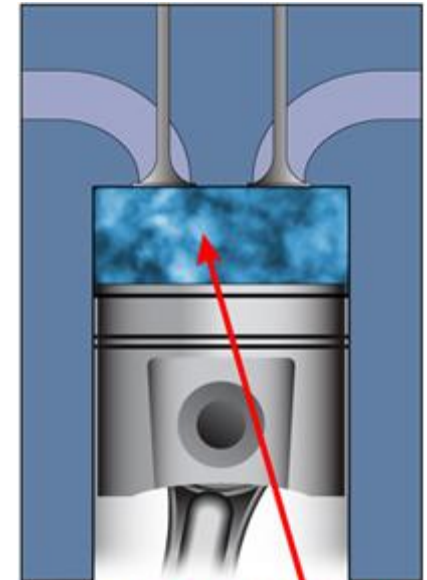
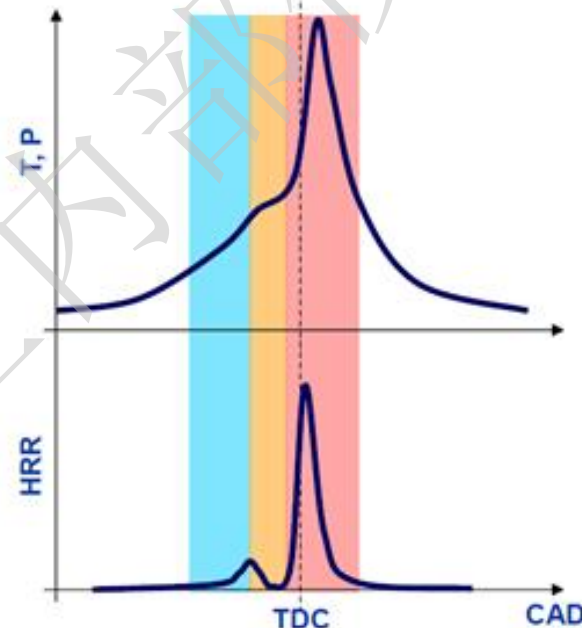
I HCCI Engine

HCCI and auto-ignition chemistry

HCCI combustion kinetics



Typical HCCI Combustion Temperature and Heat Release Rate profiles



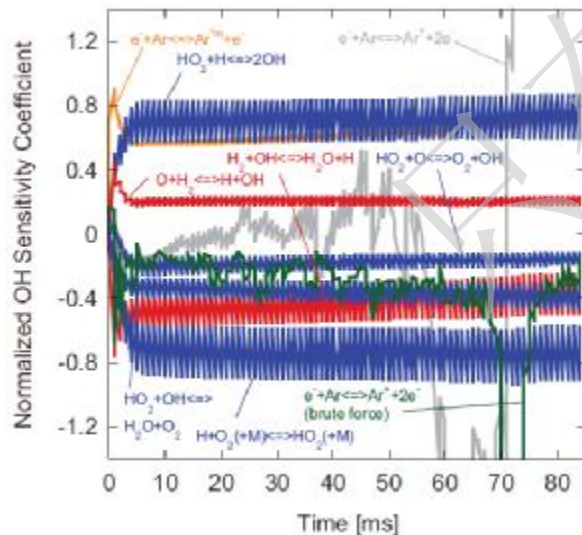
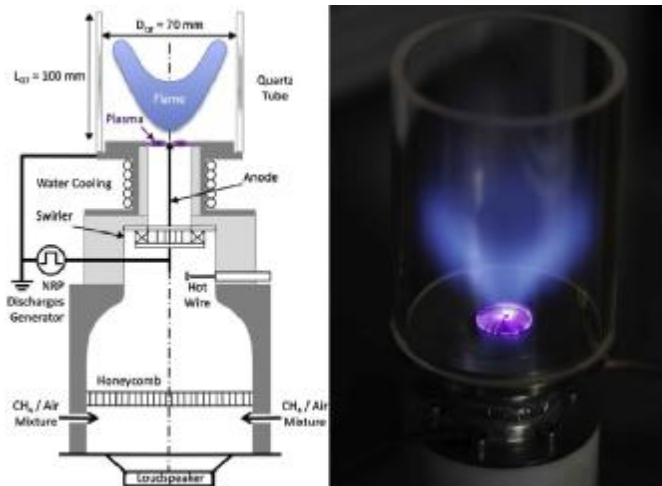
Low-Temperature Combustion:
Ultra-Low Emissions (<1900K)



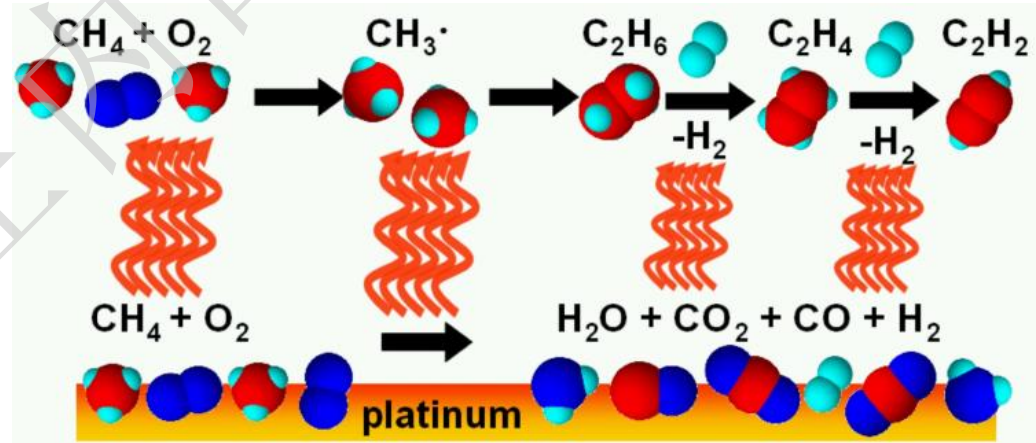
4.4. New Technology Applications

I New combustion technology

Plasma-assisted combustion



Catalytic-assisted combustion



n Chemical looping combustion

n MILD combustion, Oxy-rich.....

Y. G. Ju & W. T. Sun, Prog. Energy Combust. Sci. 48 (2015) 21

M. Reinke et al., Combust. Flame. 141 (2005) 448

5. Challenges & Opportunities

I Experimental Research

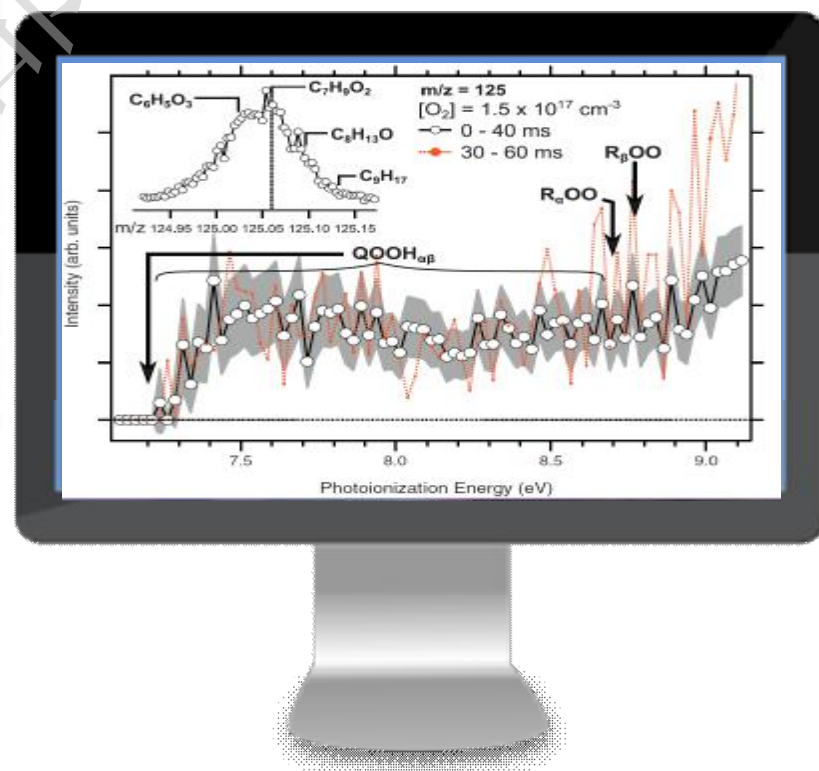
Advanced technology

ö Advanced diagnostic methods

- n Intermediate species information
- n Time-resolved information

ö Advanced experimental methods

- n Elementary reaction rate study
- n Wide pressure and temperature range
- n Ultra-high pressure, high dilution
- n Engine-like combustion processes



5. Challenges & Opportunities

I Experimental Research

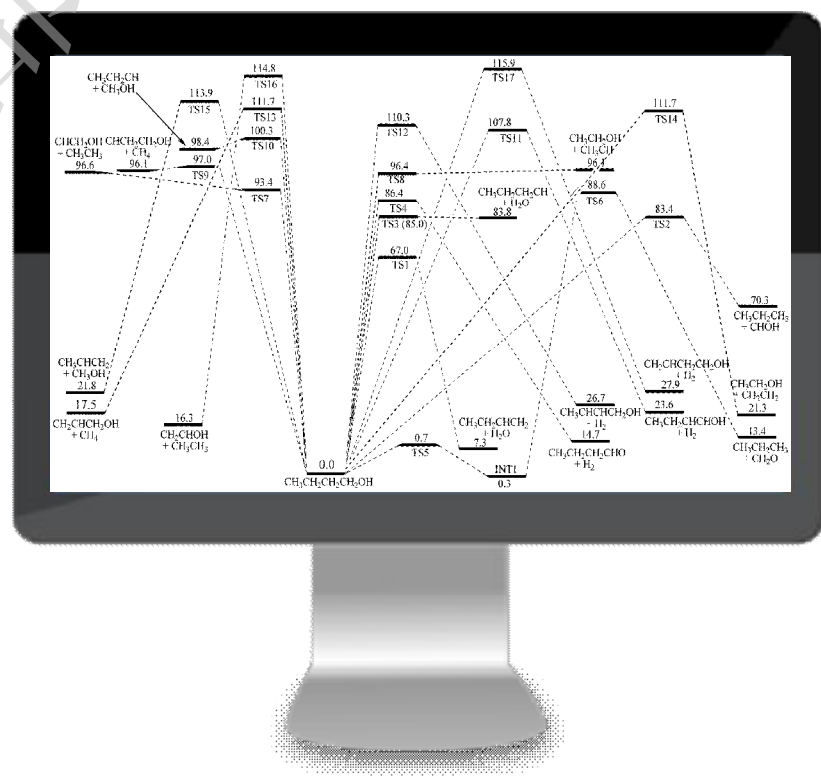
Non-conventional combustion processes

- ö Microgravity combustion
- ö Catalytic-assisted combustion
- ö Plasma-assisted combustion
- ö Chemical looping combustion
- ö MILD combustion
- ö Low temperature combustion
- ö Oxygen enriched combustion
- ö Porous media combustion
- ö



I Theoretical Research

- ö Improve calculating accuracy
- ö Quantum chemistry calculation methods of large molecules
- ö Calculation of thermodynamic and transport properties



5. Challenges & Opportunities

I Theoretical Research

Kinetic theory

ö Low temperature reaction

- n Accuracy of the barrier height calculation
- n Quantum tunnelling effect

ö Anharmonicity of vibration

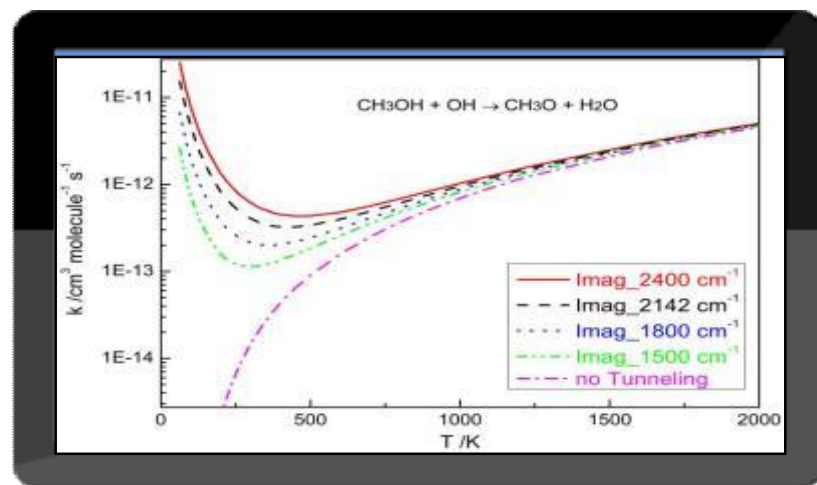
- n Quantum chemistry calculation of anharmonicity
- n Rotational-vibrational coupling

ö $1D(E) \rightarrow 2D(E, J)$

- n Collision energy transfer

$$R(E, E') \rightarrow R(E, E'; J, J')$$

- n ME (E) $\rightarrow$ (E, J)

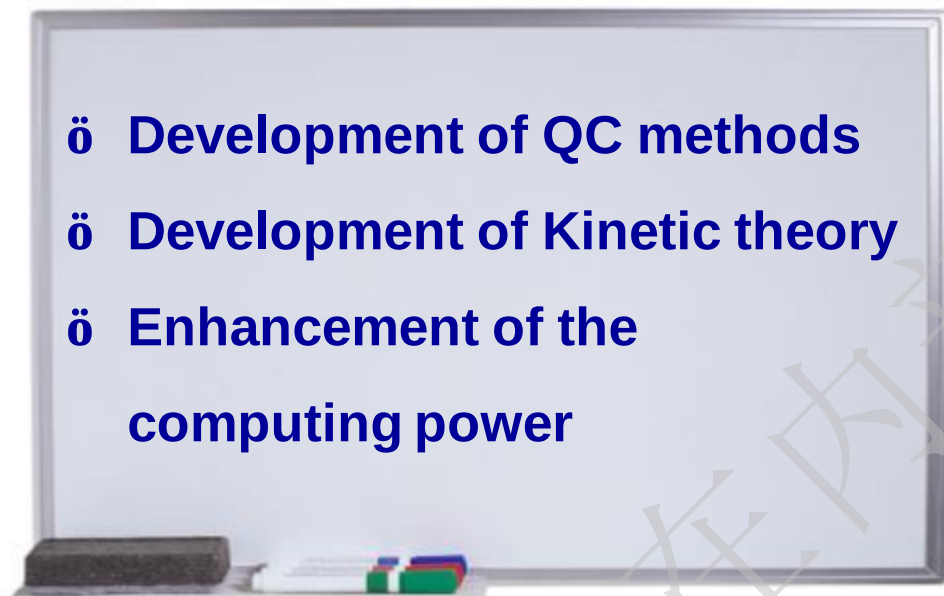


5. Challenges & Opportunities

I Theoretical Research

More accurate rate parameters for combustion

- ö Development of QC methods
- ö Development of Kinetic theory
- ö Enhancement of the computing power

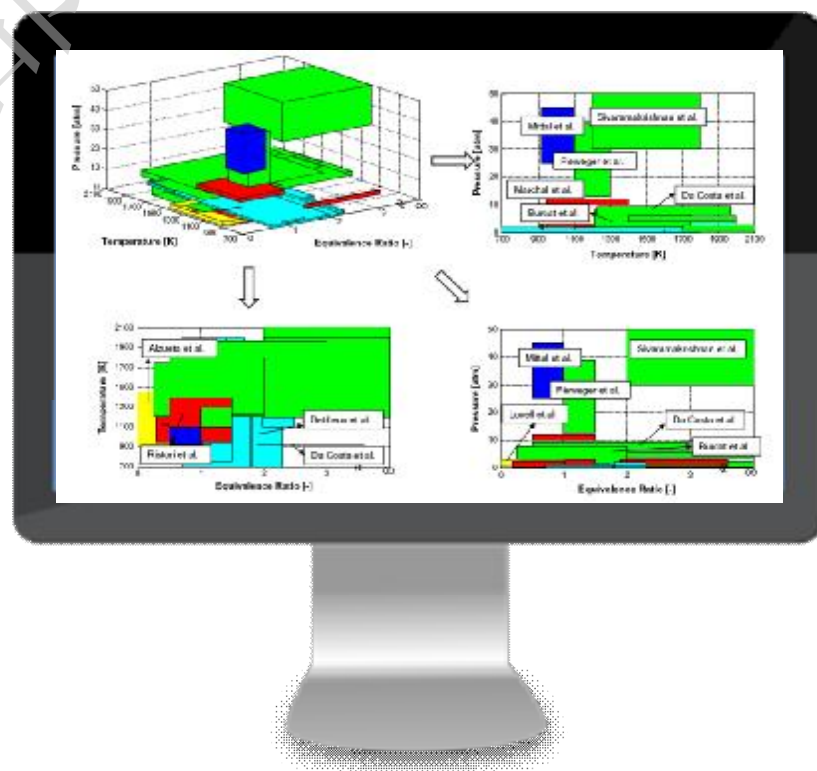


5. Challenges & Opportunities

I Kinetic Modeling Research

Chemical rigorous and comprehensiveness

- ö Accurate rate parameters
- ö Accurate thermodynamic and transport properties
- ö Reasonability of the reaction pathways
- ö Unknown reaction pathways
- ö Comprehensive validation

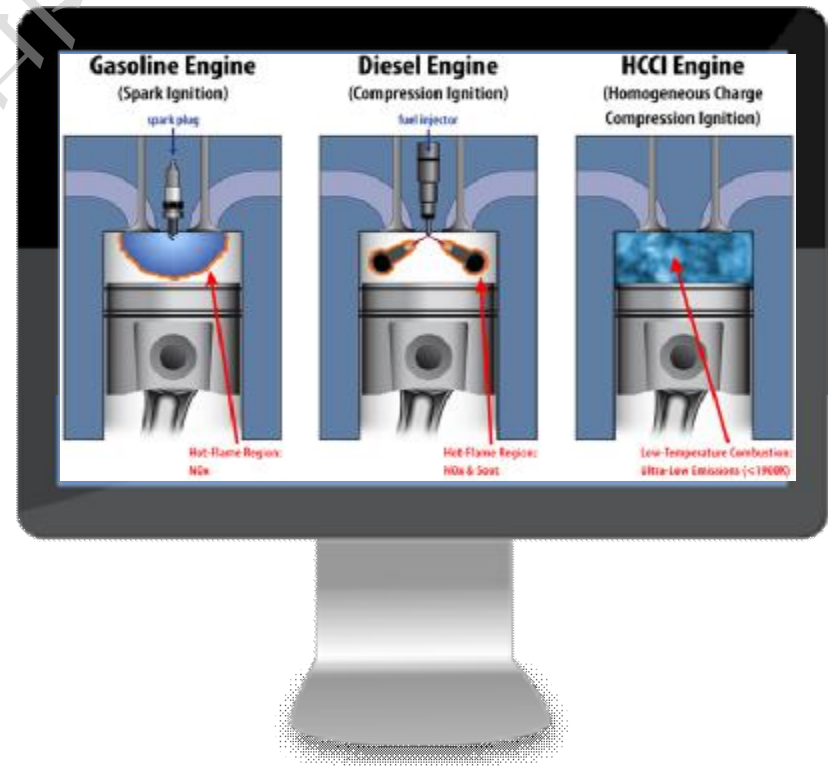


5. Challenges & Opportunities

I Kinetic Modeling Research

Kinetic modeling for engine design

- ö High pressure combustion kinetics
- ö Low temperature combustion kinetics
- ö Using for engine performance prediction and emission prediction

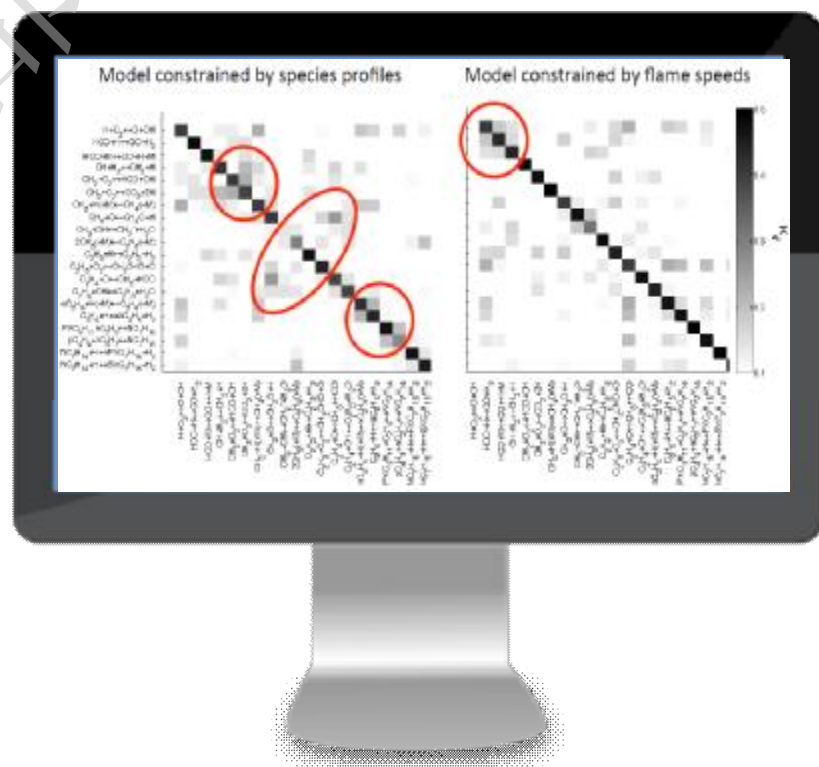


5. Challenges & Opportunities

I Kinetic Modeling Research

Uncertainty qualification/mechanism optimization

- ö Uncertainties of rate parameters, physical model, experiments
- ö Uncertainty propagation
- ö Constraint of the model uncertainty by experimental data
- ö Mechanism optimization



5. Challenges & Opportunities

I Kinetic Modeling Research

Complex reaction analysis/stiffness removal

ö Dynamic Chemical Stiffness Removal

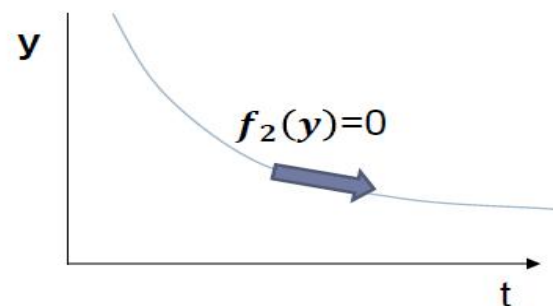
n Timescale analysis

n Separate fast and slow chemical processes using CSP & ILDM

n Chemical stiffness be removed: dynamic QSSA, dynamic PEA

ö Dynamic adaptive chemistry simulation

$$\frac{dy}{dt} = \mathbf{g}(y) = \mathbf{f}_1(y) + \epsilon^{-1} \mathbf{f}_2(y)$$

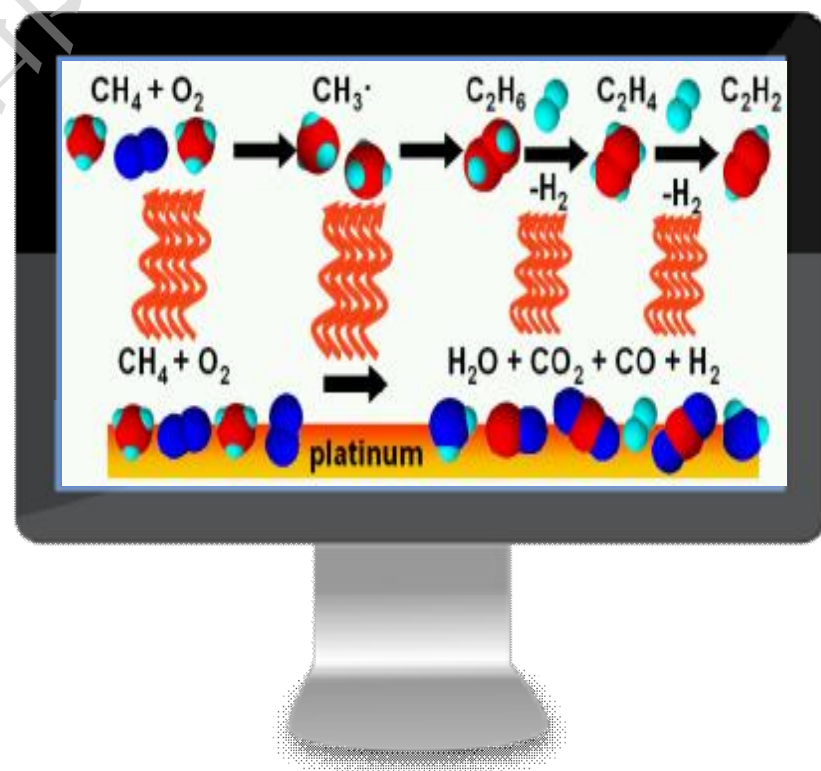


5. Challenges & Opportunities

I Kinetic Modeling Research

Kinetics for more complex combustion systems

- ö Plasma-assisted combustion
- ö Heterogeneous catalytic combustion
- ö CVD mechanism
- ö NO_x/soot formation mechanism
- ö Inorganic chemistry

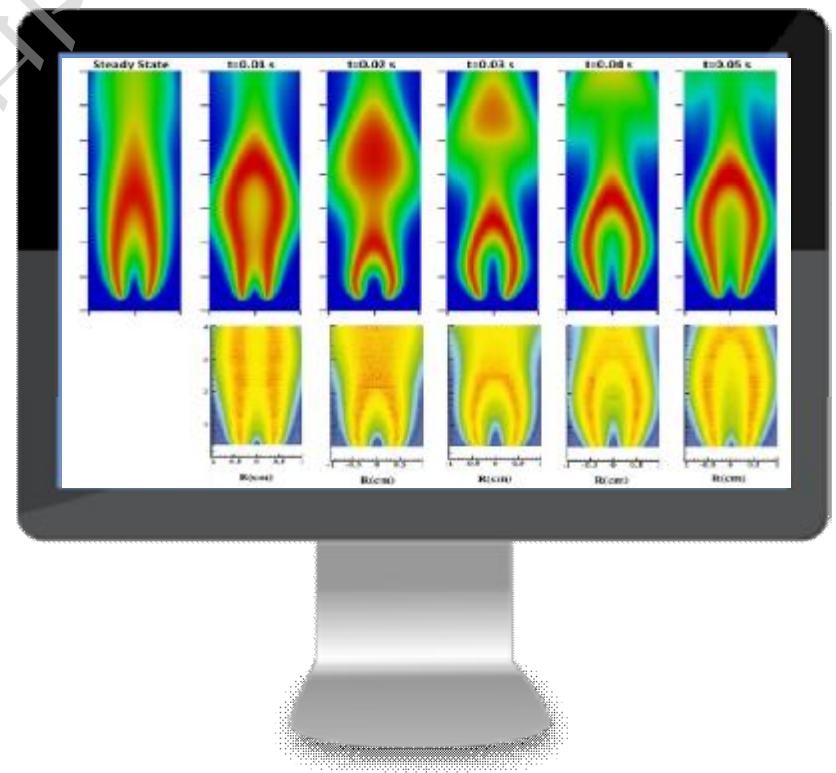


5. Challenges & Opportunities

I Kinetic Modeling Research

Complex reacting flow simulation

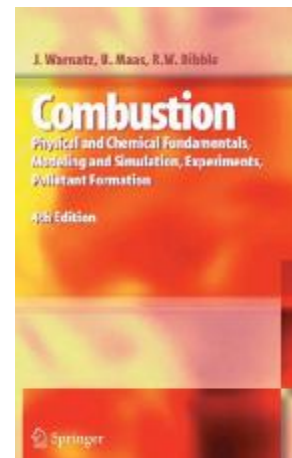
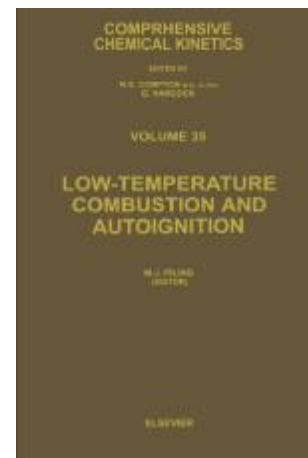
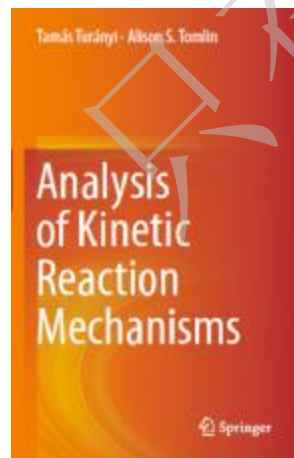
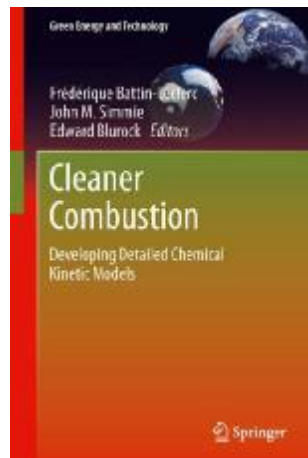
- ö 2D even 3D reacting flow system simulation using detailed chemistry
- ö Turbulence-chemistry interaction



6. Recommended Literature

I Books

1. F. Battin-Leclerc, J. M. Simmie, E. Blurock ed., Cleaner Combustion: Developing Detailed Chemical Kinetic Models, Springer-Verlag London (2013).
2. T. Turányi, A. S. Tomlin, Analysis of Kinetic Reaction Mechanisms, Springer (2014).
3. I. Glassman, R. A. Yetter, Combustion (4th Edition), Academic Press (2008).
4. M. J. Pilling ed., Comprehensive Chemical Kinetics (Volume 35): Low-Temperature Combustion and Autoignition, Elsevier Science Ltd (1997).
5. J. Warnatz, U. Maas, R. W. Dibble, Combustion: Physical and Chemical Fundamentals, Modeling and Simulation, Experiments, Pollutant Formation (4th Edition), Springer-Verlag Berlin Heidelberg (2006)



6. Recommended Literature

I Review Papers

1. F. N. Egolfopoulos, N. Hansen, Y. Ju, K. Kohse-Höinghaus, C. K. Law, F. Qi, Advances and challenges in laminar flame experiments and implications for combustion chemistry. *Progress in Energy and Combustion Science* 43 (2014) 36-67. *For students who want to know more about the experiments for combustion chemistry studies.*
2. (a) J. M. Simmie. Detailed chemical kinetic models for the combustion of hydrocarbon fuels. *Progress in Energy and Combustion Science* 29 (2003) 599–634. (b) H. J. Curran. Developing detailed chemical kinetic mechanisms for fuel combustion. *Proceedings of the Combustion Institute* 37 (2019) 57-81. *For students who want to know more about the detailed combustion kinetic model.*
3. J. Zádor, C. A. Taatjes, R. X. Fernandes. Kinetics of elementary reactions in low-temperature autoignition chemistry. *Progress in Energy and Combustion Science* 37 (2011) 371-421. *For students who want to know more about the computational chemistry methods.*

Have Fun!

Companion in Learning and Romance



Provider of Light, Warmth and Joy



Power for Mobility



Furnace for Industry, Pleasure, and Food



Combustion

Agent of Destruction



Origin of Environment Problems

